ARTICLE

Learning across diseases and outcomes for risk prediction in lymphoid cancers

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Abstract

Prognostic tools are built for one disease and one outcome, so each learns only from the events of its own patients, which for many clinical questions are few. Learning across related diseases and outcomes could recover the signal this leaves unused, but how much comes from other diseases, how much from other outcomes, and whether a model can learn what to borrow, and from whom, is largely unknown. We studied this in the lymphoid cancers, many small diseases with many important but individually rare complications, still served by simple prognostic indices. In 37,531 Danish adults starting first-line treatment for one of 24 lymphoid cancers, including plasma-cell neoplasms, sequence models read each patient's records before treatment and predicted 69 outcomes over 30 days to two years. Viewing the tasks as a grid of diseases by outcomes, we compared otherwise identical models that learned from a single cell, from one outcome across all diseases, from all outcomes of one disease, or from the whole grid; a router that learns, for each patient and outcome, how much to draw on each disease; and encoders pretrained on the records of the entire Danish population. More than **[1,750]** sequence and **[20,000]** tabular models were evaluated on patients treated from 2023, against gradient-boosted trees trained and tuned for every task and against the prognostic indices in clinical use. OPERA, **[the configuration: scope, routing and encoder]**, differed from the tuned trees by **[X]** in AUC over all cells and horizons **[(claim label)]** and had the higher AUC in **[X of Y]** disease–outcome cells, in place of **[N]** separately trained models. Learning from other diseases changed AUC by **[X]**, from other outcomes by **[Y]**, learning what to borrow by **[Z]**, and population pretraining by **[W] [, which matched the cohort-pretrained model with X% of the labelled patients]**; **[the gains were largest where a prediction had least data of its own]**. **[Own disease corrections helped where a prediction had many training events and borrowing where it had few, the pattern of partial pooling, and switching diseases off in the fitted model showed that it borrowed broadly rather than from a few related diseases.]** Ten of the 24 diseases have no validated prognostic index, and the index was missing for 58% of test patients; OPERA predicted every outcome for every patient **[and, where an index was recorded, placed X more per 1,000 patients at low risk with no higher observed risk]**. **[Concrete attribution finding, e.g. which pre-treatment findings drove infection risk and which drove treatment failure.] [If the results hold: prognosis in lymphoid cancers need not be built one disease and one outcome at a time, and a model pretrained on the whole population opens the same route to every specialty that shares the national registers.]**

Key findings [(confirm each against the locked test-set analysis)]

- One model predicted 69 outcomes in 24 lymphoid cancers, differed from per-task XGBoost by **[X]** in AUC **[(claim label)]** and had the higher AUC in **[X of Y]** disease–outcome cells, covering the tasks of **[N]** separately trained models.
- **[What learning across diseases, learning across outcomes and learning what to borrow each added, with their claim labels, and where the gains were largest.]**
- **[The routed model behaved like partial pooling: corrections for a patient's own disease helped where a prediction had many training events, borrowing from other diseases where it had few, and it borrowed broadly rather than from a few related diseases.]**
- In DLBCL, per 1,000 patients, it flagged **[N]** more of the patients who died within two years at the specificity of the IPI high-risk group, and, at a cutoff matched to the IPI low-risk group's observed risk, placed **[M]** more at low risk.
- **[National pretraining improved specialised prediction directly / after adaptation to the cohort, and reached the full-data cohort-pretrained model within the prespecified margin using X% of its labelled training patients.]**
- **[It relied on findings clinicians recognise, read differently for each complication, and can show which episodes of care each patient's risks rest on.]**
- All results are from one evaluation of patients treated from 2023, after every modelling decision had been fixed and while first-line treatment was changing.

Introduction

Medicine has learned to tell diseases apart. Classifications grow finer with every new entity recognised [1], and clinical research has, for good reasons, concentrated on what separates one diagnosis from the next. Prognostic tools have followed suit: each is built for one disease and one outcome, apart from the neighbouring diseases and related outcomes that could inform it. The price is paid in data. Every question learns only from the events of its own patients, and those are few even where patients are many: most outcomes are uncommon, events limit what any model can learn [2], and flexible models need many more of them than regression [3]. In clinical prediction as elsewhere, performance grows with data [4], yet a model built for one disease and one outcome leaves most of the available information unused.

Diseases and outcomes are not independent. Patients with different diseases share physiology, treatments, complications and patterns of care, and the outcomes of one patient are shaped by the same frailty, prior illness and course of treatment. Much of reading a record does not depend on the disease at all: an infection, a decline in kidney function or an emergency admission is recorded and handled in much the same way whatever the cancer behind it. What one task learns can therefore inform another, so that each task needs to learn only what is particular to it [5, 6]. This is the premise of multi-task learning and of pretraining.

The idea is not new. Multi-task learning trains related predictions together [7], meta-learning approaches such as MetaPred learn from related clinical tasks how to predict a new one with few patients [8–10], and in oncology a mortality model shared across eleven cancer types improved when it was then adapted to each type [11]. Closest to haematology, a transformer reading routine laboratory trajectories predicted 162 treatment-associated complications in multiple myeloma and ovarian cancer, with predictions that generalised across both [12]. But sharing does not always help. Multi-task studies of health records report uneven benefits and negative transfer [13, 14], and joint learning has helped rare phenotypes while hurting common ones [15]. Ideally, a model would not borrow indiscriminately but learn what to borrow, and from whom, weighing each related disease and outcome by how much it helps the prediction at hand, much as partial pooling weighs each estimate by its evidence [16]. A prediction of infection in a patient with chronic lymphocytic leukaemia should then draw on what was learned about infection in multiple myeloma, where the immune deficiency is similar, and set aside what was learned about relapse

in aggressive lymphoma. Mechanisms for selective sharing exist, from automated grouping of prediction tasks [17] to mixtures of experts that route patients to specialised subnetworks [18, 19], but how much they add across the diseases and outcomes of a clinical specialty has not been tested.

Over the past few years, the wider field of clinical prediction has turned towards exactly this philosophy: large models trained on the records of many patients, diseases and outcomes, which could replace the many models built one task at a time. Self-supervised models such as BEHRT, Med-BERT, CORE-BEHRT, MOTOR and SurvivEHR learn from recorded histories before any outcome is labelled [20–24], and Delphi-2M, life2vec and pan-cancer models learn across diseases and whole populations [25–29], as do generative models of medical events trained on a whole health system's records [30]; promptable models can be asked about any outcome [31]. Trained on the claims of more than 200 million people, one such model recently outperformed disease-specific gradient boosting across more than a thousand disease-onset tasks, with the largest gains for rare diseases [32]. One model for many questions would be easier to build, validate and maintain, and could serve questions too small for a model of their own. Yet such models are rarely evaluated on outcomes that matter for clinical decisions [33], against the tools in clinical use [34], or on how few labels they need [35]. It remains largely unknown how much is gained by learning across diseases, how much by learning across outcomes, and whether a model can learn how much to draw on each.

The lymphoid cancers are a fitting place to ask. They comprise dozens of entities [1], each uncommon [36]: in the cohort studied here, 24 diseases range from almost 10,000 patients with diffuse large B-cell lymphoma (DLBCL) to fewer than 50 with amyloidosis, and even the most common offers few events for most questions. A complication with a one-year risk of 2% gives fewer than 200 events among the patients with DLBCL treated before 2022, and in a disease of 500 patients a risk of 5% gives 25. At the same time, these patients share much of their work-up and care: the same blood counts and biochemistry, imaging, chemotherapy backbones and supportive care. At the start of treatment, a haematologist weighs treatment failure alongside infection, cytopenia, organ toxicity and unplanned admission, yet the tools for this remain simple. Prognostic indices such as the International Prognostic Index (IPI) and its NCCN refinement, the CLL-IPI and the revised International Staging System turn a handful of measurements at diagnosis into points for one disease and one endpoint [37–40], losing information that the same measurements carry when kept continuous [41]. They matter nonetheless: the Danish DLBCL guideline sets the number of cycles and the intensity of treatment by the age-adjusted IPI [42], and indices decide entry to trials of new first-line therapy [43, 44]. Machine learning on the wider record has improved on them [45–47], and studies pooling lymphoid cancers have examined single outcomes [48], but these models share the indices' basic limitation: one disease, one outcome. For many complications and diseases, no dedicated predictor exists.

DALY-CARE makes it possible to study the lymphoid cancers as a whole. The resource links ten nationwide registers, electronic health records and laboratory data for all Danish adults registered with a lymphoid cancer since 2002, including the plasma-cell neoplasms and their precursors [49]. The Danish registers, linked through a personal identifier, follow hospital contacts, prescriptions, laboratory results and deaths across the population [50, 51], and national quality registries record staging, treatment and response with near-complete coverage [52–54]. DALY-CARE was created in large part so that the lymphoid cancers could be modelled together, on the expectation that what they share would improve prediction for each of them.

Taken to its extreme, learning from related tasks need not stop at haematology. The records of every resident hold diseases, treatments and patterns of care far beyond any specialty, and an encoder pretrained on them has seen more of how records unfold than any cohort can show. It can also read records in finer detail: findings too rare to learn from in a cohort, such as specific molecular results, are common enough across a whole population to be learned. If such an

encoder can be reused for specialised prediction, one investment in learning from population records could serve many clinical questions, and specialties could build on it rather than pretrain their own. Whether it can is an open question, because a general encoder may not fit a specialised population. The established remedy is domain adaptation, continued pretraining on the target population [55–57].

We therefore asked how much each of these sources of signal is worth for the questions a haematologist faces at the start of treatment: other diseases, other outcomes, learning how much to draw on each, and the whole population; and whether, together, they improve on the tools haematology uses today. In 37,531 patients with 24 lymphoid cancers, we predicted 69 outcomes at six horizons from the start of first-line treatment. We compared the approaches in Fig. 1: the prognostic indices used in the clinic, gradient-boosted trees tuned for every prediction, as the established improvement from machine learning, and sequence models pretrained across the cohort, the current instantiation of learning across diseases and outcomes. To isolate what the additional information adds, we compared specialist models, each trained on one disease and one outcome, with models trained across diseases, across outcomes or both, with a router that learns how much to draw on each disease, and with encoders pretrained on the whole Danish population. Finally, we asked whether what the model learns to borrow follows clinical knowledge, against pairs of diseases expected to share signal and negative controls fixed before testing.

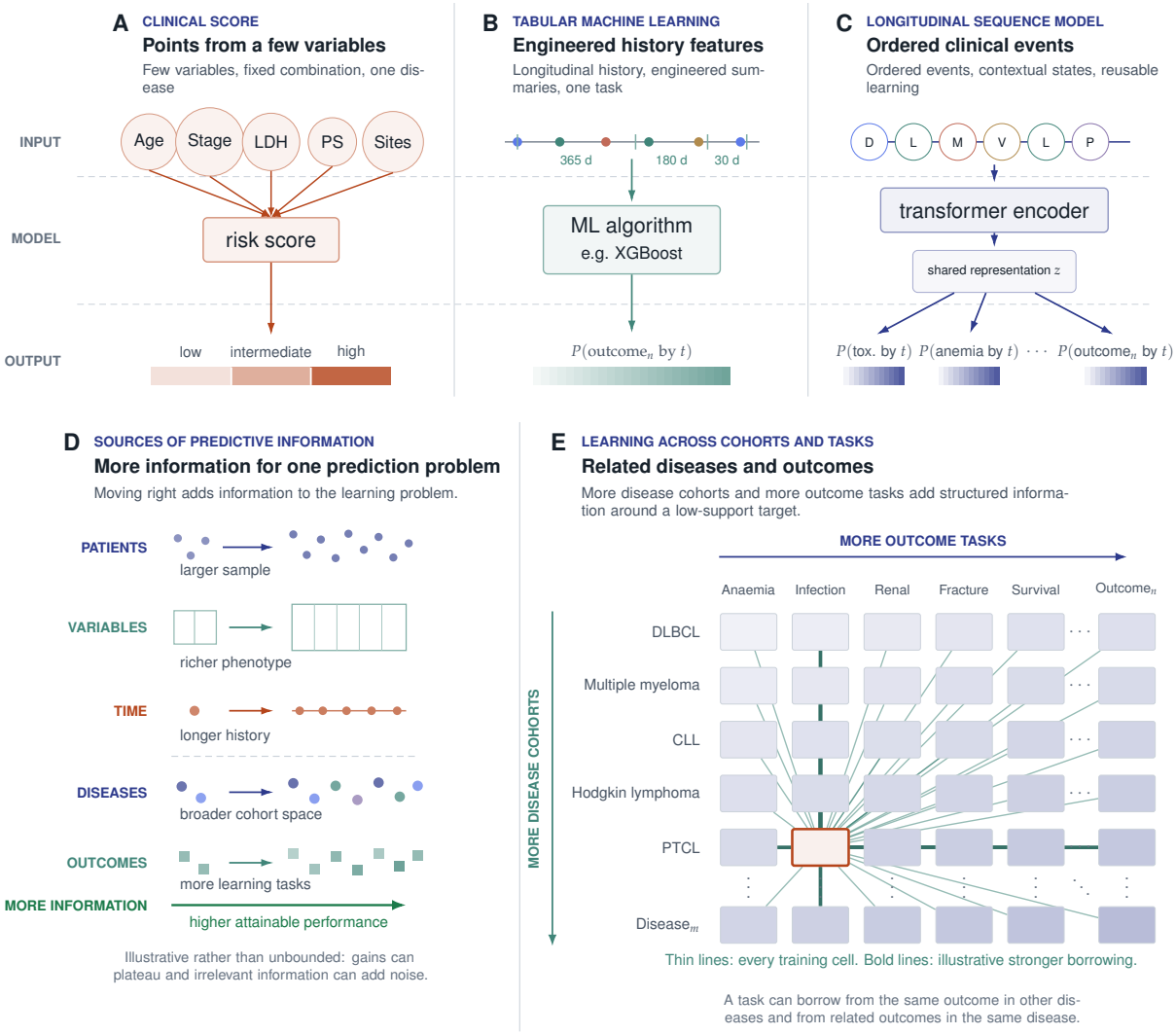


Figure 1. Approaches to risk prediction and the information each can use. Top, three approaches to risk prediction in lymphoid cancers. **A**, clinical risk scores combine a small diagnostic snapshot with a fixed rule into broad risk strata. **B**, classical machine learning uses a wider history, aggregated over time into a task-specific feature table. **C**, a longitudinal sequence model reads the ordered clinical events and can share one patient representation across outcomes; its encoder may be pretrained on the cohort or on the whole population. Bottom, sources of additional signal. **D**, conventional scaling adds patients, variables or observations to one isolated prediction problem. **E**, shared learning also draws on related diseases (rows) and outcomes (columns).

Methods

Population, prediction time and outcomes

We included adults in DALY-CARE who started first-line treatment for a lymphoid or plasma-cell malignancy [49]. Every prediction was made at the start of first-line treatment, from the diagnoses, procedures, pathology, medication, laboratory results and hospital contacts recorded before it, with age, sex and the region and specialty of care (Fig. S2). The study was approved by the Danish Health Data Authority (P-2020-561) and the National Ethics Committee (1804410).

We predicted 69 outcomes at six horizons from 30 days to two years, spanning disease course, infection, cytopenia, organ toxicity, cardiovascular events and acute care, with death as a competing event. Laboratory outcomes were defined with reference to the laboratory criteria of the Common Terminology Criteria for Adverse Events [58, 59]: 13 equal the thresholds of a CTCAE grade, 13 follow a grade with a stated difference (for example, a value exactly at a cut-off counted in the more severe grade, or a baseline arm not applied), and three, including acute kidney injury, have no CTCAE counterpart; symptoms and interventions, required by some grades, are not recorded. The correspondence of every outcome is listed in the Supplementary Methods. Laboratory outcomes were counted only in patients whose blood was monitored, so that time without testing did not pass as event-free. Outcome definitions, disease assignment, treatment lines and ascertainment rules are given in the Supplementary Methods.

Temporal design

Patients treated through 2021 formed the training set, those treated in 2022 the tuning set, on which every modelling choice was made, and those treated from 2023 the held-out test set, which was evaluated once. Pretraining used events before 1 January 2022.

Learning across diseases and outcomes

Every sequence model in this study has the same backbone: a causal transformer, building on CORE-BEHRT [22], pretrained to predict each next event in a patient's record and fine-tuned to give the cumulative incidence of every outcome at every horizon, with death competing [60, 61]. Models were trained with a binary cross-entropy at each of the six horizons, over the patients whose status at that horizon was known, averaged within each outcome and then equally over its 52 endpoints (grade variants of one finding together), and were evaluated on the full risk set with censoring weights (Evaluation). What differs is what each model learns from (Fig. S1). Arranged as a grid of diseases by outcomes, a specialist learns from a single cell, a disease-pooled model from one column (one outcome in every disease), an outcome-pooled model from one row (every outcome in one disease), and the joint model from the whole grid. Comparing them shows how much a prediction gains from other diseases, from other outcomes and from both.

Sharing the whole grid has a cost: the joint model represents each patient once, so every disease and every outcome must make do with the same representation. The routed model keeps that shared representation and lets each prediction adjust it, in the spirit of partial pooling (Fig. S15). Each disease has an adapter, or expert: a small correction learned only from its own patients' labels and shrunk towards zero when it has few events. For every patient and target outcome, a router decides how much of each disease's adapter to apply, the patient's own and those of other diseases [62–64]. The model thus learns what to borrow, and from whom. The joint model knows the patient's disease from the record, but everything it learns about a disease must be built into one encoder that all diseases share and the largest dominate, and at the readout every outcome has parameters of its own but no disease has. A disease adapter adds a patient-by-disease-by-

outcome term that the router turns up or down for each patient. Outcome adapters, which add little to an outcome's own output layer, were screened as well.

The router was configured on the tuning set in three steps: which adapters (disease only, outcome only or both), their size (rank 8, 16 or 32), and how the router combines patient, disease and outcome (additive, or interacting with a width of 128 or 512). Each step took the option with the highest tuning AUC, averaged over estimable cells with each disease weighing equally, each outcome equally within a disease and all six horizons. Options within the difference between two fits of the same model counted as tied and went to the simplest; that difference was measured as the router trained with every source masked, which computes the joint model, against the joint model itself. An option whose IPA fell short of the simplest option's by more than the same kind of noise floor was not eligible. The screen chose disease adapters only **[rank and interaction]**. The sources were screened at the smallest rank, and the choice may differ at another rank or query; the four adapter banks (none, disease, outcome, both) are therefore all trained with the locked rank and query and compared on the test set as a secondary analysis. Three controls separate borrowing from simpler explanations: a joint model told the patient's disease, which tests whether disease-specific baseline risks explain the gain, and routers limited to the patient's own disease adapter (the own-disease router) or own outcome adapter, which test whether corrections help through their own parameters, without borrowing. A sensitivity fit starts the router with a larger share on the shared representation (about 15–18% of the weight rather than about 1%); a rule fixed before its tuning results were seen decided between the two starts (Supplementary Methods).

The widest grid is the whole population. A model pretrained on the registers of every Danish resident has learned to read records from far more patients, diseases and treatments than any specialty can offer. We started the joint and routed models from PHAIR **[PHAIR reference]**, an encoder pretrained by its developers on the registers of the entire Danish population, either as delivered or after continued pretraining on the cohort's records [55]. Since the population contains the cohort, the first comparison measures what the rest of the population adds, and the second what refocusing on the cohort adds. If the population supplies much of what a model needs, the model should also need fewer labelled patients: we retrained the joint model from the cohort encoder and the adapted national encoder on 10%, 25% and 50% of the training patients, and compared the two encoders at each share and each with the joint model and the specialists trained on all patients. Randomly initialised and frozen encoders are references for pretraining and fine-tuning; architecture, training and the national data are described in the Supplementary Methods.

OPERA (Outcome Prediction with Expert Routing Across diseases and outcomes) is the configuration of scope, routing and encoder with the highest AUC on the tuning set, averaged over diseases, outcomes and all six horizons; configurations within the difference between two fits of the same model counted as tied, and the simplest was chosen, the joint before the routed model and the cohort before the national encoder **[selected configuration]**. The routed candidate is the router variant whose routing was verified to work as intended, chosen by an amendment fixed before the test set was opened (Supplementary Methods, router variants). OPERA is the model compared with tabular models and clinical indices. Because OPERA was chosen on tuning data, the joint model, fixed before any tuning result was seen, is compared with the tabular models as well; both comparisons are reported. The contrasts between scopes, routing and encoders use their fixed arms.

Comparators

We compared all models with tabular comparators, which remain hard to beat on routinely collected data [65]. XGBoost [66] and ridge logistic regression were tuned for every disease,

outcome and horizon on summaries of the same register histories, built within each disease with the pipeline of our earlier DLBCL study [47, 67]. XGBoost pooled across diseases, on features built across all patients, tests whether pooling also helps tabular prediction, and XGBoost given the quality-registry variables shows what curated clinical fields add. Clinical indices were compared with OPERA and the tabular models on the patients with the index recorded, on the endpoint each index was developed for: the IPI and NCCN-IPI in DLBCL [37, 38] and every other index recorded for at least 50 patients of a disease. Model risk groups were fixed on tuning data to match the index's low-risk observed risk and high-risk specificity.

Evaluation

The primary metric was the time-dependent area under the ROC curve (AUC) at each of the six horizons. Reported differences and Table 2 pool all six horizons, each disease–outcome–horizon cell weighing equally, and so do the figures: a cell's gain is its mean over the horizons at which it was evaluated, with an interval from the same patient resamples. How performance varies with the horizon is shown separately (Fig. 2, Fig. S8), and reliability curves, which need a single horizon, are shown at one year. The index of prediction accuracy (IPA), Brier score and calibration measured probability accuracy [68–71]; calibration slopes and intercepts came from a weighted logistic recalibration, and were reported as not estimable where the predictions separated events from non-events, so that no finite fit existed. Every contrast was paired on the same patients, with intervals from 1,000 patient resamples within disease, shared across models, outcomes and horizons. Each pooled difference was labelled by a rule fixed before testing: better (or worse) if its 95% interval lay above (below) zero, noted as beyond the practical margin when the difference exceeded 0.01 AUC; otherwise equivalent if the interval lay within ± 0.01; otherwise inconclusive; each with a paired bootstrap P value. The headline comparisons form one family, with P values adjusted by Holm's method: sharing across diseases, across outcomes and both (each against the specialists), the joint model and OPERA against per-task XGBoost, borrowing by the routed model, the national encoder against the cohort encoder, and OPERA against the clinical indices. Everything else, including results for individual diseases, outcomes, cells and disease pairs, is reported as estimates with intervals and is descriptive. Across disease–outcome cells, effects were also summarised, descriptively, by hierarchical models that estimate the typical effect, its variation between diseases and outcomes, and its relation to the number of training events; estimates for individual cells are shown in the Supplementary figures. Decision curves, in patients with DLBCL and the IPI recorded, compare the net benefit of acting on OPERA, the IPI and XGBoost when patients above a predicted two-year risk of death receive closer monitoring and supportive care, across thresholds from 5% to 40% [72], and at two prespecified points: the model's low- and high-risk cutoffs, fixed on tuning data by matching the IPI's low-risk observed risk and high-risk specificity. Training time, compute and storage are reported for one OPERA model and for the separately trained models it replaces. Integrated gradients [73] describe what OPERA relies on, by source, time and clinical finding and for individual patients, checked by event removal and weight randomisation [74]. The choices fixed before the test set was evaluated are listed in the Supplementary Methods (Prespecification). Reporting follows TRIPOD+AI [75] **[checklist in the Supplement]**.

Which diseases the model borrows from

The routed model's readout adds one routed correction per source disease and its risk head is linear, so a source can be switched off exactly, with every other weight as trained: its correction is replaced by its average over tuning patients with the same disease and outcome, which removes what the source says about the individual patient but keeps the average shift the model was trained with (mean ablation; setting it to zero is a sensitivity analysis; Supplementary Methods).

One evaluation of the locked model scored, on the same test patients, the model with all sources, with only the patient's own disease (no borrowing), with the own disease and one other (the source alone) and with all sources but one (the source left out). The source alone shows what a disease contributes on its own, which similar diseases cannot mask; the source left out shows what only that disease provides, so a disease that adds alone but costs little when left out shares its information with others. How much the model relies on borrowing overall is the AUC it loses when all borrowed corrections are switched off, for each disease, each clinical domain and over all diseases. Within one fitted model this carries no noise of refitting, but it measures what the fitted model uses, not what borrowing adds when a model is trained without it; that is the comparison with the same router trained with its own sources only. Whether borrowing favours related diseases was described by each disease's share of its borrowing gain (the positive source-alone gains) that comes from diseases of its own lineage, against the share that lineage holds among the other sources, which is what borrowing blind to lineage would give, and how broadly it borrows by the effective number of sources it gains from. Candidate pairs of diseases with a possible reason to share signal for a domain of outcomes (shared biology, treatment or complications) were listed before the test set was opened; because the routed model borrowed broadly on the tuning set, they are reported descriptively rather than tested (Supplementary Methods). The map of all pairs is descriptive and partially pooled. AUC differences were primary and IPA secondary. The analysis concerns borrowing at the readout; sharing through the common encoder is measured by the joint against the specialist models.

Results

Cohort characteristics and outcomes

The pretraining population comprised all 78,668 adults in DALY-CARE, which includes every Danish adult diagnosed with a lymphoid cancer or a precursor condition (MGUS, MBL) since 2002 [49], whose records before 2022 pretrained the cohort encoder. Of these, 37,531 started first-line treatment with an assigned disease and form the final cohort (Fig. S3 and Table 1): 29,105 treated through 2021, 1,980 in 2022 and 6,446 from 2023. The median age was 69 years (IQR 60–77), 41.7% were women, and the largest diseases were diffuse large B-cell lymphoma (26.5%), multiple myeloma (23.0%), chronic lymphocytic leukaemia (10.9%) and follicular lymphoma (7.8%).

The three periods differed in ways that a temporal evaluation has to absorb. National laboratory capture is complete only from 2017, so patients treated earlier had far fewer recorded results (median 119 in the training period against 524 in the test period), and later patients had longer recorded histories (median 11.6 against 22.0 years). Performance status and the disease-specific index category, recorded in the quality registries, were missing for 47% and 58% of test patients against 24% and 36% in training, presumably because registration of the most recent years is incomplete. The disease mix shifted too: DLBCL made up 27.5% of training and 22.7% of test patients, while indolent B-cell lymphomas rose from 13.4% to 18.8%, partly through more unspecified B-cell and splenic marginal-zone lymphoma, and test patients were older (median 72 against 69 years). The test set therefore measures performance under clear data shift. Of the 24 diseases and 69 outcomes, **[N]** cells had enough test events to be evaluated at one year. Event rates ranged from **[range]**, and **[N]** cells had fewer than 20 events to train on (Figs. S4 and S5). The tuning set selected **[configuration]** as OPERA.

Table 1. Baseline characteristics at first-line treatment. One record per patient, by temporal split; *n* (%) or median (IQR).

Characteristic	Overall *All years* *n* = 37,531	Training *Through 2021* *n* = 29,105	Tuning *2022* *n* = 1,980	Test *From 2023* *n* = 6,446
Demographics				
Age at treatment, years	69 (60–77)	69 (59–77)	71 (61–78)	72 (62–79)
Age ≥70 years	18,108 (48.2%)	13,470 (46.3%)	1,065 (53.8%)	3,573 (55.4%)
Female	15,665 (41.7%)	12,195 (41.9%)	808 (40.8%)	2,662 (41.3%)
Disease				
Large B-cell lymphomas	10,388 (27.7%)	8,356 (28.7%)	500 (25.3%)	1,532 (23.8%)
Diffuse large B-cell (DLBCL)	9,935 (26.5%)	7,996 (27.5%)	476 (24.0%)	1,463 (22.7%)
Transformed follicular (tFL)†	264 (0.7%)	210 (0.7%)	10 (0.5%)	44 (0.7%)
Richter transformation (RT)†	189 (0.5%)	150 (0.5%)	14 (0.7%)	25 (0.4%)
Burkitt and lymphoblastic lymphomas	1,365 (3.6%)	1,074 (3.7%)	86 (4.3%)	205 (3.2%)
Lymphoblastic (LBL)†	1,073 (2.9%)	838 (2.9%)	71 (3.6%)	164 (2.5%)
Burkitt (BL)	292 (0.8%)	236 (0.8%)	15 (0.8%)	41 (0.6%)
Mantle-cell lymphoma (MCL)	1,467 (3.9%)	1,117 (3.8%)	86 (4.3%)	264 (4.1%)
Indolent B-cell lymphomas	5,414 (14.4%)	3,891 (13.4%)	308 (15.6%)	1,215 (18.8%)
Follicular (FL)	2,929 (7.8%)	2,242 (7.7%)	152 (7.7%)	535 (8.3%)
Lymphoplasmacytic (LPL)	1,580 (4.2%)	1,168 (4.0%)	101 (5.1%)	311 (4.8%)
Extranodal marginal-zone (EMZL)	500 (1.3%)	318 (1.1%)	32 (1.6%)	150 (2.3%)
Unspecified B-cell (BCL)†	234 (0.6%)	87 (0.3%)	21 (1.1%)	126 (2.0%)
Splenic marginal-zone (SMZL)	144 (0.4%)	71 (0.2%)	2 (0.1%)	71 (1.1%)
Nodal marginal-zone (NMZL)†	27 (0.1%)	5 (0.0%)	0 (0.0%)	22 (0.3%)
CLL and SLL	5,183 (13.8%)	4,102 (14.1%)	254 (12.8%)	827 (12.8%)
Chronic lymphocytic leukaemia (CLL)	4,091 (10.9%)	3,230 (11.1%)	195 (9.8%)	666 (10.3%)
Small lymphocytic lymphoma (SLL)	1,092 (2.9%)	872 (3.0%)	59 (3.0%)	161 (2.5%)
Hairy-cell leukaemia (HCL)†	424 (1.1%)	295 (1.0%)	29 (1.5%)	100 (1.6%)
Hodgkin lymphoma (HL)	2,636 (7.0%)	2,072 (7.1%)	141 (7.1%)	423 (6.6%)
T-cell lymphomas	1,646 (4.4%)	1,287 (4.4%)	70 (3.5%)	289 (4.5%)
Peripheral T-cell, NOS (PTCL)	521 (1.4%)	443 (1.5%)	18 (0.9%)	60 (0.9%)
Angioimmunoblastic (AITL)	484 (1.3%)	352 (1.2%)	26 (1.3%)	106 (1.6%)
Anaplastic large-cell (ALCL)†	417 (1.1%)	343 (1.2%)	11 (0.6%)	63 (1.0%)
Unspecified T-cell (TCL)†	224 (0.6%)	149 (0.5%)	15 (0.8%)	60 (0.9%)
Plasma-cell neoplasms	8,962 (23.9%)	6,889 (23.7%)	499 (25.2%)	1,574 (24.4%)
Multiple myeloma (MM)	8,630 (23.0%)	6,656 (22.9%)	470 (23.7%)	1,504 (23.3%)
Plasma-cell leukaemia (PCL)†	201 (0.5%)	143 (0.5%)	20 (1.0%)	38 (0.6%)
Solitary plasmacytoma†	131 (0.3%)	90 (0.3%)	9 (0.5%)	32 (0.5%)
Amyloidosis	46 (0.1%)	22 (0.1%)	7 (0.4%)	17 (0.3%)
Performance status (ECOG)				
0–1	22,795 (60.7%)	18,539 (63.7%)	1,301 (65.7%)	2,955 (45.8%)
2–4	4,303 (11.5%)	3,572 (12.3%)	289 (14.6%)	442 (6.9%)
Missing	10,433 (27.8%)	6,994 (24.0%)	390 (19.7%)	3,049 (47.3%)
Risk group by disease-specific index‡				
Low	3,938 (10.5%)	3,187 (11.0%)	249 (12.6%)	502 (7.8%)
Intermediate	13,020 (34.7%)	10,797 (37.1%)	744 (37.6%)	1,479 (22.9%)
High	5,809 (15.5%)	4,656 (16.0%)	398 (20.1%)	755 (11.7%)
Missing	14,764 (39.3%)	10,465 (36.0%)	589 (29.7%)	3,710 (57.6%)
Pre-index history				
Recorded history, years	13.9 (8.2–18.9)	11.6 (6.9–15.8)	20.2 (19.5–20.6)	22.0 (21.0–23.0)
Recorded events	886 (416–1,819)	695 (341–1,353)	2,213 (1,322–3,575)	1,922 (1,095–3,318)
Distinct diagnoses	21 (13–32)	20 (12–30)	24 (15–37)	25 (15–38)
Distinct medications	21 (12–34)	19 (10–30)	32 (21–46)	34 (22–49)
Laboratory results	197 (0–487)	119 (0–349)	484 (276–801)	524 (295–881)
Days with hospital contact	19 (10–35)	16 (9–27)	37 (22–61)	42 (24–70)
Distinct contact specialties	8 (5–11)	7 (5–9)	10 (8–13)	11 (8–14)
Days from last record to treatment	2 (1–5)	2 (1–5)	2 (1–4)	2 (1–4)
Region (most contacts)				
North Denmark	3,977 (11.4%)	3,081 (11.6%)	207 (10.6%)	689 (10.8%)
Central Denmark	6,189 (17.7%)	4,291 (16.2%)	455 (23.2%)	1,443 (22.5%)
Southern Denmark	8,908 (25.5%)	7,072 (26.7%)	442 (22.5%)	1,394 (21.8%)
Capital Region	11,046 (31.7%)	8,615 (32.5%)	560 (28.5%)	1,871 (29.2%)
Zealand	4,776 (13.7%)	3,470 (13.1%)	298 (15.2%)	1,008 (15.7%)
Follow-up				
Follow-up (reverse Kaplan–Meier), years	9.1 (4.4–14.5)	11.4 (7.7–16.1)	3.7 (3.4–4.0)	1.6 (0.8–2.4)
Died during follow-up	18,847 (50.2%)	17,378 (59.7%)	539 (27.2%)	930 (14.4%)

History summaries use records before the index date. †No dedicated, externally validated prognostic index (Supplementary Table S4). ‡Category recorded with each disease's registry index (e.g. FLIPI, CLL-IPI, ISS), collapsed to low, intermediate and high; for DLBCL the NCCN-IPI computed from registry components (low 0–1; intermediate 2–5, its low- and high-intermediate groups; high ≥6), or the recorded index where it could not be computed. Missing includes conflicting records.

Which risks can be anticipated at the start of treatment?

[At one year, OPERA discriminated outcomes in the X and Y domains well (AUC A–B) and those in Z poorly (AUC C–D)] (Fig. 2). **[Change in AUC from 30 days to two years, and variation across diseases]** (Fig. S6).

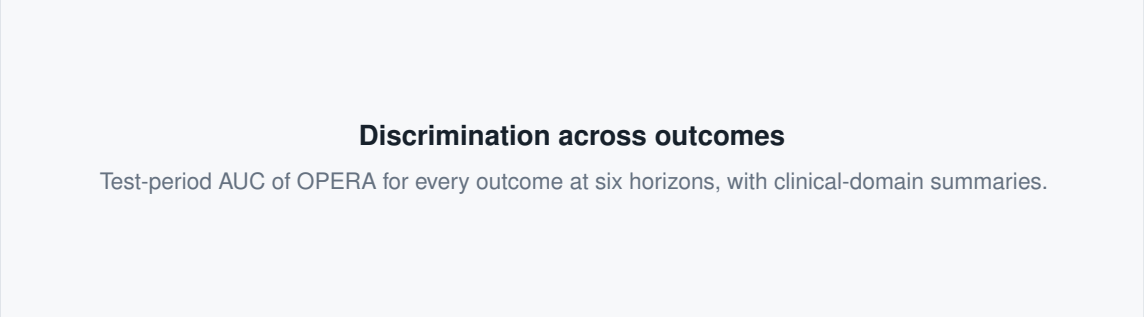


Figure 2. Discrimination of OPERA across outcomes and horizons. A, Test-period AUC for each of the 69 outcomes at each horizon (colour), against its training events. **B**, The same results summarised by clinical domain. Outcomes or horizons without enough test events are not shown, and every outcome in every disease is in Fig. S6.

Does one model outperform the methods in use?

Pooled over cells and horizons, OPERA differed from per-task XGBoost by **[X]** in AUC and **[Y]** in IPA **[(claim labels)]**. It had the higher AUC in **[X of Y]** disease–outcome–horizon cells (median difference **[Z]**) and in **[X of Y]** against XGBoost given the quality-registry variables, and the joint model had the higher AUC than the specialist trained on each cell alone in **[X of Y]** (Fig. 3A–B, Table 2). **[Its median gain over XGBoost was positive / not at every horizon from 30 days to two years]** (Fig. 3D) **[and its patient-level Brier improvement did / did not differ with the number of events recorded before treatment]** (Fig. 3E). **[At the specificity of each clinical index's high-risk category, the model's high-risk group contained X to Y percentage points more of the patients with the event]** (Fig. 3C).

Table 2. What each source of signal adds, across all horizons. Paired differences on the test set (first model minus second), pooled over all six horizons with equal weight per disease–outcome–horizon cell, with 95% paired bootstrap intervals; the same table at each horizon is in Supplementary Table [S]. Cells: cells estimable for both models. Rare diseases: fewer than 500 training patients. Cells better: share of cells with a higher AUC point estimate, irrespective of uncertainty. Bold: the interval lies above zero; red: below zero. Claim: the prespecified label from the paired bootstrap of the pooled difference: *better* (*worse*) if the interval lies above (below) zero, marked *above margin* when the estimate also exceeds the practical margin of 0.01; *equivalent* if the interval lies within ±0.01; otherwise *inconclusive*. Paired bootstrap *p*-values and the hierarchical estimates, which weight diseases and endpoints equally, are in Supplementary Table [S].

Comparison	Cells	ΔAUC (95% CI)	ΔIPA (95% CI)	ΔAUC, rare diseases	Cells better	Claim
Learning across diseases and outcomes (against the specialist)						
Other diseases (disease-pooled)	[N]	[x]	[x]	[x]	[x%]	[]
Other outcomes (outcome-pooled)	[N]	[x]	[x]	[x]	[x%]	[]
Both (joint)	[N]	[x]	[x]	[x]	[x%]	[]
Learning how much to borrow						
Routed vs joint	[N]	[x]	[x]	[x]	[x%]	[]
Population pretraining (joint model)						
National vs cohort encoder	[N]	[x]	[x]	[x]	[x%]	[]
Adapted national vs cohort encoder	[N]	[x]	[x]	[x]	[x%]	[]
Sequence models against tabular models						
Specialist vs per-task XGBoost	[N]	[x]	[x]	[x]	[x%]	[]
Pooled vs per-task XGBoost	[N]	[x]	[x]	[x]	[x%]	[]
OPERA vs per-task XGBoost	[N]	[x]	[x]	[x]	[x%]	[]
OPERA vs pooled XGBoost	[N]	[x]	[x]	[x]	[x%]	[]
OPERA vs XGBoost with registry variables	[N]	[x]	[x]	[x]	[x%]	[]



Figure 3. Performance of OPERA relative to reference methods. A, AUC gain over each reference, pooled over the six horizons, one point per disease–outcome cell with enough test events. Diamonds show medians, bars interquartile ranges, and the figures on the right the share of cells in which OPERA was better. **B**, AUC of OPERA (pooled over horizons) against per-task XGBoost in each cell, with point size reflecting training events. **C**, Share of patients with the event placed in the high-risk group by each clinical index and by OPERA at the index's specificity (cutoffs fixed on tuning data). **D**, Median AUC gain over per-task XGBoost of each neural model at each horizon. **E**, Improvement in Brier score, pooled over horizons, over the specialist and over XGBoost by the number of events recorded before treatment (×100, 95% CI). Within each contrast, models are evaluated on the same test patients.

Against XGBoost trained and tuned for each disease, outcome and horizon on the same patients, OPERA's AUC, averaged over all horizons, differed by [X] and its IPA by [Y] (Fig. 4). Most of this comes in two steps. The change from tabular summaries to the sequence model itself, with no sharing, is the specialist against XGBoost on the same task: [X] in AUC. Sharing is the joint model against the specialist, [Y]; routing and the choice of encoder make up the rest (Table 2). Pooling helped the tabular models less: XGBoost pooled across diseases differed from per-disease XGBoost by [Y], and OPERA from pooled XGBoost by [X]. XGBoost differed from logistic regression by [Y]. **[Clinical domains and diseases in which XGBoost remained better.]** At a specificity of 90% fixed on tuning data, OPERA's median sensitivity was [X] against [Y] for XGBoost, and its median MCC [X] against [Y]. **[Whether the ordering held at the other**

decision points.] Adding the quality-registry variables changed XGBoost's AUC by **[X]**, and OPERA, without them, differed from this reference by **[Y]**. **[Calibration slopes and calibration-in-the-large by model and clinical domain]** (Fig. 4B, Fig. S9).

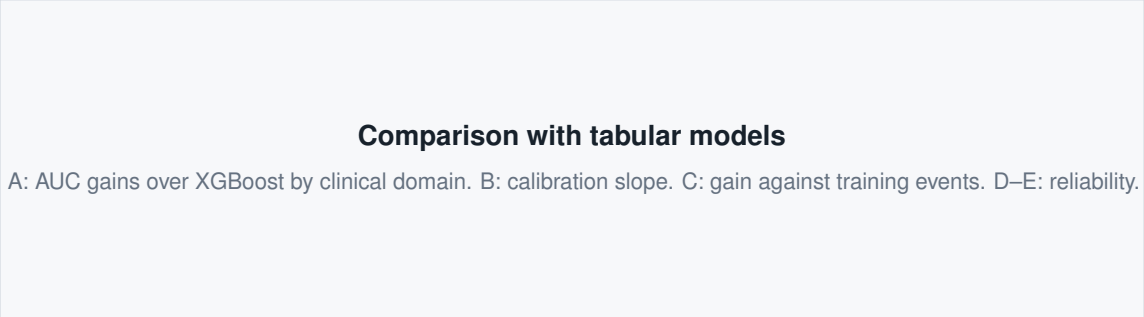


Figure 4. Comparison with tabular models: discrimination and calibration. A, AUC gain over per-task XGBoost, pooled over the six horizons, of OPERA, the joint model and the specialist, and of the joint model over XGBoost pooled across diseases and XGBoost given quality-registry variables, by clinical domain (median and interquartile range across cells). **B**, Calibration slope of OPERA, the joint, specialist and XGBoost models by domain, over every cell and horizon (ideal 1). **C**, Gain of OPERA over XGBoost against the target's training events, with the support curve; points are cells with at least 10 training events and a 95% interval narrower than 0.4 (all cells enter the curve). **D–E**, Reliability of OPERA, the joint model and XGBoost at one year for the two example cells, chosen by a rule on training counts (the cell with the most training events in its domain among cells with at least 200 training events and an event share of at least 10%). Observed risks are censoring-weighted and point size reflects patients per bin; bins with fewer than 20 patients are not drawn. No model was recalibrated on test data. The full calibration analysis is in Fig. S9.

One OPERA model replaced **[N]** separately trained specialists and **[N]** tabular models, with **[training time, GPU-hours and storage for one model against the portfolio it replaces]**. Against choosing, for each cell, whichever of the specialist, disease-pooled and outcome-pooled models was best on the tuning set, the joint model's AUC differed by **[X]** over **[N]** cells (Fig. S20).

Comparison with the prognostic indices in clinical use

An index can only be used where it is recorded. **[Across diseases, the share of patients with their disease's index recorded or computable ranged from X% to Y% (Supplementary Table S)]**, and 10 of the 24 diseases have no dedicated, externally validated index (Table S4); OPERA gave an estimate for every patient. **[Across the N indices recorded for at least 50 patients of a disease, each compared in its own population and on the endpoint it was developed for, the model's risk groups separated outcomes more clearly than the index in X of N (Fig. S14); at a cutoff matched to each index's low-risk observed risk, the model placed X more per 1,000 patients at low risk.]**

The IPI in DLBCL shows the comparison in detail. **[N of N patients with DLBCL had a recorded IPI; how they differed from those without.]** Per 1,000 patients, using a threshold matched on tuning data to the IPI high-risk category's specificity, OPERA identified **[N]** more of the patients who died within two years, and, at a cutoff matched separately to the IPI low-risk group's observed risk, placed **[M]** more at low risk, with an observed risk of **[X%]** against **[Y%]** in the IPI low-risk group (Fig. 5). Its highest-risk 5% had an observed two-year risk of **[X%]**, against **[Y%]** in the IPI high-risk group. **[At that operating point, recall and precision were X and Y for the model and A and B for the IPI, and average precision across thresholds was X versus Y]** (Fig. S11). **[Outcomes of the patients the two systems classified differently, including older patients whom the model placed at low risk]** (Fig. 5D, Fig. S12). **[Across thresholds of 5% to**

40%, acting on OPERA gave a net benefit of X more true positives per 1,000 than the IPI and Y more than XGBoost] (Fig. S13). **[Against the NCCN-IPI, result; XGBoost and logistic regression on the same patients.]** Median follow-up, by the reverse Kaplan–Meier method, was **[X years (IQR)]**.

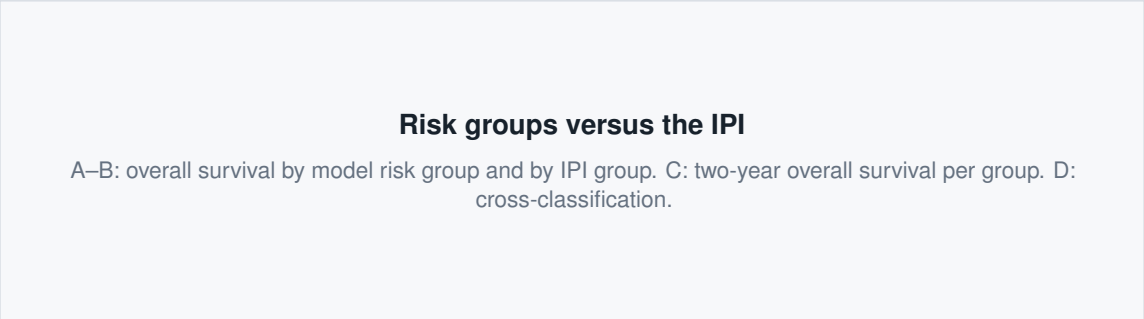


Figure 5. Risk groups defined by OPERA and by the IPI in DLBCL. Patients with diffuse large B-cell lymphoma and an IPI recorded at diagnosis, compared on overall survival, the endpoint the IPI was developed for. **A**, Kaplan–Meier overall survival by model risk group, with cutoffs fixed on tuning data to match the IPI's low-risk observed risk and high-risk specificity (**[cutoffs]**). Groups are fixed at baseline, and numbers at risk are shown below the curves. **B**, The same by IPI category. Shaded bands are 95% bootstrap intervals and the dashed line marks two years. **C**, Two-year overall survival per group with 95% CI, labelled with the number of deaths by two years and the number of patients. **D**, Cross-classification of the two systems. Rows are the IPI category and columns the model risk group, so each cell holds the patients with that combination, with their two-year survival and 95% bootstrap interval above and the number of patients below. Outlined cells are patients placed in the same group by both systems. Cells to the left of the outline are patients the model places at lower risk than the IPI, and cells to the right at higher risk. Reading along a row shows whether the model separates outcomes within one IPI category.

What does sharing across diseases and outcomes contribute?

[Sharing across diseases carried the gain; sharing across outcomes added little.] Over the cells evaluable for every model, pooled over all six horizons, a model trained on every disease for one outcome changed AUC over the specialist by **[X]**, a model trained on every outcome within one disease by **[Y]**, and the joint model, which shares both, by **[Z]** (IPA **[A]**, **[B]** and **[C]**). Seen the other way round, adding the other diseases to an outcome-pooled model changed AUC by **[X]**, whereas adding the other outcomes to a disease-pooled model changed it by **[Y]**, and the interaction of the two was **[Z] [(so the joint model's gain over the specialist was largely the gain from seeing other diseases)]** (Fig. 6A–B, every cell in Fig. S7). The pattern held across clinical domains **[(confirm)]** (Fig. 6D) and across horizons (Fig. S8). Part of the gain over the specialist reflects how little a specialist can learn from a rare subtype, so the comparison with per-task and pooled XGBoost below is the stricter test of what sharing adds.

[The gain was associated with / not related to the data a prediction had of its own: shape of the support curve with its interval] (Fig. 6C). This curve covers the tasks in which a specialist could be trained; the sparsest tasks, where it could not, appear as added coverage below. Trained on 10%, 25% and 50% of the training patients, the joint model differed from specialists trained on all of them by **[X]**, **[Y]** and **[Z]** in AUC, and started from the adapted national encoder by **[X]**, **[Y]** and **[Z]**. In the subtypes with fewer than 500 training patients, OPERA's AUC differed from per-task XGBoost by **[X]** and from pooled XGBoost by **[Y]**, and the joint model's from the specialist by **[Z]** (Table 2). In the ten subtypes without a dedicated, externally validated index (Table S4), which are assessed today with an index borrowed from another disease, most often the IPI [37], OPERA differed from the two XGBoost models by **[X]** and **[Y]**, and the joint model from the specialist by **[Z]**, over **[N]** cells, and the IPI computed from its registry components was compared with

OPERA on the same patients **[result]** (**[Supplementary Table]**). **[Number and share of cells in which sharing was worse than the specialist, with examples.]** Pooled models also predicted **[N]** cells for which no specialist could be trained, including every outcome in nodal marginal-zone lymphoma, which had no patients in the tuning year and so no model of its own, with **[absolute performance]** (Figs. S7 and S8).

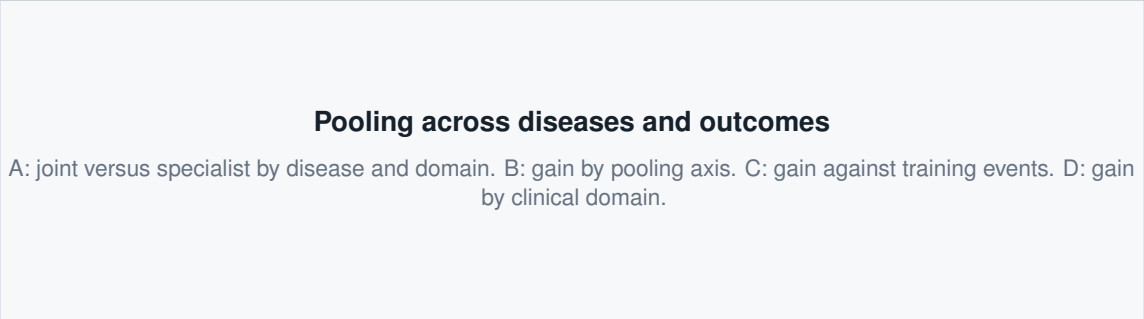


Figure 6. Effect of pooling across diseases and outcomes on discrimination. A, AUC gain of the joint model over the specialist, pooled over the six horizons, by disease (rows) and clinical domain (columns), median over the domain's outcomes. Diseases run from fewest to most training patients, given beside each name with a log-scale bar, and domains from fewest to most training events. **B**, For each disease, the median gain from pooling over diseases, pooling over outcomes and both (joint), on the same rows. **C**, Gain against the number of training events of the target cell (at one year), with the hierarchical support curve and 95% credible band; points are cells with at least 10 training events and a 95% interval narrower than 0.4 (all cells enter the curve). **D**, Gains by clinical domain (median and interquartile range across cells). Positive values favour the pooled model, and grey cells could not be estimated. Cell-level maps for each pooling axis, their interaction and all horizons are in Figs. S7 and S8.

Does learning what to borrow make sharing work?

The joint model gives every prediction the same shared signal. The routed model learns, for each patient and target, which diseases to draw on (Fig. S15); on the tuning set, disease adapters were chosen over outcome adapters and both. Routing changed AUC by **[X]** and IPA by **[Y]** relative to the joint model **[(claim labels)]**, and had the higher AUC in **[N of M]** cells (Fig. S16A). The joint model and the router trained with every source masked, two fits of the same model, differed by **[X]**, which sets the noise floor for these comparisons. **[The gain was not knowing the diagnosis or the adapters' added capacity: the joint model told the disease and the own-disease router accounted for X and Y of the routing gain, so Z came from borrowing from other diseases.]** **[On the test set, at the locked rank and query, disease adapters, outcome adapters and both changed AUC by X, Y and Z, so the tuning choice did / did not hold]** (Fig. S16B). **[In cells where the joint model had been worse than the specialist on the tuning set, routing changed test AUC by X]** (Fig. 7A, Fig. S17), **[so learning what to borrow recovered where sharing everything had hurt.]** **[Routing weights concentrated on the patient's own disease / related diseases]** (Fig. S18).

The routed model adds two corrections to the shared prediction: one learned for the patient's own disease and outcome, and one borrowed from other diseases and outcomes. Fitting the same architecture with only the shared signal, with the own corrections added, and with borrowing added separates the two (Fig. 7). The routed model is the variant whose routing we could verify: in the first router, routing collapsed onto a few sources with small corrections, and a global adapter outside the routed mix with two epochs of router warm-up restored it (Supplementary Methods, router variants). We chose this model, prespecified as an amendment before any test data were opened, although it scored slightly below the first router on the tuning set (AUC -0.004, 95% interval -0.011 to 0.002), because its routing works as intended and can be explained; the first router is reported as a sensitivity analysis (**[X]** against the routed model on the test set). Own

corrections changed AUC by [X] and borrowing by [Y], [Z] together, pooled over horizons. **[The two depended on training support in opposite directions. Own corrections helped where the target had many training events and hurt where it had few; borrowing helped where it had few and was neutral where it had many, so the two crossed at about N training events]** (Fig. 7B–C). **[A prediction with ample data of its own is pulled towards its own estimate, and one with little is pulled towards what related tasks have learned: the behaviour of partial pooling in a hierarchical model, learned here by the router rather than fixed by a prior.] [The borrowing was broad rather than selective: diseases drew on similar mixtures of sources (median correlation of borrowing profiles X)]**, so its value lay less in which tasks were borrowed from than in how much a task leaned on others.

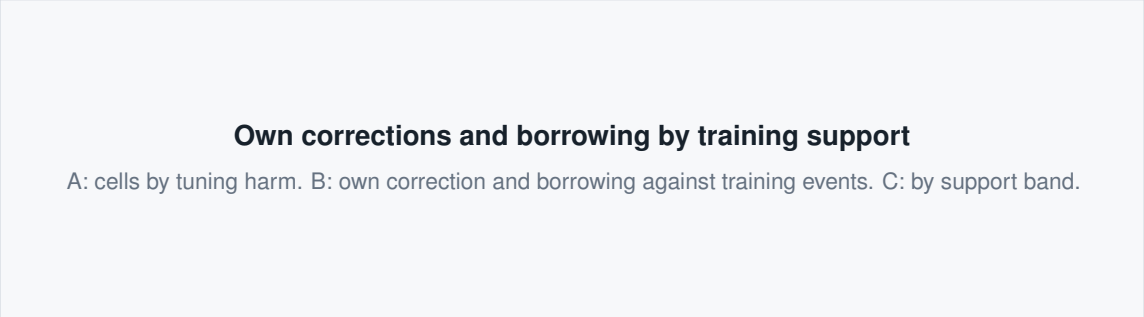


Figure 7. Own corrections and borrowing in the routed model, by training support. Own corrections are the routed architecture with only the patient's own disease and outcome corrections against the same architecture with only the shared correction; borrowing is the full routed model against the own-only model; both is the full model against the shared-only model. Gains are test-set AUC pooled over the six horizons. **A**, Each disease–outcome cell by the joint model's gain over the specialist on the tuning set (left of the dashed line, sharing had hurt) against the routed model's gain on the test set, coloured by the cell's training events. **B**, Own correction and borrowing against the target's training events at one year (log scale): quadratic fits in log events with 95% bootstrap bands over cells. Points are cells with at least 10 training events and a 95% interval narrower than 0.4 (all cells enter the fits); the strip along the axis marks the bands of C. **C**, Median and interquartile range across cells within each band of training events, with the number of cells in each band.

From whom does the model borrow?

Switching source diseases off in the fitted routed model shows what it borrows (Fig. 8). Because each source adds its own correction, a source can be removed from the trained model without retraining or changing anything else, so these comparisons are within one model on the same patients. **[Most single sources added little; the largest gains from one source were for X from Y]** (Fig. 8A). **[With all borrowed corrections switched off, the model lost X AUC points over all diseases, from Y in the largest disease to Z in the smallest]** (Fig. 8B), **[and the loss was larger in diseases with fewer training patients]** (Fig. S19B). **[The diseases borrowed from similar mixtures of sources (median correlation of their borrowing profiles X), rather than each from a few related diseases]** (Fig. 8C), so the router's choice lay more in how much a disease leaned on the others than in which it leaned on, which is the behaviour of the partial pooling described above. **[Borrowing mattered most for X outcomes and least for Y]** (Fig. 8D), **[and IPA agreed / differed]** (**[Supplementary Table]**). Pairs of diseases expected to share signal were listed before the test set was opened; because the router borrowed broadly, they are reported descriptively (**[Supplementary Table]**) rather than as a test.

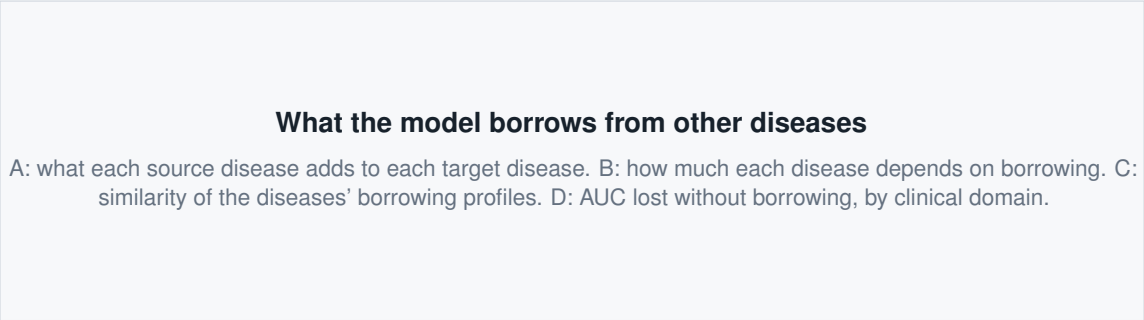


Figure 8. What the model borrows from other diseases. The routed model's readout adds one correction per source disease, so sources can be switched off in the trained model without retraining or changing anything else. Each comparison scores the same test patients over all outcomes and horizons (AUC points; 95% intervals from paired patient resampling). **A**, The gain for each target disease (rows, with its number of patients) when one other disease (columns) is added to its own, partially pooled across pairs. Dots mark gains whose interval lies above zero; the diagonal is grey; diseases are grouped by lineage for reading only, and grey labels mark diseases with few estimable outcomes. AL, amyloidosis; SP, solitary plasmacytoma; tFL, transformed follicular lymphoma. **B**, On the same rows, the AUC the trained model loses when all its borrowed corrections are switched off; dashed line, all diseases weighing equally. **C**, **[Similarity of the diseases' borrowing profiles: for each pair of target diseases, the correlation of the shares of borrowed correction they draw from each source.]** **D**, The AUC lost without borrowing, as in B, by clinical domain, all diseases weighing equally; dashed line, all outcomes. This loss beside the gains over separately trained models, by disease size, the source left out and every clinical domain are in Fig. S19. The figure shows how the fitted model uses its corrections, not how diseases are related.

Can population-scale pretraining extend the shared signal?

Replacing the encoder pretrained on the cohort with the national encoder, which had also learned from the rest of the population, changed the joint model's AUC by **[X]** when transferred directly and by **[Z]** after adaptation to the cohort, and the routed model's by **[Y]** and **[W]** (Fig. 9A). **[The incremental effect of adaptation, and whether population pretraining and routing were additive.]** **[The effect by the share of a patient's history covered by the national vocabulary, by training events and by disease]** (Fig. 9C–D, Fig. S21). With 10%, 25% and 50% of the training patients, the adapted national encoder differed from the cohort encoder by **[A]**, **[B]** and **[C]** (Fig. 9B). Against the cohort-encoder joint model trained on all patients, the adapted national encoder differed by **[X]**, **[Y]** and **[Z]** with 10%, 25% and 50% of them, **[so the smallest share that matched it (claim label equivalent or better) was X%]** As references, pretraining on the cohort changed the joint model's AUC by **[X]** over random initialisation, and fine-tuning by **[Y]** over a frozen encoder.

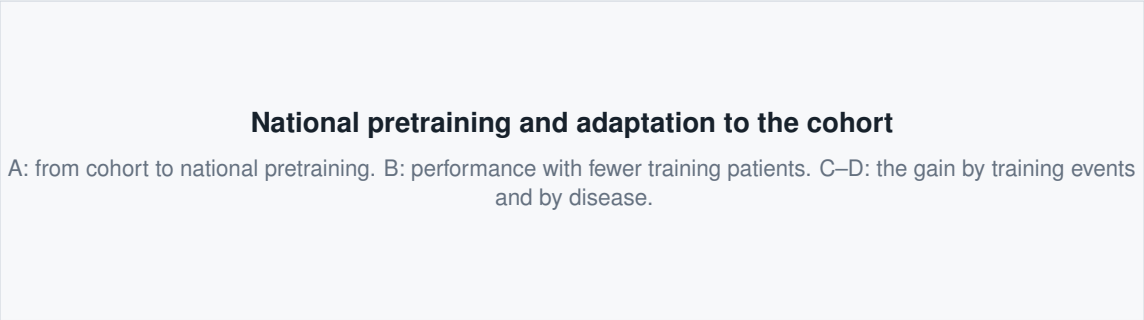


Figure 9. National pretraining, adaptation to the cohort and training-set size. A, One-year AUC of the joint model (filled) and the routed model (outline) initialised from the encoder pretrained on the cohort alone, from the national encoder (pretrained on the whole population, the cohort included) used directly, and from the national encoder adapted to the cohort by continued pretraining on its records (95% CI, averaged over the cells evaluated for all six models). Arrows give the paired effect of each step, and the bracket the two together. **B**, Median one-year AUC over the same cells of the joint model from the cohort encoder and from the adapted national encoder, trained on 10%, 25%, 50% or all of the training patients (nested random subsets with the same patients for every outcome). Numbers give the median paired difference at each share; the dashed line marks the cohort-encoder model trained on all patients, and the open circle the share at which the national model reaches it (interpolated on the log scale). **C**, Paired effect of the adapted national versus the cohort encoder for each cell, by the number of training events in the cell (line and band: median and interquartile range within bins). **D**, The same effects by disease (diamond and bar: median and interquartile range). Tuning and test patients were never subsampled.

What the predictions rest on

Sharing raises an obvious concern: a model trained on every outcome might learn one general measure of frailty and apply it to all. We decomposed each held-out prediction of OPERA with integrated gradients into the contributions of the recorded events, after checking that the attributions reflect the model: removing the five most credited events lowered the predicted risk more than removing five random events for [[n]] of [[n]] check outcomes, attributions of a model with randomised weights agreed little with those of the trained model (median rank agreement [[x]]), and attributions were stable to the choice of reference ([[x]]).

Across [[n]] outcomes in [[n]] held-out patients, the timing of care alone (when and how often a patient was seen, with age and subtype, but not what was recorded) carried a median [[x]] of the differences in predicted risk between patients. What was recorded carried the rest, and which record mattered depended on the complication: **[laboratory results for cytopenias and organ toxicity, medication for infection and diagnoses for cardiovascular events and death]** (Fig. 10A, every outcome in Fig. S32). Records from the last 30 days before treatment carried [[x]]. Outcomes of one clinical domain rested on the same episodes of a patient's history (median agreement [[x]]), and outcomes of different domains on different ones ([[x]]) (Fig. S33A)**[; if so: the shared model reads one history differently for each complication]**. Of [[n]] prespecified effects of clinical findings, [[n]] were seen and [[n]] went the other way (Fig. 10C). **[Examples, such as high LDH raising predicted death, low neutrophils severe neutropenia and recent antibacterials serious infection, and any strong pattern that was not expected.]** Molecular findings recorded as pathology codes moved predictions **[as expected / not as expected]**: **[the change in predicted log-odds of treatment failure and death with a recorded TP53 aberration, MYC rearrangement or unmutated IGHV]** (Fig. 10C).

Read on the timing of care alone, without retraining, held-out AUC changed by [[x]], without laboratory results by [[x]], without medication by [[x]], without diagnoses by [[x]], without the region and specialty of care by [[x]] and with only the last year of history by [[x]] (Fig. 10B, by outcome in Fig. S33B). For a single patient, the same computation shows which episodes of care

move each risk and when they happened (Fig. 10D, Fig. S34): in a synthetic patient constructed for illustration (no real patient is shown), a recent pneumonia moves the risk of serious infection, the findings of the diagnostic work-up the risk of death, and low neutrophils days before treatment the risk of severe neutropenia.

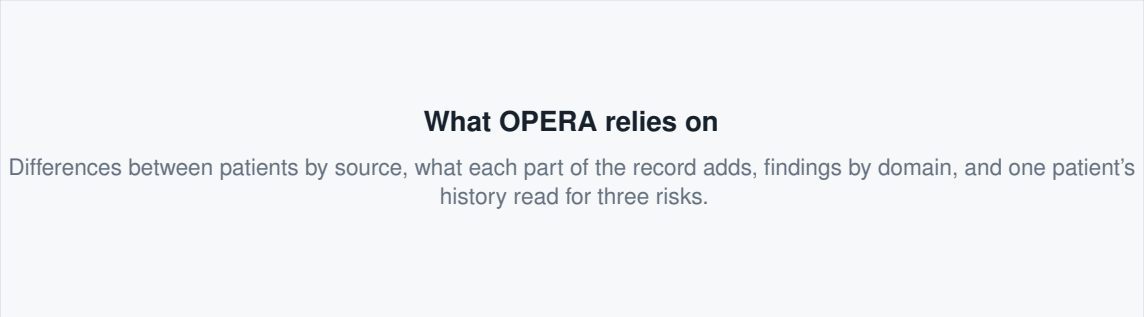


Figure 10. What OPERA relies on. Rows in A–C are clinical domains. Integrated gradients split each held-out patient's prediction into a reference, which keeps the timing of care, age and subtype but replaces every recorded event by an average one, and the contribution of each event. **A**, The differences in predicted one-year risk between patients carried by the timing of care alone (grey) and by each source of records, averaged over the outcomes of each domain. **B**, Change in held-out AUC ($\times 100$) when the fitted model reads only part of the record, without retraining: only the timing of care, no laboratory results, no medication, no diagnoses, no region or specialty of care, or only the last year of history. **C**, Mean change in predicted log-odds among the patients with each finding, averaged over the outcomes of each domain and shown as the corresponding odds ratio, with findings grouped by source. Boxed cells contain a prespecified expected effect, drawn solid where it was seen and dashed where it was not; grey cells had too few patients with the finding to report. **D**, One synthetic patient's history read for three risks, with the episodes that move them most labelled above the tracks. Each track is one outcome, and each circle is an episode of care at its time before treatment, sized by how much it changes the predicted one-year risk (percentage points) and coloured by direction. The predicted risk is given at the right, with the risk for the same timing of care and average findings. Sienna and indigo mark effects throughout (raising risk or losing AUC, and lowering risk), and the other colours identify sources of records. Every outcome is shown in Figs. S32 and S33, and the patient's history read for six complications in Fig. S34.

How does performance vary over time, across regions and between patients?

The held-out test set spans patients treated from 2023, a period in which first-line treatment and register coverage changed. **[One-year AUC and calibration-in-the-large of OPERA and XGBoost in each calendar year of treatment, and whether the gain was stable from the first to the last year.] [The joint model's patient-level Brier gain did / did not depend on characteristics known at the start of treatment: amount of recorded history, age, sex, region and recorded prognostic index]** (Figs. S10 and S22). **[Results with the known-status estimand, excluding the 209 patients whose disease was assigned after day 14, with size-matched clinical cutoffs, and with equal weight per clinical domain.]**

Discussion

A single model that learned across 24 lymphoid cancers and 69 outcomes **[discriminated better than gradient-boosted trees trained and tuned for every disease, outcome and horizon in X of Y predictions, separated outcomes more clearly than the prognostic indices in their own populations and on their own endpoints, and gave predictions for diseases and complications that no index covers]**, and it did so as one model in place of **[N]**. **[Principal findings with uncertainty: what other diseases, other outcomes, learning how much to borrow and population pretraining**

each added; where the gains were largest; whether what the model borrowed followed clinical expectation; and the most informative null or harmful result.] To our knowledge, this is the first study to learn across the diseases and outcomes of the lymphoid cancers as a whole, and the first to separate how much is gained from other diseases, from other outcomes, from learning how much to draw on each and from population pretraining, against both tuned task-specific models and the indices used in the clinic. The principles come from multi-task learning and pretraining; what this study shows is how much they are worth for the questions a haematologist faces at the start of treatment. Recent work has shown the value of shared models for disease onset across claims [32] and for complications in myeloma and ovarian cancer [12]; our results carry this from the onset of disease to the outcomes of its treatment, across every lymphoid cancer and the full record, against the clinical tools of the specialty, and show where the gain comes from. A recent review of clinical AI found no comparison of the large models of medical histories with the clinical scores in use, and judged that the most credible progress so far is on narrowly defined tasks with clear endpoints [34]. This study aims at both: its 69 outcomes were prespecified and fixed at the start of treatment, and each was compared with models tuned for it and, where one exists, with the index used in the clinic.

The controlled comparisons show where shared signal comes from. Another disease can supply examples of the same complication; another outcome can teach the encoder the patient's underlying vulnerability; the two can complement each other, repeat the same information or conflict. **[Which pattern occurred here, and how the gain varied with the data a prediction had of its own.]** The asymmetry between the axes matters for how such models should be built: as described in Methods, the joint model gives every outcome parameters of its own but no disease any, and disease adapters were chosen on the tuning set **[and added more on the test set]**, so the room for improvement lies in letting each disease adjust a shared representation rather than in pooling more outcomes. Sharing everything is not guaranteed to help: multi-task studies of health records have reported uneven benefits and negative transfer [13–15], and pooling related aggressive lymphomas has matched but not clearly improved on a model trained on DLBCL alone [47]. **[Whether and where sharing everything hurt here, and whether learning how much to borrow recovered it.]**

This runs against how haematology has organised prognosis. The field has advanced by telling diseases apart: each new entity in the classification, each index built for one disease and validated in its trials, and each registry and study group organised by disease rests on the assumption that prognosis is disease-specific. That assumption is often right about treatment and biology. **[If the results hold: but for predicting what happens to a patient after treatment starts, the patients of other diseases were informative, and a prediction for one disease improved when the model learned from all of them, also in the common diseases, where a specialised model had the most data of its own.] [If the borrowing analysis agrees: what the model drew on included diseases far apart in the classification that share complications or treatment, such as chronic lymphocytic leukaemia and myeloma for infection (Fig. 8C).]** The boundaries that matter for diagnosis and treatment need not be the boundaries that matter for prediction, and treating them as such leaves information unused, most of all in the diseases with the fewest patients.

Routing is one way to let a model learn how much to borrow; meta-learning [8], automated task grouping [17], mixtures of experts [18, 19] and priors over related tasks are others. The routing controls locate the gain **[if the results hold]**: it did not come from knowing the diagnosis or from the adapters' added capacity, but from borrowing **[(with its claim label)]**. Where it helped is as informative as how much, and two mechanisms predict different places. A disease's own correction should matter most for outcomes whose meaning differs between diseases, such as progression, second-line treatment and transformation; borrowing from other diseases should matter most where complications are shared, such as infection and cytopenia after similar treatment. **[Which domains relied most on borrowing]** (Fig. 8D, Fig. S19D). Switching sources off in the fitted model goes further than route weights, which show where the router sends

weight but not what that weight contributes. That what the model borrowed **[followed]** the pairs expected from biology, treatment and complications, tested as a set against expectations fixed before the test set was opened and against negative controls, **[suggests that it found shared prognostic signal rather than arbitrary correlation]**; pairs it relied on that clinicians would not have predicted **[examples]** are hypotheses worth testing, not findings about the diseases. Borrowing at the readout is also only part of the sharing: **[the fitted router depended on its borrowed corrections by X AUC points, against Y for pooling across diseases (the disease-pooled against the specialist models)]**, so most of what diseases share is carried by the common representation, and routing adds a targeted layer on top **[if the results hold]**.

The most consequential change may be in coverage. Prognostic tools in haematology exist where a disease is common enough to have one and where its inputs were recorded: 10 of the 24 diseases have no dedicated, validated index, and in the test period the index category was missing for 58% of patients, because quality-registry fields arrive late. A model that reads the routine record gives every patient an estimate at the start of treatment, for every outcome, in every disease, including nodal marginal-zone lymphoma, which had five training patients and none in the tuning year, so that no model of its own could even be fitted, and **[N]** disease–outcome combinations with fewer than 20 training events. **[Performance in these cells, and in the diseases without an index (Table 2 and Fig. S7).]** **[If the smallest diseases also gained most from borrowing (Fig. 8B, Fig. S19B), their predictions rest on what other diseases taught the model rather than on their own few events.]** For such questions the choice is not between a shared model and a better specialised one, because the specialised one cannot be built.

The same holds over time. The test period brought new first-line treatment **[confirm for Denmark: e.g. polatuzumab vedotin in DLBCL, targeted agents in CLL, daratumumab combinations in myeloma]** [43], and the records themselves changed between the training and test years: the national patient register moved from LPR2 to LPR3 in 2019, laboratory capture was complete only from 2017, and quality-registry fields were increasingly missing. Shifts of this kind are a common cause of failure for clinical prediction models [76]. **[If performance held, this supports robustness over the observed period]**, although not to future changes or to another health system. New treatments also start with few treated patients. If population pretraining and sharing reduce the labelled patients a model needs **[the joint model from the adapted national encoder reached the full-data cohort-encoder model with X% of the training patients]**, a model can follow a new regimen or a new outcome after fewer patients have been treated, rather than years later.

Population pretraining takes sharing to its limit, and its result may matter most beyond haematology. **[Whether the national encoder helped directly, only after adaptation, or not at all, and with how few labelled patients it matched the cohort encoder.]** A gain need not come only from what the encoder learned from other patients: a national extract also represents each record at finer resolution, keeping specific codes, such as molecular findings, that a cohort-sized vocabulary merges. **[If the national encoder helped:]** the encoder was not built for haematology, its developers never saw these outcomes, and it improved specialised prediction in a cohort that is a small fraction of the population it learned from. That sets a precedent for every clinical specialty in Denmark. The national registers record the diagnoses, procedures, prescriptions, laboratory results and contacts of the whole population, so a model pretrained once on them could be the starting point for cardiology, nephrology, psychiatry or any specialty with data of its own, each adapting it to its patients and labelling only its own outcomes, rather than building and pretraining a model from scratch. The specialties and diseases with the fewest patients, which cannot pretrain a model of their own, would gain most, and a reduction in the labelled patients needed would make the gain larger still. Whether it holds in other specialties is for those studies to show, but the infrastructure to test it exists for the whole country. **[If it did not help: what this says about the value of specialised pretraining, and whether the finer vocabulary compensated.]** Shared models also change the cost of prediction. One OPERA model replaced **[N]** specialists **[and N tabular models, at X of their combined compute and storage]**, and is one

pipeline to validate, monitor and update rather than thousands; a new outcome needs a new output head rather than a new project. Because the registers are national, one model could serve every hospital if it works equally well in every region, which it **[did / did not]** (Fig. S10).

For patients, a shared model gives a profile of risks at the start of treatment rather than a single prognostic summary. A patient at low risk of treatment failure may be at high risk of serious infection, and prophylaxis, monitoring and dose planning differ accordingly. A profile is useful only if its probabilities are accurate, which is why calibration is reported alongside discrimination, and more useful if a clinician can see what each risk rests on: a patient's explanation can show, as in the synthetic example, that a high risk of serious infection follows a recent pneumonia while a high risk of death follows the findings of the diagnostic work-up (Fig. S34). Where the model places patients at low risk who are on standard treatment, its predictions support that treatment rather than change it. Where it identifies patients at high risk, the evidence needed depends on what is done with them [77]: closer monitoring and supportive care, the action in the decision curves, asks for little, whereas changing treatment needs a trial. The model can define who enters such a trial, as CLL-TIM selects patients for PreVent-ACaLL [78, 79], and several of the outcomes predicted here, such as serious infection and severe neutropenia, are what prophylaxis trials would aim to prevent. Whether explanations of this kind help clinicians decide needs testing with them [80].

This study has limitations. Validation was temporal within one health system, and test patients' records before 2022 contributed to self-supervised pretraining, without outcome labels, so the evaluation concerns later treatment episodes rather than unseen individuals. A memorization check found no sign that the pretrained encoder reproduces the records it was trained on: its next-event loss could not tell training from unseen patients (AUC 0.505, 95% CI 0.480 to 0.531), and its continuations of a patient's record were not systematically closer for training patients than for unseen ones (Supplementary Methods); another health system would need to adapt and validate its own model. Every choice was made on a tuning year of 1,980 patients, small for the rarest diseases, which we addressed by keeping choices few and rule-based. Each configuration was trained once; the difference between two fits of the same model **[X]** indicates the variation this omits. Outcomes were derived from registers: laboratory abnormalities are CTCAE-graded measurements rather than adjudicated adverse events, laboratory risk sets condition on monitoring, transformation is a new diagnosis rather than a histological finding, and diseases coded after treatment started were assigned up to 90 days later. Models were trained on patients with known status at each horizon, which tends to overestimate risk where censoring is common; all models share this, but calibration at long horizons should be read with it in mind. Indices could only be compared where they were recorded, mostly as categories rather than scores, and the recorded patients may differ from those without; the cutoffs matched to them were estimated on the tuning year and carry uncertainty that the intervals do not include. The tabular and sequence models used the same patients and index dates but differ in representation and history windows, so their comparison estimates the value of complete pipelines. The national and cohort encoders differ in population, register deliveries, vocabulary and training budget, so their comparison concerns the encoders as delivered rather than scale alone. Region was an input, and two regions contribute richer hospital data, so region partly marks which data a patient has; performance is reported within each region. The models predict risk under the care patients received, not the effect of changing it [81], and attributions describe the fitted model, not causes. Likewise, the borrowing analysis describes how the fitted model uses each source at its readout: removing a source alters a model trained with all of them rather than retraining without it, sharing through the encoder is not resolved by disease, and pairs involving the smallest diseases are imprecise. Finally, the study does not measure whether decisions or outcomes improve when the predictions are used, evidence that remains rare for clinical AI in general [34]; a model deployed in practice would also need its inputs from hospital systems at the time of the decision rather than from registers updated with a delay [78].

[Conclusion. Draft if the results hold: prognosis in the lymphoid cancers does not have to be built one disease and one outcome at a time. One model that learned across their diseases and outcomes, and how much to draw on each, borrowed most where clinicians would expect shared signal, discriminated better than models tuned for each task, gave predictions where no tool exists, and held up over a period of changing treatment and registers. A model pretrained on the whole population improved it further, which opens the same route to every specialty that shares the national registers. The next step is to test whether acting on its predictions improves care, beginning with the complications that prophylaxis can prevent.]

Data availability

According to Danish legislation, the health data used in this study may not be shared publicly. Within the approved DALY-CARE protocol, the data may be accessed on a collaborative basis under a data processing agreement, and all analyses must be performed within the DALY-CARE data resource at the Danish National Genome Center [49]. The protocol and the agreement template are available with the DALY-CARE documentation.

Code availability

All code, model configurations, outcome definitions and analysis recipes are publicly available at **[repository URL and archival DOI]**. The conversion of the registers to the event format, with its configurations, is available at `https://github.com/FGA-DIKU/ehr2meds` **[licence]**. Trained model weights were fitted on patient-level data and are not released.

Patient and public involvement

[Statement on patient and public involvement in the design, conduct or reporting of this research.]

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Author contributions

[CRediT statement.]

Competing interests

[Statement.]

References

[1] Rita Alaggio, Catalina Amador, Ioannis Anagnostopoulos, et al. The 5th edition of the World Health Organization classification of haematolymphoid tumours: Lymphoid neoplasms. *Leukemia*, 36(7):1720–1748, 2022. doi: 10.1038/s41375-022-01620-2.

[2] Richard D. Riley, Joie Ensor, Kym I. E. Snell, Frank E. Harrell, Glen P. Martin, Johannes B. Reitsma, Karel G. M. Moons, Gary Collins, and Maarten van Smeden. Calculating the sample size required for developing a clinical prediction model. *BMJ*, 368:m441, 2020. doi: 10.1136/bmj.m441.

[3] Tjeerd van der Ploeg, Peter C. Austin, and Ewout W. Steyerberg. Modern modelling techniques are data hungry: a simulation study for predicting dichotomous endpoints. *BMC Medical Research Methodology*, 14:137, 2014. doi: 10.1186/1471-2288-14-137.

[4] Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md. Mostofa Ali Patwary, Yang Yang, and Yanqi Zhou. Deep learning scaling is predictable, empirically. *arXiv preprint arXiv:1712.00409*, 2017.

[5] Jonathan Baxter. A model of inductive bias learning. *Journal of Artificial Intelligence Research*, 12:149–198, 2000. doi: 10.1613/jair.731.

[6] Andreas Maurer, Massimiliano Pontil, and Bernardino Romera-Paredes. The benefit of multitask representation learning. *Journal of Machine Learning Research*, 17(81):1–32, 2016.

[7] Rich Caruana. Multitask learning. *Machine Learning*, 28:41–75, 1997. doi: 10.1023/A:1007379606734.

[8] Xi Sheryl Zhang, Fengyi Tang, Hiroko H Dodge, Jiayu Zhou, and Fei Wang. MetaPred: meta-learning for clinical risk prediction with limited patient electronic health records. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 2487–2495, 2019. doi: 10.1145/3292500.3330779.

[9] Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1126–1135, 2017. doi: 10.48550/arXiv.1703.03400.

[10] Hao Liu, Muhan Zhang, Zehao Dong, Lecheng Kong, Yixin Chen, Bradley Fritz, Dacheng Tao, and Christopher King. Time associated meta learning for clinical prediction. *arXiv preprint arXiv:2303.02570*, 2023. doi: 10.48550/arXiv.2303.02570. URL `https://arxiv.org/abs/2303.02570`.

[11] E. J. Oh, R. B. Parikh, C. Chivers, and J. Chen. Two-stage approaches to accounting for patient heterogeneity in machine learning risk prediction models in oncology. *JCO Clinical Cancer Informatics*, 5:1015–1023, 2021. doi: 10.1200/CCI.21.00077.

[12] Jannik Lübberstedt, Krischan Braitsch, Jacqueline Lammert, Christof Winter, Florian Gabriel, Tristan Lemke, Christopher Zirn, Markus Graf, Friedrich Puttkammer, Hartmut Häntze, Johannes Moll, Anirudh Narayanan, Andrei Zhukov, Fabian Drexel, Zeineb Ben Chaaben, Sebastian Ziegelmayer, Su Hwan Kim, Marion Högner, Jan Kirschke, Florian Bassermann, Marcus Makowski, Christian Wachinger, Lisa Adams, and Keno Bressem. Routine laboratory trajectories encode the onset of organ-level complications in cancer. arXiv preprint 2606.08538, 2026. URL `https://arxiv.org/abs/2606.08538`.

[13] Hrayr Harutyunyan, Hrant Khachatrian, David C Kale, Greg Ver Steeg, and Aram Galstyan. Multitask learning and benchmarking with clinical time series data. *Scientific Data*, 6:96, 2019. doi: 10.1038/s41597-019-0103-9.

[14] Matthew B. A. McDermott, Bret Nestor, Evan Kim, Wancong Zhang, Anna Goldenberg, Peter Szolovits, and Marzyeh Ghassemi. A comprehensive evaluation of multi-task learning and multi-task pre-training on EHR time-series data. *arXiv preprint arXiv:2007.10185*, 2020.

[15] Daisy Yi Ding, Chloé Simpson, Stephen Pfohl, Dave C. Kale, Kenneth Jung, and Nigam H. Shah. The effectiveness of multitask learning for phenotyping with electronic health records data. In *Pacific Symposium on Biocomputing*, volume 24, pages 18–29, 2019. URL `https://psb.stanford.edu/psb-online/proceedings/psb19/ding.pdf`.

[16] Bradley Efron and Carl Morris. Data analysis using Stein's estimator and its generalizations. *Journal of the American Statistical Association*, 70(350):311–319, 1975. doi: 10.1080/01621459.1975.10479864.

[17] Suhan Cui and Prasenjit Mitra. Automated multi-task learning for joint disease prediction on electronic health records. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL `https://arxiv.org/abs/2403.04086`.

[18] Jeong Min Lee and Milos Hauskrecht. Learning to adapt clinical sequences with residual mixture of experts. In *Artificial Intelligence in Medicine (AIME)*, 2022. URL `https://arxiv.org/abs/2204.02687`.

[19] Xiaoyang Wang and Christopher C. Yang. MoE-Health: A mixture of experts framework for robust multimodal healthcare prediction. In *Proceedings of the 16th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB)*, 2025. URL `https://arxiv.org/abs/2508.21793`.

[20] Yikuan Li, Shishir Rao, José Roberto Ayala Solares, Abdelaali Hassaine, Rema Ramakrishnan, Dexter Canoy, Yajie Zhu, Kazem Rahimi, and Gholamreza Salimi-Khorshidi. BEHRT: transformer for electronic health records. *Scientific Reports*, 10(1):7155, 2020. doi: 10.1038/s41598-020-62922-y.

[21] Laila Rasmy, Yang Xiang, Ziqian Xie, Cui Tao, and Degui Zhi. Med-BERT: pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *npj Digital Medicine*, 4(1):86, 2021. doi: 10.1038/s41746-021-00455-y.

[22] Mikkel Fruelund Odgaard, Kiril Vadimovic Klein, Martin Sillesen, Sanne Møller Thysen, Espen Jimenez-Solem, and Mads Nielsen. CORE-BEHRT: A carefully optimized and rigorously evaluated BEHRT. In *Proceedings of the 9th Machine Learning for Healthcare Conference*, volume 252 of *Proceedings of Machine Learning Research*. PMLR, 2024. URL `https://proceedings.mlr.press/v252/odgaard24a.html`.

[23] Ethan Steinberg, Jason Alan Fries, Yizhe Xu, and Nigam H. Shah. MOTOR: A time-to-event foundation model for structured medical records. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL `https://proceedings.iclr.cc/paper_files/paper/2024/file/b36554b97da741b1c48c9de05c73993e-Paper-Conference.pdf`.

[24] Charles Gadd, Krishna Gokhale, Aditya Acharya, Jennifer Cooper, Francesca Crowe, Leah Fitzsimmons, Thomas Jackson, Krishnarajah Nirantharakumar, Christopher Yau, and The OPTIMAL Consortium. SurvivEHR: a competing risks, time-to-event foundation model for multiple long-term conditions from primary care electronic health records. *npj Digital Medicine*, 9:546, 2026. doi: 10.1038/s41746-026-02709-z.

[25] Artem Shmatko, Alexander Wolfgang Jung, Kumar Gaurav, Søren Brunak, Laust Hvas Mortensen, Ewan Birney, Tom Fitzgerald, and Moritz Gerstung. Learning the natural history of human disease with generative transformers. *Nature*, 647(8088):248–256, 2025. doi: 10.1038/s41586-025-09529-3.

[26] Germans Savcisens, Tina Eliassi-Rad, Lars Kai Hansen, Laust Hvas Mortensen, Lau Lilleholt, Anna Rogers, Ingo Zettler, and Sune Lehmann. Using sequences of life-events to predict human lives. *Nature Computational Science*, 4(1):43–56, 2024. doi: 10.1038/s43588-023-00573-5.

[27] Davide Placido, Bo Yuan, Jessica X Hjaltelin, Chunlei Zheng, Amalie D Haue, Piotr J Chmura, Chen Yuan, Jihye Kim, Renato Umeton, Gregory Antell, Alexander Chowdhury, Alexandra Franz, Lauren Brais, Elizabeth Andrews, Debora S Marks, Aviv Regev, Siamack Ayandeh, Mary T Brophy, Nhan V Do, Peter Kraft, Brian M Wolpin, Michael H Rosenthal, Nathanael R Fillmore, Søren Brunak, and Chris Sander. A deep learning algorithm to predict risk of pancreatic cancer from disease trajectories. *Nature Medicine*, 29(5):1113–1122, 2023. doi: 10.1038/s41591-023-02332-5.

[28] J. Jee, C. Fong, K. Pichotta, T. N. Tran, A. Luthra, M. Waters, et al. Automated real-world data integration improves cancer outcome prediction. *Nature*, 636(8043):728–736, 2024. doi: 10.1038/s41586-024-08167-5.

[29] S. Bjerregaard-Michelsen, L. Ø. Poulsen, A. Bjerrum, M. Bøgsted, and C. Vesteghem. Machine learning for prediction of 30-day mortality in patients with advanced cancer: comparing pan-cancer and single-cancer models. *ESMO Real World Data and Digital Oncology*, 8: 100146, 2025. doi: 10.1016/j.esmorw.2025.100146.

[30] Shane Waxler, Paul Blazek, Davis White, Daniel Sneider, Kevin Chung, Mani Nagarathnam, et al. Generative medical event models improve with scale. *arXiv preprint arXiv:2508.12104*, 2025. doi: 10.48550/arXiv.2508.12104.

[31] Payal Chandak, Gregory Kondas, Liat Antwarg Friedman, Isaac Kohane, and Matthew McDermott. EveryQuery: A promptable foundation model for clinical prediction tasks over electronic health records. *arXiv preprint arXiv:2603.07900*, 2026. doi: 10.48550/arXiv.2603.07900. URL `https://arxiv.org/abs/2603.07900v3`. Version 3, revised 26 September 2026.

[32] Fan Ma, Yuntian Liu, Xiang Lan, Weipeng Zhou, Jun Ni, Mauro Giuffrè, Lingfei Qian, Xueqing Peng, Yujia Zhou, Ruey-Ling Weng, Huan He, Lu Li, Huiyuan Wang, Qingyu Chen, Andrew Loza, Laila Rasmy, Degui Zhi, Yuan Lu, Chenjie Zeng, Joshua C. Denny, Lee Schwamm, Daniella Meeker, Lucila Ohno-Machado, Yong Chen, and Hua Xu. Foundation models to unlock real-world evidence from nationwide medical claims. arXiv preprint 2605.02740, 2026. URL `https://arxiv.org/abs/2605.02740`.

[33] Michael Wornow, Yizhe Xu, Rahul Thapa, Birju Patel, Ethan Steinberg, Scott Fleming, Michael A Pfeffer, Jason Fries, and Nigam H Shah. The shaky foundations of large language models and foundation models for electronic health records. *npj Digital Medicine*, 6:135, 2023. doi: 10.1038/s41746-023-00879-8.

[34] John Emmett Worth, Anastasia Perez, David J. H. Wu, Peter Brodeur, Emily Tat, Arjun Manrai, David Wu, Chase Walton, Liam G. McCoy, Priyank Jain, Rebecca Handler, Jason Hom, Laura Zwaan, Vishnu Ravi, Brian Han, Kevin Schulman, Kathleen Lacar, Kameron Black, Adrian Haimovich, Adam Rodman, Ethan Goh, and Jonathan H. Chen. State of clinical AI in 2026. *BMJ Digital Health & AI*, 2(1):e000113, 2026. doi: 10.1136/bmjdh-2026-000113.

[35] Michael Wornow, Rahul Thapa, Ethan Steinberg, Jason A. Fries, and Nigam H. Shah. EHRSHOT: An EHR benchmark for few-shot evaluation of foundation models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. URL `https://proceedings.neurips.cc/paper_files/paper/2023/hash/d42db1f74df54cb992b3956eb7f15a6f-Abstract-Datasets_and_Benchmarks.html`.

[36] Milena Sant, Claudia Allemani, Carmen Tereanu, Roberta De Angelis, Riccardo Capocaccia, et al. Incidence of hematologic malignancies in europe by morphologic subtype: results of the HAEMACARE project. *Blood*, 116(19):3724–3734, 2010. doi: 10.1182/blood-2010-05-282632.

[37] International Non-Hodgkin's Lymphoma Prognostic Factors Project. A predictive model for aggressive non-Hodgkin's lymphoma. *New England Journal of Medicine*, 329(14):987–994, 1993. doi: 10.1056/NEJM199309303291402.

[38] Zheng Zhou, Laurie H Sehn, Alfred W Rademaker, Leo I Gordon, Ann S LaCasce, Allison Crosby-Thompson, Ann Vanderplas, Andrew D Zelenetz, Gregory A Abel, Maria A Rodriguez, Auayporn Nademanee, Mark S Kaminski, Myron S Czuczman, Michael Millenson, Joyce Niland, Randy D Gascoyne, Joseph M Connors, Jonathan W Friedberg, and Jane N Winter. An enhanced International Prognostic Index (NCCN-IPI) for patients with diffuse large B-cell lymphoma treated in the rituximab era. *Blood*, 123(6):837–842, 2014. doi: 10.1182/blood-2013-09-524108.

[39] International CLL-IPI working group. An international prognostic index for patients with chronic lymphocytic leukaemia (CLL-IPI): a meta-analysis of individual patient data. *The Lancet Oncology*, 17(6):779–790, 2016. doi: 10.1016/S1470-2045(16)30029-8.

[40] Antonio Palumbo, Hervé Avet-Loiseau, Stefania Oliva, et al. Revised international staging system for multiple myeloma: A report from international myeloma working group. *Journal of Clinical Oncology*, 33(26):2863–2869, 2015. doi: 10.1200/JCO.2015.61.2267.

[41] Jorne Biccler, Sandra Eloranta, Peter de Nully Brown, et al. Simplicity at the cost of predictive accuracy in diffuse large B-cell lymphoma: a critical assessment of the R-IPI, IPI, and NCCN-IPI. *Cancer Medicine*, 7(1):114–122, 2018. doi: 10.1002/cam4.1271.

[42] Dansk Lymfomgruppe. Diffust storcellet B-celle lymfom. Klinisk retningslinje, version 4.0. DMCG.dk, Danish Multidisciplinary Cancer Groups, 2025. Professionally approved 4 December 2025. `https://www.dmcg.dk/kliniske-retningslinjer/kliniske-retningslinjer-opdelt-paa-dmcg/lymfom/diffust-storcellet-b-celle-lymfom/`.

[43] Hervé Tilly, Franck Morschhauser, Laurie H. Sehn, Jonathan W. Friedberg, Marek Trněný, Jeff P. Sharman, et al. Polatuzumab vedotin in previously untreated diffuse large B-cell lymphoma. *New England Journal of Medicine*, 386(4):351–363, 2022. doi: 10.1056/NEJMoa2115304.

[44] Petra Langerbeins, Can Zhang, Sandra Robrecht, et al. The CLL12 trial: ibrutinib vs placebo in treatment-naïve, early-stage chronic lymphocytic leukemia. *Blood*, 139(2):177–187, 2022. doi: 10.1182/blood.2021010845.

[45] J. L. Biccler, S. Eloranta, P. de Nully Brown, H. Frederiksen, M. Jerkeman, J. Jørgensen, L. H. Jakobsen, K. E. Smedby, M. Bøgsted, and T. C. El-Galaly. Optimizing outcome prediction in diffuse large B-cell lymphoma by use of machine learning and nationwide lymphoma registries: a Nordic Lymphoma Group study. *JCO Clinical Cancer Informatics*, 2:1–13, 2018. doi: 10.1200/CCI.18.00025.

[46] Rudi Agius, Christian Brieghel, Michael A Andersen, Alexander T Pearson, Bruno Ledergerber, Alessandro Cozzi-Lepri, Yoram Louzoun, Christen L Andersen, Jacob Bergstedt, Jakob H von Stemann, Mette Jørgensen, Man-Hung Eric Tang, Magnus Fontes, Jasmin Bahlo, Carmen D Herling, Michael Hallek, Jens Lundgren, Cameron Ross MacPherson, Jan Larsen, and Carsten U Niemann. Machine learning can identify newly diagnosed

patients with CLL at high risk of infection. *Nature Communications*, 11:363, 2020. doi: 10.1038/s41467-019-14225-8.

[47] Mikkel Werling, Alexander D Fuglkjær, Peter Brown, Carsten U Niemann, and Rudi Agius. Machine learning enhances risk stratification and treatment failure prediction in diffuse large B-cell lymphoma. *HemaSphere*, 10(4):e70336, 2026. doi: 10.1002/hem3.70336.

[48] Mehdi Parviz, Christian Brieghel, Mikkel Werling, Thomas Lacoppidan, Emelie Rotbain, Carsten U. Niemann, and Rudi Agius. Post-treatment infection prediction in CLL using domain adaptation of lymphoma electronic health records. *Acta Oncologica*, 65:109–118, 2026. doi: 10.2340/1651-226X.2026.44569.

[49] Christian Brieghel, Mikkel Werling, Casper Møller Frederiksen, Mehdi Parviz, Thomas Lacoppidan, Tereza Faitova, Rebecca Svanberg Teglgaard, Noomi Vainer, Caspar da Cunha-Bang, Emelie Curovic Rotbain, Rudi Agius, and Carsten Utoft Niemann. The Danish Lymphoid Cancer Research (DALY-CARE) data resource: the basis for developing data-driven hematology. *Clinical Epidemiology*, 17:131–145, 2025. doi: 10.2147/CLEP.S479672.

[50] Morten Schmidt, Lars Pedersen, and Henrik Toft Sørensen. The Danish Civil Registration System as a tool in epidemiology. *European Journal of Epidemiology*, 29(8):541–549, 2014. doi: 10.1007/s10654-014-9930-3.

[51] Morten Schmidt, Sigrun Alba Johannesdottir Schmidt, Kasper Adelborg, Jens Sundbøll, Kristina Laugesen, Vera Ehrenstein, and Henrik Toft Sørensen. The Danish health care system and epidemiological research: from health care contacts to database records. *Clinical Epidemiology*, 11:563–591, 2019. doi: 10.2147/CLEP.S179083.

[52] Bente Arboe, Pär Josefsson, Judit Jørgensen, et al. Danish National Lymphoma Registry. *Clinical Epidemiology*, 8:577–581, 2016. doi: 10.2147/clep.S99470.

[53] Caspar da Cunha-Bang, Christian H. Geisler, Lisbeth Enggaard, et al. The Danish National Chronic Lymphocytic Leukemia Registry. *Clinical Epidemiology*, 8:561–565, 2016. doi: 10.2147/clep.s99486.

[54] Peter Gimsing, Morten O. Holmström, Tobias W. Klausen, et al. The Danish National Multiple Myeloma Registry. *Clinical Epidemiology*, 8:583–587, 2016. doi: 10.2147/clep.s99463.

[55] Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A. Smith. Don't stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8342–8360, 2020.

[56] Lin Lawrence Guo, Jason Fries, Ethan Steinberg, et al. A multi-center study on the adaptability of a shared foundation model for electronic health records. *npj Digital Medicine*, 7: 171, 2024. doi: 10.1038/s41746-024-01166-w.

[57] Joshua Lemmon, Lin Lawrence Guo, et al. Self-supervised machine learning using adult inpatient data produces effective models for pediatric clinical prediction tasks. *Journal of the American Medical Informatics Association*, 30(12):2004–2011, 2023. doi: 10.1093/jamia/ocad175.

[58] National Cancer Institute. *Common Terminology Criteria for Adverse Events (CTCAE), Version 5.0*, 2017. URL `https://dctd.cancer.gov/research/ctep-trials/for-sites/adverse-events/ctcae-v5-5x7.pdf`. Published 27 November 2017.

[59] National Cancer Institute. *Common Terminology Criteria for Adverse Events (CTCAE), Version 6.0*, 2025. URL https://dctd.cancer.gov/research/ctep-trials/for-sites/adverse-events/ctcae-v6.pdf. Published 22 July 2025.

[60] Michael F Gensheimer and Balasubramanian Narasimhan. A scalable discrete-time survival model for neural networks. *PeerJ*, 7:e6257, 2019. doi: 10.7717/peerj.6257.

[61] Changhee Lee, William Zame, Jinsung Yoon, and Mihaela van der Schaar. DeepHit: A deep learning approach to survival analysis with competing risks. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 2018. doi: 10.1609/aaai.v32i1.11842.

[62] Jiaqi Ma, Zhe Zhao, Xinyang Yi, Jilin Chen, Lichan Hong, and Ed H. Chi. Modeling task relationships in multi-task learning with multi-gate mixture-of-experts. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 1930–1939, 2018. doi: 10.1145/3219819.3220007.

[63] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for NLP. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2790–2799, 2019. URL https://proceedings.mlr.press/v97/houlsby19a.html.

[64] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. URL https://arxiv.org/abs/2106.09685.

[65] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on tabular data? *arXiv preprint arXiv:2207.08815*, 2022. doi: 10.48550/arXiv.2207.08815.

[66] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

[67] Martin Bernstorff, Kenneth Enevoldsen, Jakob Damgaard, Andreas Danielsen, and Lasse Hansen. timeseriesflattener: a Python package for summarizing features from (medical) time series. *Journal of Open Source Software*, 8(83):5197, 2023. doi: 10.21105/joss.05197.

[68] Paul Blanche, Jean-François Dartigues, and Hélène Jacqmin-Gadda. Estimating and comparing time-dependent areas under receiver operating characteristic curves for censored event times with competing risks. *Statistics in Medicine*, 32(30):5381–5397, 2013. doi: 10.1002/sim.5958.

[69] Thomas A. Gerds and Martin Schumacher. Consistent estimation of the expected brier score in general survival models with right-censored event times. *Biometrical Journal*, 48(6): 1029–1040, 2006. doi: 10.1002/bimj.200610301.

[70] Rotraut Schoop, Jan Beyersmann, Martin Schumacher, and Harald Binder. Quantifying the predictive accuracy of time-to-event models in the presence of competing risks. *Biometrical Journal*, 53(1):88–112, 2011. doi: 10.1002/bimj.201000073.

[71] Michael W. Kattan and Thomas A. Gerds. The index of prediction accuracy: an intuitive measure useful for evaluating risk prediction models. *Diagnostic and Prognostic Research*, 2: 7, 2018. doi: 10.1186/s41512-018-0029-2.

[72] Andrew J. Vickers and Elena B. Elkin. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making*, 26(6):565–574, 2006. doi: 10.1177/0272989X06295361.

[73] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 3319–3328. PMLR, 2017. URL https://proceedings.mlr.press/v70/sundararajan17a.html.

[74] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://papers.nips.cc/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html.

[75] Gary S. Collins, Karel G. M. Moons, Paula Dhiman, Richard D. Riley, Andrew L. Beam, Ben Van Calster, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.

[76] Samuel G. Finlayson, Adarsh Subbaswamy, Karandeep Singh, John Bowers, Annabel Kupke, Jonathan Zittrain, Isaac S. Kohane, and Suchi Saria. The clinician and dataset shift in artificial intelligence. *New England Journal of Medicine*, 385(3):283–286, 2021. doi: 10.1056/NEJMc2104626.

[77] Robin Blythe, Melissa Baysari, Gary S. Collins, Darren Dahly, Nicholas Graves, Jessica Hullman, Tong Liu, Lian Leng Low, Yian Ma, Yilin Ning, Jasmine Chiat Ling Ong, Marcus Eng Hock Ong, Mark Sculpher, Dan Zeltzer, and Nan Liu. Towards rigorous economic evidence for clinical AI. SSRN preprint 7474800, 2026. URL https://ssrn.com/abstract=7474800.

[78] Rudi Agius, Anders C. Riis-Jensen, Bettina Wimmer, Caspar da Cunha-Bang, Daniel Dawson Murray, Christian Bjørn Poulsen, Marianne B. Bertelsen, Berit Schwartz, Jens Dilling Lundgren, Henning Langberg, et al. Deployment and validation of the CLL treatment infection model adjoined to an EHR system. *npj Digital Medicine*, 7(1):147, 2024. doi: 10.1038/s41746-024-01132-6.

[79] Rigshospitalet. Acalabrutinib and venetoclax treatment of newly diagnosed patients with CLL at high risk of infection or early treatment (PreVent-ACaLL). ClinicalTrials.gov identifier NCT03868722, 2019. URL https://clinicaltrials.gov/study/NCT03868722.

[80] Marzyeh Ghassemi, Luke Oakden-Rayner, and Andrew L. Beam. The false hope of current approaches to explainable artificial intelligence in health care. *The Lancet Digital Health*, 3(11):e745–e750, 2021. doi: 10.1016/S2589-7500(21)00208-9.

[81] Nan van Geloven, Sonja A. Swanson, Chava L. Ramspek, Kim Luijken, Merel van Diepen, Tim P. Morris, Rolf H. H. Groenwold, Hans C. van Houwelingen, Hein Putter, and Saskia le Cessie. Prediction meets causal inference: the role of treatment in clinical prediction models. *European Journal of Epidemiology*, 35(7):619–630, 2020. doi: 10.1007/s10654-020-00636-1.

[82] Maria Elkjær Montgomery, Christian Igel, Mikkel Fruelund Odgaard, Martin Sillesen, and Mads Nielsen. How should transformers encode numeric values in electronic health records? In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 90118–90133. PMLR, 2026. URL https://proceedings.mlr.press/v306/montgomery26a.html.

[83] Maria Elkjær Montgomery, Kiril Klein, Mikkel Odgaard, Stephan Sloth Lorenzen, and Zahra Sobhaninia. BONSAI: A framework for processing and analysing electronic health records (EHR) data using transformer-based models. *Journal of Open Source Software*, 10 (114):8869, 2025. doi: 10.21105/joss.08869.

[84] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL `https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html`.

[85] Seyed Mehran Kazemi, Rishab Goel, Sepehr Eghbali, et al. Time2Vec: Learning a vector representation of time. *arXiv preprint arXiv:1907.05321*, 2019. URL `https://arxiv.org/abs/1907.05321`.

[86] Philippe Solal-Céligny, Pascal Roy, Philippe Colombat, et al. Follicular lymphoma international prognostic index. *Blood*, 104(5):1258–1265, 2004. doi: 10.1182/blood-2003-12-4434.

[87] Eva Hoster, Martin Dreyling, Wolfram Klapper, et al. A new prognostic index (MIPI) for patients with advanced-stage mantle cell lymphoma. *Blood*, 111(2):558–565, 2008. doi: 10.1182/blood-2007-06-095331.

[88] Philip R. Greipp, Jesus San Miguel, Brian G. M. Durie, et al. International staging system for multiple myeloma. *Journal of Clinical Oncology*, 23(15):3412–3420, 2005. doi: 10.1200/JCO.2005.04.242.

[89] Dirk Hasenclever and Volker Diehl. A prognostic score for advanced Hodgkin's disease. *New England Journal of Medicine*, 339(21):1506–1514, 1998. doi: 10.1056/NEJM199811193392104.

[90] Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87, 1991. doi: 10.1162/neco.1991.3.1.79.

[91] Jonas Pfeiffer, Aishwarya Kamath, Andreas Rücklé, Kyunghyun Cho, and Iryna Gurevych. AdapterFusion: Non-destructive task composition for transfer learning. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 487–503, 2021. doi: 10.18653/v1/2021.eacl-main.39.

[92] Matthew Stephens. False discovery rates: a new deal. *Biostatistics*, 18(2):275–294, 2017. doi: 10.1093/biostatistics/kxw041.

[93] Gary S. Collins, Emmanuel O. Ogundimu, and Douglas G. Altman. Sample size considerations for the external validation of a multivariable prognostic model: a resampling study. *Statistics in Medicine*, 35(2):214–226, 2016. doi: 10.1002/sim.6787.

[94] Norbert Schmitz, Samira Zeynalova, Maike Nickelsen, et al. CNS international prognostic index: A risk model for CNS relapse in patients with diffuse large B-cell lymphoma treated with R-CHOP. *Journal of Clinical Oncology*, 34(26):3150–3156, 2016. doi: 10.1200/JCO.2015.65.6520.

[95] Apostolia M. Tsimberidou, Susan O'Brien, Issa Khouri, et al. Clinical outcomes and prognostic factors in patients with Richter's syndrome treated with chemotherapy or chemoimmunotherapy with or without stem-cell transplantation. *Journal of Clinical Oncology*, 24(15):2343–2351, 2006. doi: 10.1200/JCO.2005.05.0187.

[96] Adam J. Olszewski, Lasse H. Jakobsen, Graham P. Collins, et al. Burkitt lymphoma international prognostic index. *Journal of Clinical Oncology*, 39:1129–1138, 2021. doi: 10.1200/JCO.20.03288.

[97] Pierre Morel et al. International prognostic scoring system for Waldenström macroglobulinemia. *Blood*, 113:4163–4170, 2009.

[98] Catherine Thieblemont et al. MALT lymphoma prognostic index: a scoring system for patients with extranodal marginal-zone lymphoma. *Blood*, 130:1409–1417, 2017.

[99] Carlos Montalbán et al. Simplification of risk stratification for splenic marginal zone lymphoma: a point-based score for practical use. *British Journal of Haematology*, 159:164–171, 2012.

[100] Luca Arcaini, Côme Bommier, Juan Pablo Alderuccio, Michele Merli, et al. Marginal zone lymphoma international prognostic index: a unifying prognostic index for marginal zone lymphomas requiring systemic treatment. *eClinicalMedicine*, 72:102592, 2024. doi: 10.1016/j.eclinm.2024.102592.

[101] Michael R. Grever et al. Consensus guidelines for the diagnosis and management of patients with classic hairy cell leukemia. *Blood*, 129:553–560, 2017.

[102] Andrea Gallamini et al. Peripheral T-cell lymphoma unspecified (PTCL-U): a new prognostic model from a retrospective multicentric clinical study. *Blood*, 103:2474–2479, 2004.

[103] Ranjana H. Advani et al. Outcomes and prognostic factors in angioimmunoblastic T-cell lymphoma: final report from the International T-cell Project. *Blood*, 138:213–220, 2021.

[104] David Sibon et al. Long-term outcome of adults with systemic anaplastic large-cell lymphoma treated within the Groupe d'Etude des Lymphomes de l'Adulte trials. *Journal of Clinical Oncology*, 30:3939–3946, 2012.

[105] Artur Jurczyszyn, Jakub Radocha, Julio Davila, et al. Prognostic indicators in primary plasma cell leukaemia: a multicentre retrospective study of 117 patients. *British Journal of Haematology*, 180(6):831–839, 2018. doi: 10.1111/bjh.15092.

[106] David Dingli, Robert A. Kyle, S. Vincent Rajkumar, et al. Immunoglobulin free light chains and solitary plasmacytoma of bone. *Blood*, 108:1979–1983, 2006.

[107] Shaji Kumar et al. Revised prognostic staging system for light chain amyloidosis incorporating cardiac biomarkers and serum free light chain measurements. *Journal of Clinical Oncology*, 30:989–995, 2012.

[108] Daniel Runfola, Austin Anderson, Heather Baier, Matt Crittenden, Elizabeth Dowker, Sydney Fuhrig, Seth Goodman, Grace Grimsley, Rachel Layko, Graham Melville, Maddy Mulder, Rachel Oberman, Joshua Panganiban, Andrew Peck, Leigh Seitz, Sylvia Shea, Hannah Slevin, Rebecca Youngerman, and Lauren Hobbs. geoBoundaries: A global database of political administrative boundaries. *PLOS ONE*, 15(4):e0231866, 2020. doi: 10.1371/journal.pone.0231866.

Supplementary Methods: model and analysis details

Data sources and processing

All data were held and processed in the DALY-CARE environment at the Danish National Genome Center [49]. Registers were linked through the civil registration number and converted to the Medical Event Data Standard (MEDS) with ehr2meds (`https://github.com/FGA-DIKU/ehr2meds`). MEDS stores every record as an event with a patient, a time, a code and an optional value, so all sources share one format. Table S1 lists the sources, what each contributes and when its events are dated, and Fig. S2 shows how they form one sequence per patient.

Table S1. Sources of the event sequences. Register names follow the DALY-CARE resource [49]. Coverage gives the years with national data in the extract.

Source	Events	Time of the event	Coverage
Civil registration	sex, date of birth, date of death	fixed per patient	**[years]**
National Patient Register (LPR2 to 2019, LPR3 from 2019)	diagnoses, procedures, and the reason, priority, type, specialty and region of hospital contacts	diagnoses at contact end, contact attributes at contact start, procedures when performed	**[years]**
National Pathology Register	SNOMED codes, including molecular findings such as TP53 aberrations **[confirm]**; specialty and region of the requesting department	receipt of the requisition	**[years]**
Register of Pharmaceutical Sales	redeemed prescriptions (ATC)	dispensing	**[years]**
Hospital Medication Register	hospital-administered medication (ATC), with specialty and region	start of the order	**[years]**
Laboratory database	results by NPU code, as value bin or text result	sampling	**[years, varies by region]**

The registers arrive as separate tables with their own identifiers and conventions for time, so the conversion first gives them a common form. Diagnoses and procedures are recorded against a hospital contact rather than a patient, so they were joined to their contact. Contacts, pathology requisitions and hospital medication orders were assigned a region and a primary specialty from the national register of health-care organisations, using each organisation as it was registered on the date of the record, because departments merge, move and change specialty over time. The change of patient register in 2019 also changed how some contact attributes are coded; where a documented correspondence exists, older values were mapped to their newer equivalents, so that a token means the same before and after the change.

The time given to an event determines what the model can see at the start of treatment. The patient registers do not record when during a contact a diagnosis was made, so diagnoses were placed at the end of their contact, the first moment they are certain to have been known; a diagnosis made during an admission that spans the start of treatment is therefore not visible to the model. Attributes known when a contact opens were placed at its start, procedures when performed, prescriptions at dispensing, hospital medication at the start of the order and laboratory results at sampling.

Rare codes give the model too few examples to learn from, so codes with fewer than 100 events were merged into the nearest ancestor in their classification that reached 100 events (a shorter diagnosis, procedure, ATC or SNOMED code); laboratory and contact codes were not merged. Laboratory values were binned separately for each test: values were clipped at the 1st and 99th percentiles and assigned to quantile bins, with more bins for tests with more distinct

values, following the power law that Montgomery et al. fitted for transformers on health records ($B = \text{round}(1.14\, N_{\text{distinct}}^{0.237})$, between 2 and 100) [82]. Each result became one token combining the test with its bin, or with its text value for non-numeric results such as "<0.1". Bins and code counts were fitted on a random 90% of patients; this split serves only these statistics and is unrelated to the study's training, tuning and test periods.

BONSAI [83] then turned each patient's events into the sequence the models read. The vocabulary was fitted on events before 1 January 2022, and codes first seen later were read as an unknown token, so no code that appears only in the test period has a learned embedding. Sex and age opened the sequence, a separator marked each change of time so that events recorded together stay grouped, and every event kept its time. For each prediction the sequence was cut strictly before the start of treatment and, if longer than 8,192 tokens, shortened to the sex and age tokens and the most recent events. The conversion can be rerun on the same registers inside the DALY-CARE environment from the configurations in the ehr2meds repository.

Model development

Encoder and pretraining. The encoder builds on CORE-BEHRT [22], adapted from a bidirectional to a causal transformer [84] that predicts each next event from the events before it, with every event's time encoded by a scaled Time2Vec representation [85] (Table S2). It was pretrained on the histories before 1 January 2022 of the pretraining population, every DALY-CARE patient: adults diagnosed with a lymphoid cancer or a precursor condition (MGUS, MBL) since 2002 (78,668), for 10 epochs, the budget of the national encoder. An encoder twice as wide was considered before the analysis was locked and not adopted: fine-tuned as the joint model, it improved tuning AUC by 0.003, within the difference between two fits of the same model, so the narrower encoder was kept. This encoder initialised every supervised model, each trained, tuned and tested on the final cohort (37,531 patients).

Sampling and loss weighting. Patients were sampled in their natural proportions, without oversampling patients with events or reweighting classes, so that the predicted risks keep the base rates of the population and stay calibrated. Each training patient has events for a few of the 69 outcomes and none for most, so no single weighting of patients could balance all disease–outcome–horizon tasks at once; balance between tasks instead comes from the loss, averaged so that each endpoint group weighs equally. As a sensitivity analysis fixed before the test set was opened, the routed and the joint model were refitted with importance-weighted event sampling: batches were enriched towards patients with events at the six horizons, and each patient's loss was divided by their sampling weight, so the likelihood remained that of the natural population (**[X]** and **[Y]** in AUC against natural sampling).

Memorization check. Test patients' records before 2022 were part of pretraining, so we checked whether the encoder reproduces the records it was trained on. The pretrained model was given the first X events of a patient's record (X = 16 to 4,096) and continued it greedily for Y events (Y = 1 to 64); the continuation was compared with the real record by four scores: all Y events reproduced in order, the share of the next Y events reproduced in any order, the same within 30 days of when they occurred, and the share of the patient's rare codes (held by fewer than 1 in 1,000 training patients) reproduced. Records are partly predictable, so each score was compared between up to 2,000 pretraining patients and up to 2,000 patients the encoder never saw, and memorization was read as the excess of the former, considering only combinations of X and Y with at least 50 scored patients in each group. The next-event loss could not tell the two groups apart (AUC 0.505, 95% CI 0.480 to 0.531), and **[the largest excess of any score was X (CI) at X = , Y =]** (code: `bonsai/run/memorization_check.py` in the BONSAI repository).

Table S2. Model and training specification.

Encoder	causal transformer; hidden size 256, 4 layers, 4 attention heads of 64 dimensions; **[feed-forward width, parameters]**
Input	up to 8,192 tokens; scaled Time2Vec time encoding; laboratory results as test–bin tokens
Pretraining	next-event prediction; events before 1 January 2022; 10 epochs; **[optimiser, learning rate, batch, tokens]**
Prediction head	linear; piecewise-constant hazards for each outcome and for competing death; cumulative incidence at 30, 60, 90, 180, 365 and 730 days
Supervised training	effective batch of 256 patients; up to 30 epochs, stopping after 5 without improvement in tuning loss; tuning-best checkpoint

Prediction head and loss. Every supervised model consisted of the pretrained encoder, trained further, and a linear head that produced, for each outcome, piecewise-constant hazards for the outcome and for competing death, from which the cumulative incidence at each horizon was derived [60, 61]. The loss was a binary cross-entropy at each horizon over the patients whose status was known: those with the event before the horizon were positive, those who died of another cause or were followed beyond it negative, and those censored earlier did not contribute at that horizon. Censoring weights were used in evaluation, not in training. Losses were averaged within each outcome and then across endpoints, treating the grade variants and nested definitions of one finding as one endpoint (69 outcomes, 52 endpoints), so that a finding defined at two grades did not count twice.

Supervision scopes. The four scopes, described in Methods, differed only in which cells of the disease-by-outcome grid provided labels (Fig. S1). Every model predicted all six horizons and was trained with the same settings (Table S2) on the same target patients, labels and horizons. Each selected its checkpoint on the tuning labels within its own scope, so a contrast between scopes includes what the extra labels do to training as well as to the final prediction. A disease without tuning patients therefore cannot have a model of its own: nodal marginal-zone lymphoma had five training patients and none in 2022 (Table 1), so its specialists and its outcome-pooled model were not estimable by design and were recorded as such, not as failures; its predictions come only from the disease-pooled, joint and routed models. Because equal epochs do not mean equal computation, optimisation steps and compute are reported for each scope.

Outcome panel

We predicted 69 outcomes, each at 30, 60, 90, 180, 365 and 730 days, in 12 clinical domains (definitions supplement): disease course and later malignancy (including second-line treatment, treatment failure, transformation, new malignancy and death); infection and antimicrobial use; erythroid findings and transfusion; neutrophil, white-cell and platelet findings; lymphocyte and eosinophil counts; renal injury and support; liver injury and coagulation; sodium and potassium; calcium and magnesium; metabolic and cell-turnover markers; cardiovascular and thrombotic events; and acute care and other events. Laboratory outcomes were defined with reference to the CTCAE laboratory criteria; for each outcome, the definitions supplement states whether it equals a CTCAE grade, follows one with a stated difference, or has no counterpart [58]. Infection was also captured through antimicrobial treatment, labelled as such. A patient contributed to an outcome only if it could be ascertained for them, for example if laboratory follow-up was available, and was otherwise left out rather than counted as negative. Death was a competing event for every non-fatal outcome. Definitions, code lists and risk-set sizes are given in the definitions supplement.

Tabular comparators: features and tuning

For each disease–outcome cell and horizon, we trained gradient-boosted trees (XGBoost) [66] and ridge-penalised logistic regression on the same patients, labels and competing-event handling as the neural models. Their features came from the pipeline of our earlier DLBCL study [47]: using timeseriesflattener [67], prescriptions, laboratory results, procedures and referrals, pathology codes and diagnoses were summarised over 90-day, one-year and three-year windows before the index date (counts and distinct counts for codes; latest, minimum, maximum, variance and slope for laboratory values). Variables recorded for fewer than 10% of training patients were represented by per-source counts of such rare occurrences, and comorbidity by these counts rather than a weighted index such as Charlson. Features were constructed separately within each disease, so each disease's models used a feature set suited to its own patients. The XGBoost pooled across diseases used one matrix constructed in the same way across all supervised patients and, like the neural pooled models, was not given the disease. To align the register sources of the tabular and sequence models, quality-registry variables such as stage and performance status were removed, and age and sex at the index date were added. Numeric features were median-imputed with missingness indicators and standardised, and categorical features one-hot encoded, with all preprocessing fitted on training data. Hyperparameters were chosen for each cell by log loss on the tuning set from a prespecified grid: for XGBoost, shallow trees (depth 1–3), deeper trees (depth 4 and 6) and the setting of the earlier study (depth 8, learning rate 0.01, up to 3,000 trees), with the number of trees also chosen on the tuning set; for logistic regression, seven penalty strengths. Neither model was class-reweighted or recalibrated. **[Confirm against the delivered feature export: anchoring at first-line treatment, source list and number of features.]**

As a reference outside the information-matched comparison, XGBoost was also trained for each cell with the quality-registry predictors of the earlier study kept: variables recorded at diagnosis (such as performance status, stage, lactate dehydrogenase, extranodal involvement and cytogenetic risk, where the registry for the disease records them) and registry-recorded adverse events before the index date, left missing where not recorded. The sequence models never saw these variables, so this arm estimates what the curated registry adds, not a like-for-like comparison.

National encoder: data and representation

The national (PHAIR) encoder was pretrained on the registers of the whole Danish population, converted to MEDS with the same pipeline. The registers are the same, but the national extract uses older deliveries of some of them: pathology without the requesting department's specialty and region, and an earlier research extract of the laboratory database; it also includes psychiatric contacts. Its vocabulary and laboratory bins were fitted on the national data. To use it, our patients' records were taken from the national extract and tokenised with its vocabulary, so that it read them in the form it was trained on; outcomes, cohort membership and splits were unchanged. Because the national population contains the cohort, the national encoder had seen the cohort's pre-2022 records as well as everyone else's, and continued pretraining on the cohort adds no patients: it is domain adaptation. The national and cohort encoders therefore differ in population, register deliveries and training budget, and the comparisons concern the encoders as delivered.

With the same threshold of 100 events, the national conversion keeps specific codes, such as particular molecular findings or rare diagnoses, that the cohort's conversion merged into their parents, and bins laboratory results more finely. Finer resolution is one route by which the national encoder, or any larger dataset, could improve prediction, separate from what it learns from other patients. In the national conversion, rare codes were counted over the whole period, including events after 2021. No outcomes were involved, and everything the models learned

from was restricted to events before 2022, but which codes kept their own token depends on these later frequencies; we note it as a limitation.

The national encoder was pretrained for 10 epochs with a 1 January 2022 cutoff. Continued pretraining on the pretraining population (adaptation) and pretraining on it from scratch ran for the same 10 epochs; on tuning data, encoders pretrained for 10 and 20 epochs gave fine-tuned performance within the difference between two fits of the same model. As references, a randomly initialised encoder of the same architecture was trained with the same supervised protocol, and the pretrained cohort encoder was kept frozen while only the prediction head was trained. Patients, tokens and optimisation steps are reported separately, because equal epochs do not guarantee equal exposure. **[National training population, observation years, events and tokens, encoder version and PHAIR reference.]**

Data efficiency of national pretraining

To test whether population-scale pretraining is most useful when local labels are scarce, the joint model was retrained from the cohort encoder and from the national encoder adapted to the cohort on 10%, 25% and 50% of the training patients, with otherwise identical training. The subsets are nested (10% within 25% within 50%), identical for every outcome, and keep each disease's share. Tuning and test patients were never subsampled, model selection used the full tuning set, and national and cohort encoders were compared at each share with the paired bootstrap of the main analysis.

Comparison with clinical prognostic indices

The quality registries record, for each patient, the prognostic index scored at diagnosis (for example the IPI, FLIPI, MIPI, CLL-IPI or ISS) [37, 39, 86–88] and its risk category. Every index recorded for at least 50 patients of a subtype was compared with OPERA's prediction for the endpoint the index was developed for, in the patients who had that index, and every index used in several subtypes also across all of them, since the IPI, for instance, is applied to many rare lymphomas without a score of their own. For DLBCL we also computed the NCCN-IPI from its components in the lymphoma registry, following the original definition [38]: age (41–60, 61–75 and over 75 years: 1, 2 and 3 points), LDH relative to the upper limit of normal (above 1 and above 3 times: 1 and 2 points), Ann Arbor stage III–IV, involvement of bone marrow, central nervous system, liver, gastrointestinal tract or lung, and performance status $\geq$2 (1 point each), grouped as low (0–1), low-intermediate (2–3), high-intermediate (4–5) and high ($\geq$6). Patients missing a component were not scored.

Model risk groups were fixed on tuning patients with known two-year status and applied unchanged to the test set, anchored to the index at both ends. The low cutoff was the highest predicted risk at which the group below it, of at least 20 patients, had an observed risk no higher than the index's low-risk category. The high cutoff matched the specificity of the index's high-risk category, so that both flagged the same share of patients without the event; this is also the operating point marked in the precision–recall comparison. Patients in between were divided in the proportions of the index's intermediate categories. A model that ranks patients better can then place more patients at low risk at the same observed risk, and capture more events at the same specificity. For each index we report group sizes, observed risks, and the sensitivity and specificity of classifying a patient in a group or higher, and, for the model, the observed risk in its highest-risk 5% and 10%, which the index cannot resolve within its top category. As a sensitivity analysis, model groups were given exactly the index's group sizes.

Overall survival was the endpoint for the IPI, NCCN-IPI, CLL-IPI, ISS, R-ISS, FLIPI and MIPI. The IPS, built for freedom from progression [89], was compared on treatment failure, a proxy

that is reported separately as endpoint-approximate. The horizon was two years where at least a third of the index's test patients could have been followed for two years (judged from index dates and the data cutoff alone, without reference to outcomes), and one year otherwise; one year is reported alongside. Censoring before the horizon is handled by inverse probability of censoring weighting, so the criterion concerns precision, not bias. The original indices were developed for longer horizons, usually five years, so these are shorter-horizon evaluations, not reproductions of the original analyses. Discrimination used the index category (or NCCN-IPI points) as an ordinal score.

Decision curves ask whether acting on a prediction does more good than harm [72]. For each threshold probability p, net benefit is the share of patients correctly flagged minus the share wrongly flagged, weighted by the odds $p/(1-p)$ at which a clinician would trade one for the other. The action was closer monitoring and supportive care for patients above a predicted two-year risk of death, at thresholds from 5% to 40%, fixed before the test set was opened. The action changes how closely patients are followed, not which treatment they receive; a prediction that changed treatment would have to meet the evidence standards of that treatment [77]. Many test patients were followed for less than two years, so true and false positives were weighted by the inverse probability of remaining uncensored, as for the primary metrics; dropping them instead would overstate the risk and the benefit. An index has categories rather than probabilities, so each category entered as its observed two-year risk among tuning patients, giving a curve fixed before testing. OPERA, the index, per-task XGBoost, acting for everyone and acting for no one were compared on the same patients, with 95% intervals for the differences in net benefit from 1,000 patient resamples.

Routing formulation and selection

To test whether a joint model benefits from conditioning its predictions on disease beyond ordinary shared learning, we compared it with a routed model trained on the same panel. In the routed model, the patient representation h_i is adjusted for each target outcome k by a weighted combination of small residual adapters, one per source s:

$$\begin{aligned} h_{ik} &= h_i + \lambda \sum_s \pi_{iks}\, c_s\, A_s(h_i), \\ A_s(h) &= U_s\, \mathrm{GELU}(V_s h), \qquad c_s = \frac{e_s}{e_s + \tau}. \end{aligned} \tag{1}$$

Here A_s is a low-rank adapter, $\lambda = 0.7$ a fixed residual scale and c_s shrinks sources with few training events e_s ($\tau = 10$ events). The weights π_{iks} come from a softmax over the sources and one further slot that applies no adapter (its logit starting at 0), so the router can leave the shared representation unchanged; they depend on the patient, the patient's disease and the target outcome, and the patient's own disease always receives at least a floor weight of 0.05. The adapted representation enters the same linear head as in the joint model. Each adapter is updated only by the supervision of its own source; when another disease borrows it, its parameters are used but not updated. The sources can be diseases, outcomes or both; the locked routed model uses diseases only. The approach relates to mixtures of experts and parameter-efficient adapters [62–64, 90, 91].

The router was configured on the tuning set in three steps, each ordered from the simplest option: which sources (disease only, outcome only, both), the adapter rank (8, 16, 32), and the routing query (additive with width 128, interacting with width 128, interacting with width 512). Each step chose the option with the highest panel AUC: the AUC difference averaged over estimable cells (at least five cases and five controls), each disease weighing equally, each outcome equally within a disease and each of the six horizons equally. Options whose panel AUC lay within the noise floor of the best were treated as tied and the simplest of them was chosen; an option whose

panel IPA was worse than the simplest option's by more than the IPA noise floor was not eligible. The noise floor is the difference between two fits of the same model: the router trained with every source masked, which computes exactly the joint model's prediction from a different initialisation, against the joint model (**[about 0.004]** tuning AUC); a difference within 0.005 also confirms that the routing machinery changes nothing by itself. The 0.01 margin applies only to claims on the test set, not to these choices. The source step chose disease adapters only; **[the rank and query steps chose X and Y]**. OPERA was then chosen among the scopes, routing and encoders by the same rule, with the joint model simpler than the routed one and the cohort encoder simpler than the national.

Three controls separate borrowing from simpler explanations. The joint model told the patient's disease adds a learned, zero-initialised shift per disease to the representation, so it can express disease-specific baseline risks but borrows nothing. The own-disease router and the own-outcome router are routed models whose router may use only the patient's own disease adapter, or only the target outcome's own adapter, so they add the adapters' capacity without borrowing. All controls are trained with the same protocol as the routed model. A fourth control keeps the selected source banks but lets each patient use only their own disease's and own outcome's adapter, so the difference from the routed model is the borrowing of the routed model itself. A sensitivity fit, added before stage two, starts the slot that applies no adapter with a larger share of the routing weight (logit 3.0, the code's default, against 0; about 15–18% rather than 1% with 94 sources) and is otherwise identical, to test whether the starting point changes how selectively the router borrows. A rule fixed before either fit finished decided between them (Supplementary Methods, prespecification): the routed model stays primary unless the sensitivity fit is better on the tuning panel by more than the noise floor in AUC, passes the IPA guard and stays better in a second seed of each; the other is reported as the sensitivity analysis. Route entropy, the effective number of sources and the share of weight on a patient's own adapters are reported for both (**[values]**). Route weights describe how the model allocates corrections and are not interpreted as relationships between diseases; what each source contributes is measured by removing it (Supplementary Methods, borrowing map).

Router variants and the primary routed model

On the tuning set, the routed model fixed by the steps above routed poorly. In most patients most of the routing weight sat on one or two sources whose corrections were small (the overall-survival adapter carried 40% of the routed contribution), while the disease adapters, which enter every patient's prediction at the 0.05 floor of their own disease, had grown corrections about 20 times larger. Any weight borrowed from another disease's adapter therefore injected a large correction learned for that disease, and the router learned to park its weight on a source that changed little. The disease adapters pointed in unrelated directions (median pairwise cosine about 0), and those of related B-cell lymphomas resembled each other, so the adapters had learned disease-specific corrections; it was the routing that had collapsed.

Before any variant was trained, we fixed the variants, their controls and the rule for choosing among them (analysis lock, addenda of 26 October 2026). Two changes address the collapse: a global adapter applied to every prediction outside the routed mix (A), so that general corrections no longer have to pass through a source, and two epochs in which the router is held fixed while the adapters train (B), so that early routing does not lock in before the adapters have learned anything. A third bounds the routing scores (T, three times the cosine of query and key) and was to be used only if A and B did not route healthily; a fourth applies a patient's own disease and outcome adapters at full weight outside the mix, which then holds only borrowed sources (O). Each was trained with controls on the same protocol: the same model with only the patient's own adapters (own-only), the model with the global adapter and no sources (global-only), and a second seed of the candidate. Routing was judged healthy when, measured by contribution

rather than weight (route weight times the source's support shrinkage, or its correction size for O), the most frequent top source led in at most half of the rows, at least three different borrowed sources led across diseases, and the median effective number of sources exceeded 1.5.

The routed model with A and B (R_{AB}) routed healthily in both seeds; A alone, and O, did not. Borrowing within R_{AB}, the full model against its own-only control, changed tuning AUC by $+0.014$ (95% interval 0.001 to 0.027), above the seed-to-seed difference of 0.002. Against the first router, R_{AB} differed by -0.004 (-0.011 to 0.002) and O by -0.009 (-0.021 to 0.002). The prespecified rule kept the first router, since R_{AB} was below it by more than the seed-to-seed difference. We amended the rule before any test data were opened and made R_{AB} the primary routed model: the first router's routing is known to have collapsed, so nothing it appears to borrow could be interpreted; R_{AB} routes as intended in both seeds; choosing the model with the lower tuning score cannot bias the test results upwards; and the cost on tuning was small, with an interval including zero. The first router is reported throughout as a sensitivity analysis, and the national routed models use the same readout as R_{AB}. In R_{AB}, Eq. (1) gains a global term, $h_{ik} = h_i + \lambda \left[G(h_i) + \sum_s \pi_{iks} c_s A_s(h_i) \right]$, with G an adapter of the same form trained on every prediction, and the router is held fixed for the first two epochs.

Hierarchy for effect heterogeneity

Claims rest on the pooled paired-bootstrap differences and their labels (Table 2); the hierarchical model is descriptive. Many cells are small, so their raw estimates are noisy and rankings of individual cells are unreliable. We therefore also fit a simple Bayesian hierarchical model for partial pooling: each cell's effect is the panel mean plus a disease effect, an outcome effect and a cell-specific deviation, so that a noisy cell is shrunk towards what similar cells show, by an amount the data determine. It answers two questions: how large is the typical effect, and how much does it vary between diseases and between outcomes? Because the same patients contribute to every outcome within a disease, the cells' sampling covariance, taken from the shared bootstrap draws, enters the likelihood.

Whether effects depend on the amount of training data for the target is a separate question, and we did not assume a linear relationship: the cell effect varies smoothly with log training events through a cubic spline with few knots and a shrinkage prior towards a flat relationship. As a check that does not depend on smoothing, cells were also grouped into bins of training events fixed before test-set evaluation. Pooling contrasts were fitted as their independent parts (disease pooling, outcome pooling and their interaction), so that every reported combination is an exact sum and all are mutually consistent. Priors are weakly informative on each metric's scale and checked by prior predictive simulation; convergence and posterior predictive checks are reported, and posterior probabilities of benefit, harm and practical equivalence use a prespecified tolerance and are descriptive, not claim labels. Raw paired estimates are always shown alongside. **[Insert locked priors, tolerances, support-bin edges and sampling diagnostics.]**

Supplementary Methods: evaluation and reporting

Prespecification

Every modelling choice was made on the tuning set (patients treated in 2022), and the test set (patients treated from 2023) was evaluated once, after the analysis plan had been locked **[lock date; the plan as a supplementary file]**. Fixed before testing were: the outcomes, horizons and estimand; the primary and secondary metrics; the paired resampling scheme and the hierarchical models with their priors; the practical margin (0.01 AUC and 0.01 IPA, one point on the scale of 100 used in the figures) and the labels better, worse, equivalent and inconclusive, with the paired bootstrap test; the rules by which the router configuration was chosen on tuning data (Supplementary Methods, model details); the selection of OPERA; the risk-group cut-offs for the comparison with clinical indices; the design of the borrowing analysis and its candidate pairs, reported descriptively (below); and the sensitivity analyses (known-status estimand, excluding the 209 patients whose disease was assigned after day 14, size-matched cut-offs, equal weight per clinical domain, and the stratum without a dedicated index without nodal marginal-zone lymphoma and with angioimmunoblastic and Burkitt lymphoma added). The action and threshold range of the decision curves were fixed before testing: closer monitoring and supportive care for patients above the threshold, from 5% to 40%, with the model's risk-group cutoffs as the two prespecified operating points.

The router rule was revised once, on 21 October 2026, after rough source-screen estimates had been seen and before the test set was opened; all four source banks are compared on the test set, and the rank, query and OPERA results had not been seen. The router screens were then rerun from scratch after an audit showed that the optimiser's numerical constant (Adam's epsilon, 10^{-6}) damped the updates of the routing parameters, whose gradients were mostly smaller; every routing parameter was retrained with 10^{-10} before the router was locked and before the test set was opened, and the earlier screens are not reported as results. The rule for the sensitivity fit's starting point (Supplementary Methods, model details) was added on 25 October 2026, from training-time diagnostics of unfinished fits, before their tuning-panel results or any test result had been seen.

Weighting of summaries

Summaries over cells weigh them in one of three ways, each fixed before testing. Headline differences and Table 2 weigh every disease–outcome–horizon cell equally. Panel summaries and the hierarchical models weigh every disease equally and, within a disease, every loss endpoint (the grade variants of one finding together; 52 endpoints) equally; a sensitivity analysis weighs every clinical domain equally. Router choices and the borrowing analysis weigh every disease equally, every outcome equally within a disease and every horizon equally within an outcome. Only estimable cells enter a summary: at least five cases and five controls, and at least 80% of bootstrap replicates with a finite estimate; other cells are shown but not pooled.

Borrowing map

The routed readout adds to the shared patient representation one routed correction per source disease (routing weight times that disease's adapter output), and the risk head is linear, so each source's contribution to every hazard can be switched off exactly while all other weights stay as trained; no weight is renormalised. A switched-off source's term is replaced by its mean over tuning patients for the same target disease and outcome (mean ablation); diseases without tuning patients use the mean over all tuning patients. The router spreads its weight over many sources, so setting terms to zero would move the representation far from anything seen in training; the mean

keeps the average shift and removes only what the source contributes for the individual patient. Zero ablation is reported as a sensitivity analysis. One evaluation of the locked model scored, on the same patients, the full model, the model without borrowing (the patient's own disease only), each source alone (own disease plus that source) and each source left out (all sources but that one). Differences in AUC used one paired patient bootstrap shared by all variants: the AUC lost without borrowing (full minus none), the source alone (alone minus none; primary) and the source left out (full minus left out). The source alone is what a disease adds with no other source, the source left out what it adds with all the others present. The loss without borrowing measures what the fitted model uses; it can exceed the gain over a model trained without borrowing (the own-disease router), because the encoder was trained alongside the borrowed corrections. Per target disease, source and clinical domain, differences were averaged over estimable cells (at least five events and five non-events), each outcome weighing equally and each horizon equally within an outcome. The map shows the source alone with adaptive shrinkage [92]: within each clinical domain all pairs share one prior, a mixture of normal distributions centred on no borrowing with weights fitted to the data, so that the many near-null pairs are pulled towards zero while large, precisely estimated ones are kept. Disease groups are not used in the pooling; they order the map for reading only. Dots mark pairs whose pooled 95% interval lies above zero. The AUC lost without borrowing (full against no borrowing in the fitted router) was estimated per target disease, per outcome domain and over all diseases, each disease weighing equally (Fig. 8B, D), with the same bootstrap, and set beside routing (routed against joint model) and borrowing as trained (routed model against the router trained with its own disease adapter only), and against disease size (Fig. S19A–B). Candidate pairs (a source disease, a target disease and a clinical domain) were listed from clinical reasoning and the literature and kept only if a single-pair estimate could be informative: at least 200 training patients in the source, so that its correction learned something to remove, and at least 50 test patients in the target, so that the estimate was not inconclusive by construction. Smaller diseases remain in the descriptive map. A confirmatory test of these pairs was planned, with the expected pairs and negative controls to be rated by study haematologists without our rationale. Before any rating or test-set evaluation, the routed model was found on the tuning set to borrow broadly, with similar mixtures of sources across diseases (correlation of borrowing profiles 0.83 to 0.89), so a test of a few selected pairs would not describe how it borrows. The pairs are therefore reported descriptively, each with its unshrunk source-alone gain and 95% interval (Table S3, with the excluded candidates and the reason for each), and whether borrowing favours related diseases is described over all diseases by lineage concentration (main text, Methods). IPA differences were computed in the same way as a secondary metric.

Table S3. Prespecified borrowing pairs and negative controls. Fixed before the test set was opened and reviewed by the study haematologists. Source: the disease switched on alone; target: the disease predicted; outcomes: the clinical domain tested. Rating: the study haematologists' expected strength, given blind to the rationale and to whether the pair was proposed as a hypothesis or a control. Pairs were labelled against a practical margin of 0.25 AUC points.

	Source	Target	Outcomes	Rating	Rationale (evidence)
Expected to share signal					
H1	Chronic lymphocytic leukaemia (CLL)	Small lymphocytic lymphoma (SLL)	all outcomes	–	One disease in two presentations (blood vs lymph nodes), managed by the same criteria and treatments. (WHO 5th edition classification (Alaggio et al., Leukemia 2022))
H2	Extranodal marginal-zone lymphoma (EMZL)	Splenic marginal-zone lymphoma (SMZL)	disease course and later malignancy	–	The marginal-zone lymphomas share prognostic factors: the MZL-IPI (LDH, haemoglobin, lymphocyte and platelet counts, subtype) was built for and validated across all subtypes, and advanced disease is treated with the same rituximab-based regimens. First-line treatment often differs (radiotherapy for localised EMZL; rituximab alone or splenectomy for SMZL). (Arcaini et al., EClinicalMedicine 2024, doi:10.1016/j.eclinm.2024.102592)

	Source	Target	Outcomes	Rating	Rationale (evidence)
H3	Multiple myeloma (MM)	Chronic lymphocytic leukaemia (CLL)	infection and antimicrobial use	–	Both cause secondary humoral immune deficiency (hypogammaglobulinaemia in up to 85% of CLL during the disease course, and intrinsic or treatment-related in myeloma), with the same immunoglobulin-replacement indications. (Frontiers in Immunology 2014, doi:10.3389/fimmu.2014.00626; Frontiers in Immunology 2016, doi:10.3389/fimmu.2016.00317)
H4	Chronic lymphocytic leukaemia (CLL)	Multiple myeloma (MM)	infection and antimicrobial use	–	The same, in the other direction. (Frontiers in Immunology 2014, doi:10.3389/fimmu.2014.00626; Frontiers in Immunology 2016, doi:10.3389/fimmu.2016.00317)
H5	Burkitt lymphoma (BL)	Lymphoblastic lymphoma (LBL)	neutrophil, white-cell and platelet findings	–	Both receive short, highly intensive regimens (CODOX-M/IVAC, hyper-CVAD, ALL-type protocols) with marked myelosuppression despite growth-factor support. (REFERENCE NEEDED: toxicity of intensive regimens in adult Burkitt lymphoma)
H6	Burkitt lymphoma (BL)	Lymphoblastic lymphoma (LBL)	metabolic and cell-turnover markers	–	Rapid tumour turnover and intensive induction put both at risk of tumour lysis (potassium, phosphate, urate, creatinine). (REFERENCE NEEDED: tumour lysis in Burkitt and lymphoblastic lymphoma)
H7	Peripheral T-cell lymphoma, NOS (PTCL)	T-cell lymphoma, unspecified (TCL)	disease course and later malignancy	–	Unspecified T-cell lymphoma is managed like peripheral T-cell lymphoma, not otherwise specified (CHOP-based therapy), with a similarly poor course. (WHO 5th edition classification (Alaggio et al., Leukemia 2022))
H8	Peripheral T-cell lymphoma, NOS (PTCL)	Angioimmunoblastic T-cell lymphoma (AITL)	disease course and later malignancy	–	Angioimmunoblastic T-cell lymphoma (nodal T-follicular-helper lymphoma in WHO 5) is treated with the same CHOP-based first-line regimens as PTCL, not otherwise specified, with a similarly poor course; the two are often hard to tell apart histologically. (WHO 5th edition classification (Alaggio et al., Leukemia 2022))
H9	Chronic lymphocytic leukaemia (CLL)	Lymphoplasmacytic lymphoma (LPL)	cardiovascular and thrombotic events	–	BTK inhibitors, used in CLL and in Waldenstrom macroglobulinaemia, cause atrial fibrillation and bleeding. But first-line Waldenstrom treatment in Denmark has mostly been chemoimmunotherapy (e.g. bendamustine-rituximab), so a null result here would be unsurprising. (Brown et al., Haematologica 2017 (pooled AF in ibrutinib trials), PMC5622864)
Negative controls					
C1	Multiple myeloma (MM)	Hodgkin lymphoma (HL)	all outcomes	–	No shared biology, treatment or complication profile (young patients on ABVD-type therapy vs older patients on myeloma regimens); MM is the second-largest source, so this also tests whether a large source helps regardless of relevance. (–)
C2	Hairy-cell leukaemia (HCL)	Diffuse large B-cell lymphoma (DLBCL)	disease course and later malignancy	–	No shared biology or course (indolent, cladribine-treated vs aggressive, R-CHOP-treated). (–)
C3	Anaplastic large-cell lymphoma (ALCL)	Chronic lymphocytic leukaemia (CLL)	all outcomes	–	No shared biology, treatment or complication profile (a T-cell lymphoma treated with CHOP-type therapy vs an indolent B-cell leukaemia). (–)
Candidates excluded before rating (estimate not informative)					
X1	Diffuse large B-cell lymphoma (DLBCL)	Transformed follicular lymphoma (tFL)	disease course and later malignancy	–	target has 44 test patients
X2	Diffuse large B-cell lymphoma (DLBCL)	Richter transformation (RT)	disease course and later malignancy	–	target has 25 test patients
X3	Chronic lymphocytic leukaemia (CLL)	Richter transformation (RT)	infection and antimicrobial use	–	target has 25 test patients
X4	Multiple myeloma (MM)	Plasma-cell leukaemia (PCL)	disease course and later malignancy	–	target has 38 test patients
X5	Multiple myeloma (MM)	Amyloidosis	renal injury and support	–	target has 17 test patients
X6	Anaplastic large-cell lymphoma (ALCL)	Solitary plasmacytoma	all outcomes	–	target has 32 test patients
X7	Amyloidosis	Hodgkin lymphoma (HL)	all outcomes	–	source has 22 training patients

	Source	Target	Outcomes	Rating	Rationale (evidence)
X8	Transformed follicular lymphoma (tFL)	Hairy-cell leukaemia (HCL)	disease course and later malignancy	–	HCL has no second-line outcomes and almost no deaths, so no estimable cells

Sample size and precision

The cohort was not sized for this study. It comprises every eligible patient in Denmark over the study period, so the relevant question is what precision this fixed sample allows, not how many patients would be needed. Formal criteria for the sample size of a new prediction model [2] assume one outcome and a fixed set of predictors, and do not carry over to a model whose learned parameters are shared across 69 outcomes. We therefore report, for every disease–outcome cell, the number of training events and of test events and non-events (Fig. S5), evaluate a cell only when the test period has at least five events and five non-events, and judge each estimate by the width of its interval. For validation, about 100 events and 100 non-events are commonly recommended for precise estimates of discrimination and calibration [93]. **[N]** cells reached this at one year, and cells below this threshold are interpreted with particular caution. Pooling evidence across cells can support a panel-level conclusion but does not validate a sparsely observed cell for individual clinical use.

Risk sets and paired comparisons

The unit of evaluation was one disease–outcome–horizon combination. Its risk set was determined by eligibility and ascertainment of the outcome, not by the models compared. For each split we report eligible patients, outcome events, competing deaths and patients whose status was unknown at the horizon. Within a cell, every model was evaluated on the same patients and labels. Contrasts can cover different sets of evaluable cells, so their cell and patient counts are reported. Cells in which a model could not be fitted or a metric could not be estimated are shown with the reason (Fig. S4) and not treated as zero effects.

A pooled model could predict cells in which no specialist could be fitted. These cells have absolute performance estimates but no paired specialist comparison, so they are reported as added coverage and kept out of the paired averages. Outcome definitions and the reconstruction of treatment lines are in the definitions supplement.

Clinical prognostic indices

Which index each subtype has. Table S4 lists, for every subtype, the prognostic index derived for it, how far that index is established, and what is used in practice. A subtype was classed as having no dedicated prognostic index if no multivariable index had been derived for that entity and externally validated in an independent cohort; indices developed for other entities (such as the IPI, built for aggressive lymphoma in 1993 and applied to Richter transformation, transformed follicular lymphoma and anaplastic large-cell lymphoma) or for broader families (the MZL-IPI for all marginal-zone lymphomas) did not count. This literature criterion was preferred to guideline adoption, which differs between guidelines. Two sensitivity analyses vary it: counting family-level indices removes nodal marginal-zone lymphoma from the stratum, and requiring routine, guideline-recommended use adds angioimmunoblastic T-cell lymphoma, whose dedicated scores have limited external validation, and Burkitt lymphoma, whose validated index does not yet assign treatment.

The IPI is the main comparator because it is recorded for most patients with DLBCL and sets treatment in the Danish guideline [42]; the NCCN-IPI, developed for rituximab-treated DLBCL

Table S4. Prognostic indices by lymphoid subtype. Dedicated index: a multivariable index derived for that entity. Status: *routine*, externally validated and recommended in international guidelines; *validated*, externally validated but not the basis of routine decisions; *proposed*, derived without independent validation; –, none. In practice: what is applied when there is no dedicated index (borrowed indices in italics). †Prespecified stratum without a dedicated index. **[Check the guideline column against the current Danish (DLG, DMSG, DCLLSG) guidelines.]**

Subtype	Dedicated index	Status	In practice
DLBCL and related aggressive B-cell lymphomas			
Diffuse large B-cell lymphoma	IPI, NCCN-IPI, CNS-IPI [37, 38, 94]	Routine	IPI (Danish guideline)
Richter transformation†	Richter score [95]	Proposed	*IPI*; clonal relatedness, TP53
Transformed follicular lymphoma†	–	–	*IPI*; managed as DLBCL
Burkitt and lymphoblastic lymphoma			
Burkitt lymphoma	BL-IPI [96]	Validated	Low/high-risk criteria
Lymphoblastic lymphoma†	–	–	ALL-type risk and response
Mantle-cell lymphoma			
Mantle-cell lymphoma	MIPI [87]	Routine	MIPI
Indolent B-cell non-Hodgkin lymphoma			
Follicular lymphoma	FLIPI [86]	Routine	FLIPI
Lymphoplasmacytic lymphoma	IPSSWM [97]	Routine	IPSSWM
Extranodal marginal-zone lymphoma	MALT-IPI [98]	Routine	MALT-IPI
Splenic marginal-zone lymphoma	HPLL/ABC score [99]	Validated	HPLL/ABC score
Nodal marginal-zone lymphoma†	– (MZL-IPI covers all MZL [100])	–	*FLIPI* or *IPI*
B-cell lymphoma, unspecified†	– (category)	–	*IPI*
CLL and small lymphocytic lymphoma			
Chronic lymphocytic leukaemia	CLL-IPI [39]; Rai/Binet stage	Routine	CLL-IPI, stage
Small lymphocytic lymphoma	CLL-IPI (same disease)	Routine	CLL-IPI, stage
Hairy-cell leukaemia, Hodgkin lymphoma			
Hairy-cell leukaemia†	–	–	Treatment criteria [101]
Hodgkin lymphoma	IPS [89]; early-stage risk groups	Routine	IPS, risk groups
T-cell non-Hodgkin lymphoma			
Peripheral T-cell lymphoma, NOS	PIT [102]	Validated	*IPI* most often; PIT
Angioimmunoblastic T-cell lymphoma	AITL score [103]	Proposed	*IPI* or PIT
Anaplastic large-cell lymphoma†	– [104]	–	*IPI*; ALK status
T-cell lymphoma, unspecified†	– (category)	–	*IPI* or *PIT*
Myeloma and other plasma-cell neoplasms; amyloidosis			
Multiple myeloma	ISS, R-ISS [40, 88]	Routine	ISS, R-ISS
Plasma-cell leukaemia†	PCL score [105]	Proposed	*ISS, R-ISS*; treated as high-risk myeloma
Solitary plasmacytoma†	Risk factors only [106]	–	Light-chain ratio, marrow involvement
AL amyloidosis	Mayo stage [107]	Routine	Mayo stage

[38], is reported alongside in its complete cases. How the indices were compared, and how model risk groups were anchored to them, is described under "Comparison with clinical prognostic indices". All methods in an index comparison used the same patients, those with the index recorded at diagnosis, and how often it was recorded is reported by disease and calendar period. Because the models read the wider history, a gain over an index is a gain of the whole pipeline, not of the model alone. Index categories are not probabilities, so they were used for ranking and stratification only, not for Brier scores or calibration. **[Index, version, population and availability inventory from the secure analysis.]**

Risk groups, age and operating points

Continuous model risks were converted to groups fixed at baseline with cutoffs selected on tuning data, while index groups kept their original definitions. Kaplan–Meier estimates describe endpoints that include death, and Aalen–Johansen estimates non-fatal endpoints with death as a competing event. Group sizes, events and 95% bootstrap intervals accompany every estimate. Patients without a recorded index were not included in index comparisons.

In the age figures, patient counts, mean predicted risk and observed risk are shown as separate rows on the same age bands and groups. The predicted row is the sequence model's mean probability, also within index categories. The index itself supplies no probability. Estimates are suppressed for cells with fewer than ten patients, while counts remain visible.

Operating points used thresholds fixed on tuning data, so the specificity matched on tuning data is not exactly matched in the test period. We report the achieved sensitivity, specificity, positive and negative predictive values and patient counts. Confusion counts use patients with known status at the horizon, and the number excluded by censoring is reported. Censoring-weighted quantities are labelled as such.

For the panel as a whole, every model was also evaluated at six decision points, each fixed per model, cell and horizon on that model's tuning predictions. The first flagged as many tuning patients as had the event, so that every model raised the same number of alerts. The second and third maximised Youden's index and the MCC. The remaining three were the lowest thresholds with a tuning specificity of at least 80%, 90% and 95%. The test period then shows how far the achieved specificity drifts from its tuning target. For each decision point we report the threshold, the confusion matrix, sensitivity, specificity, positive and negative predictive values, MCC and the share of patients flagged, with 95% intervals from the shared patient bootstrap and paired differences between OPERA and XGBoost, the joint model and the specialist, and the routed and joint models. Confusion counts below five are suppressed in exported tables, as required by the secure environment. Cells with fewer than five tuning events or non-events have no decision point.

Censoring, calibration and robustness

The primary metrics weight patients by the inverse probability of remaining uncensored, estimated separately by disease, outcome and split; an analysis on patients with known status, without weights, is a prespecified sensitivity analysis. We report the censoring survival at each horizon and the distribution of weights, and flag long horizons where little follow-up makes estimates unstable. The weights are held fixed across bootstrap resamples, so intervals do not include their uncertainty, and they assume censoring independent of prognosis, which may not hold where monitoring follows the patient's condition.

Calibration is reported as calibration-in-the-large, the calibration slope where estimable, and reliability curves for outcomes chosen by a fixed rule. No model was recalibrated on test data. Sensitivity analyses by horizon, calendar year, amount of recorded history and timing of the

subtype diagnosis are reported on their own risk sets. Raw paired effects are reported with the hierarchical summaries, and the support curve, unsmoothed and binned analyses are treated as views of the same data rather than independent replications.

Regional performance groups each patient by the region with the most pre-treatment contacts. The models were trained nationally, so this describes variation within the development health system rather than validation in regions excluded from training.

Patient representations

The representation figures show the same test patients at five stages: pretraining on the cohort, the joint model from that encoder, national pretraining, national pretraining adapted to the cohort, and the nationally initialised joint model. For each stage, the projection was fitted on tuning patients and applied unchanged to test patients, so coordinates are comparable within a stage but not between stages. Clusters and changes in separation are descriptive. They do not show differences in prediction, biological mechanisms or the effect of transfer. Only kernel-smoothed summaries over at least [N] patients are shown, following the disclosure rules of the secure environment.

Numerical tables

The printed supplementary tables summarise the comparisons that the figures show: the panel-level comparisons behind Table 2, paired differences by comparison and clinical domain, calibration by model and domain, the evaluated cells, events and non-events by domain and horizon, threshold metrics at the six decision points, and, where available, the comparisons with each index on its original endpoint and the computational resources of each model. Each model was trained once, so intervals describe sampling of test patients only. The complete cell-level results, including every confusion matrix, accompany the paper as machine-readable files, with counts below five suppressed.

[Reviewed aggregate summary tables have not yet been exported. Complete cell-level numerical tables will accompany the figures.]

What the model relies on: attributions and input restriction

This secondary analysis asks what OPERA reads in a patient's history, so that clinicians can judge whether its predictions rest on sensible information. It is computed once on the model selected on tuning data, before any attribution is inspected, and every figure and table is a summary of that one computation. Outcomes, sampling, the reference, the expectations and the checks were specified beforehand.

Attribution. Let $f_k(E; C)$ be the prediction for outcome k at one year, where E holds the content embeddings of the recorded events (which diagnosis, drug or laboratory result and value bin each event was) and C the context held fixed (each event's time, the age at it, its position and the patient's subtype). The reference E^0 replaces every event's content by the mean content embedding of training events, an average event, while keeping the timeline. The encoder was pretrained to predict the next event, so the mask token was never trained and is not a suitable reference. Integrated gradients [73] move the record step by step from this average record to the patient's own, and add up how much each event's content changes the prediction along the way; the integral below is that running sum, and α runs from the average record ($\alpha = 0$) to the

patient's record ($\alpha = 1$). Event j thus receives the credit

$$a_{jk} = \sum_d (E_{jd} - E^0_{jd}) \int_0^1 \frac{\partial f_k(E^0 + \alpha(E - E^0); C)}{\partial E_{jd}} \, d\alpha, \tag{2}$$

and the credits add up to $f_k(E; C) - f_k(E^0; C)$. For every outcome f_k is the log-odds of the predicted cumulative incidence, so that contributions are comparable across rare and common outcomes and across low- and high-risk patients. For death, serious infection and severe neutropenia the absolute risk is also attributed, for the removal checks and the explanations of single patients. All targets share one forward pass along the integration path, and the attribution passes through the router, so it explains the prediction the model actually makes. For death the attributed quantity is the risk of death, so a positive contribution raises predicted mortality.

The path integral is evaluated by Gauss–Legendre quadrature with 8 nodes and repeated with 32 when the absolute gap between the summed credits and $f_k(E; C) - f_k(E^0; C)$, divided by $\max(|f_k(E; C) - f_k(E^0; C)|, 0.001)$, exceeds 0.05. Patient–outcome pairs that still fail are reported by outcome and excluded from the summaries. Every held-out test patient was explained, and patient-level outputs stay in the secure environment.

Episodes and concepts. The unit of interpretation is the episode of care: events within one day of each other, such as one admission or one diagnostic work-up, because the diagnoses, results and drugs of one admission are recorded at different times. Concepts group codes by structure only: ICD-10 to three characters, ATC to the pharmacological subgroup, procedures to four characters, pathology SNOMED codes to three, and laboratory results to the test and the third of its value range (low, middle or high) of the latest result. Codes were named from the official classifications (ICD-10 in its Danish SKS form, the WHO ATC index, the NPU terminology for laboratory tests, SKS for procedures and ICD-O morphology for pathology), at the finest level of the code's hierarchy that had a name, and grouped into the chapters of each classification and into clinical groups whose diagnosis codes are those of the outcome definitions (main-text Outcomes section), so a finding and the outcome it resembles are defined by the same codes. **[The names of the concepts shown in the figures were reviewed by [reviewer].]**

What makes patients differ. Because the credits add up, each patient's prediction is $f_k(E^0; C) + \sum_g c_g + r$, with c_g the summed credit of source (or time window) g and r the numerical residual. The between-patient variance of the predictions therefore divides exactly into $\mathrm{cov}(f_k(E^0; C), f_k) + \sum_g \mathrm{cov}(c_g, f_k) + \mathrm{cov}(r, f_k)$. Each share, a covariance divided by the variance, is signed and the shares sum to one. The share of the reference is the part of the differences between patients carried by the timing of care, age and subtype rather than by what the events were. Intervals come from 1,000 weighted patient-bootstrap resamples.

Outcome-specific use of the history. For each patient and pair of outcomes, the episode contributions were ranked within each outcome and correlated (Spearman). Patient correlations were averaged on the Fisher-z scale with the sampling weights, for patients with at least five episodes and pairs with at least 20 such patients. A model that had learned only a general measure of frailty would give high agreement between all outcomes.

Explanations for one patient. For death, serious infection and severe neutropenia, explained in absolute risk, each patient's episodes of care are summed and labelled with their timing and main concepts. Two records are extracted. For every patient and outcome, the three episodes that most raise and the three that most lower the predicted risk, each with its contribution in percentage points, is the format a clinical tool could show beside each prediction. For one patient, the five

episodes that move any of the three risks most are shown with what each adds for every outcome, so that the reference, the episodes and the rest of the history add up to the prediction. Concepts are named from a reviewed list of codes, with laboratory results named by the value third of the result. Both records stay in the secure environment and pairs that failed the completeness check are left out. The manuscript shows a synthetic patient.

Findings and expected effects. Thirteen findings (seven laboratory, three diagnostic and three medication findings) are summarised for every outcome as the mean change in predicted log-odds among the patients who have them, with 95% patient-bootstrap intervals, suppressed below 20 patients. For 21 finding–outcome cells a direction was fixed before the analysis (for example, high LDH raising predicted death and treatment failure, and low neutrophils raising predicted severe neutropenia and serious infection). Each is seen or opposite when the interval excludes zero on the expected or other side, and unclear otherwise. The other cells are descriptive. Agreement with expectations shows plausibility, not faithfulness.

Checks. For faithfulness, the k events with the largest attributions were replaced by the reference (and, separately, deleted) and the drop in predicted risk was compared with removing k random events and the k least attributed events, in 200 patients, with $k = 5$ prespecified. Deletion can create unfamiliar inputs, which is why replacement is primary. In 100 patients the attributions were recomputed with the readout, and with the readout and encoder layers, redrawn at their original scale [74] (agreement should be low), and with a per-source reference (agreement should be high). Intervals describe the explanation sample given the fitted model and the reference, not training-run uncertainty.

Input restriction. Separately, the same model was evaluated on the whole test set with restricted input and without retraining: the timing of care only (every event replaced by the average event, as in the reference), without laboratory results, without medication, without diagnoses, without the region and specialty of care (two regions contribute richer hospital-record data, so region partly marks which data a patient has), and with the last year of history only. For each outcome the held-out AUC at one year was compared with the full input, using paired patient-bootstrap resamples shared across outcomes and restrictions, and averaged over outcomes and clinical domains. This measures what the fitted model relies on, not what a model trained on less information could achieve.

Supplementary figures

Supplementary figures group study design, performance, clinical comparisons, representations and explanation diagnostics; all are cross-referenced in the text.

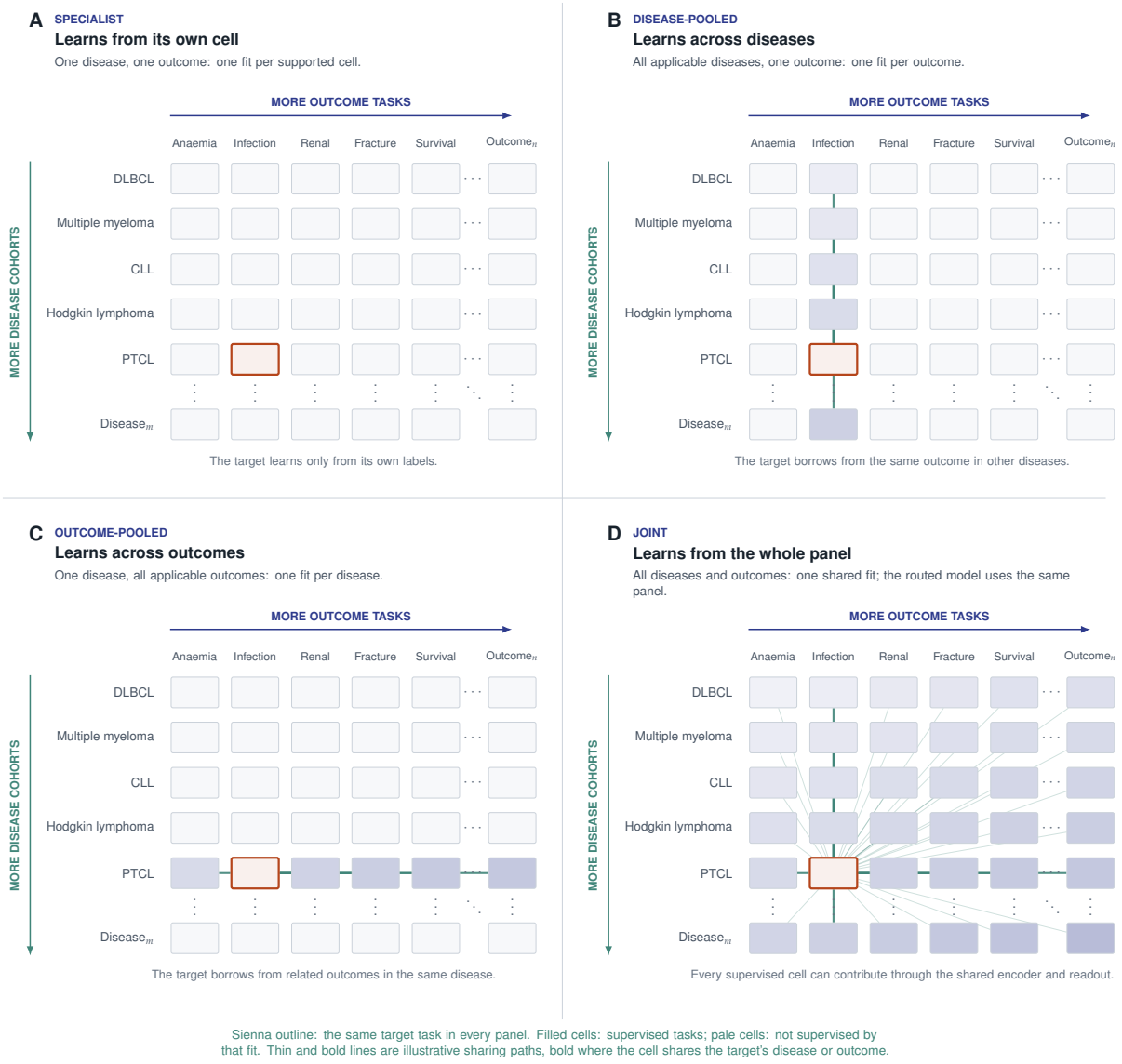


Figure S1. Supervision scopes on the disease–outcome matrix. The matrix of Fig. 1E, with one target task (outlined) evaluated identically in every panel. Diseases are rows and outcomes are columns. **A**, A specialist learns one outcome in one disease. **B**, A disease-pooled model learns one outcome across all diseases. **C**, An outcome-pooled model learns all outcomes within one disease. **D**, The joint model learns all outcomes in all diseases. Filled cells indicate supervision, not performance. The routed model uses the same panel as the joint model and changes only how predictions are read out. Every model predicts all six horizons.

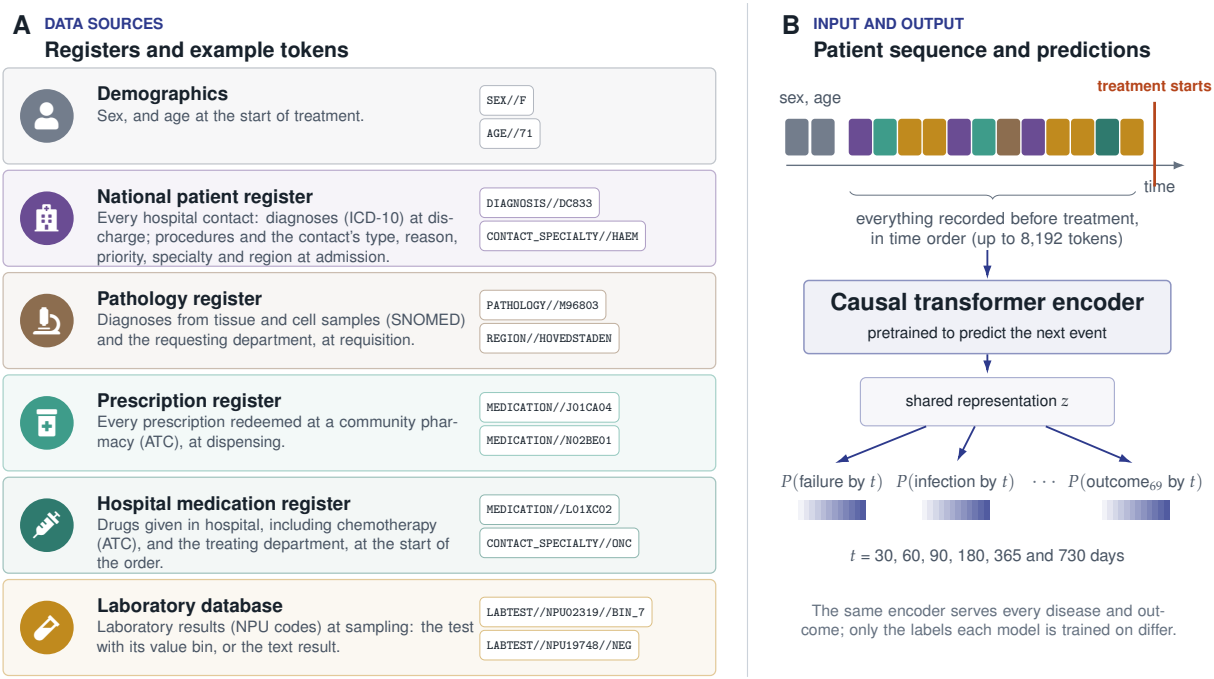


Figure S2. Data sources and input representation of the sequence model. A, the six sources linked through the civil registration number, what each contributes and when its events are dated, with two example tokens per source. Codes are illustrative, and laboratory values enter as quantile bins fitted on events before 2022. **B**, each patient's history becomes one token sequence, coloured by source, in time order and ending at the start of first-line treatment. The pretrained encoder reads it once into a shared representation, from which the model gives the probability of every outcome by each of six horizons.

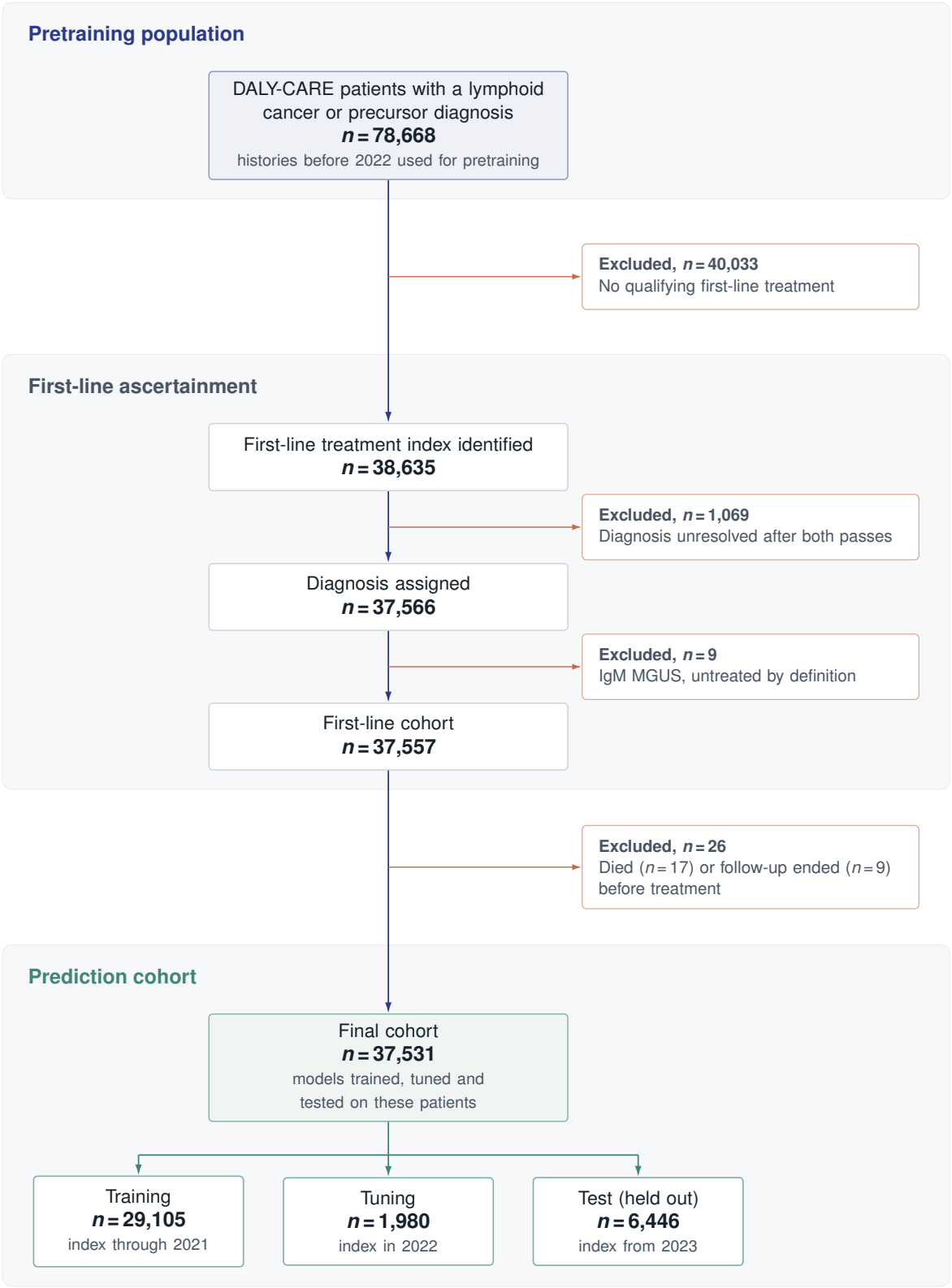


Figure S3. CONSORT diagram. Of the pretraining population of 78,668 adults in DALY-CARE with a lymphoid cancer or precursor (MGUS, MBL) diagnosis, whose pre-2022 histories were used for pretraining, 38,635 started first-line treatment. A diagnosis was assigned to 37,566 of them: to 37,357 from diagnoses recorded up to 14 days after treatment start, and to a further 209 on reassessment up to day 90 for late-coded diagnoses. The 37,531 with an assigned lymphoid disease who were alive and followed at treatment start form the final cohort, the patients whose outcomes were predicted: every prediction model was trained, tuned and tested on them, by treatment start in the training (to 2021), tuning (2022) or test (from 2023) period. Counts are unique patients. Not every patient contributes to every endpoint: outcome-specific eligibility, competing events and follow-up are applied within each split, and the resulting risk sets are given in the Supplementary Methods.



Figure S4. Estimability of disease–outcome cells. Status of every disease–outcome cell at one year: all models evaluated, too few training events to train a specialist, no events in the test period, fit failed, or not applicable (sex- or disease-specific outcomes). Counts in the legend.



Figure S5. Training events per disease–outcome cell. A, Training events by one year in each disease–outcome cell (log scale, grey for none). **B**, Cumulative distribution across cells, with the support-bin edges used in the analysis marked.

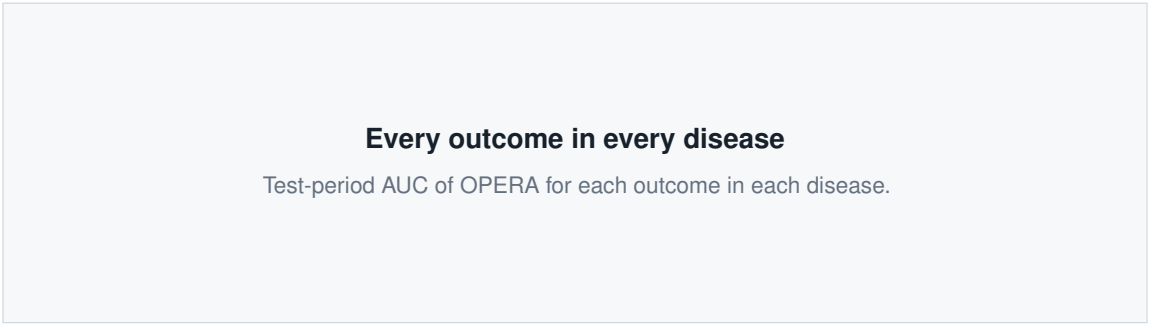


Figure S6. Discrimination of OPERA for each outcome in each disease. A, Test-period one-year AUC of OPERA for each outcome (rows, by clinical domain) in each disease (columns, by disease group). Grey cells had too few events to estimate. **B**, The same outcomes over all diseases at each horizon.

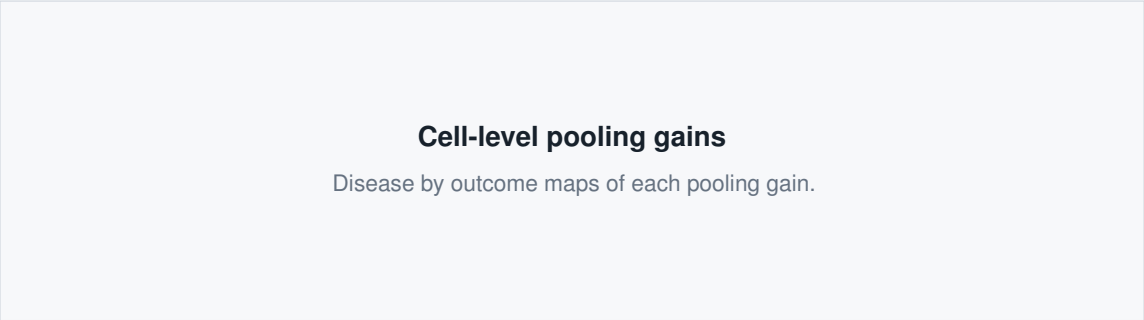


Figure S7. Pooling gains for each disease–outcome cell. As Fig. 6A–D at the resolution of single cells. Diseases (rows) run from fewest to most training patients and outcomes (columns) from fewest to most training events, with log-scale bars showing both. Dots mark cells whose 95% interval excludes zero: white for a gain, an open ring for a loss. Most cells are small, so colours alone do not establish benefit or harm. Dots mark unadjusted pointwise intervals, not multiplicity-corrected discoveries. Grey cells could not be estimated.

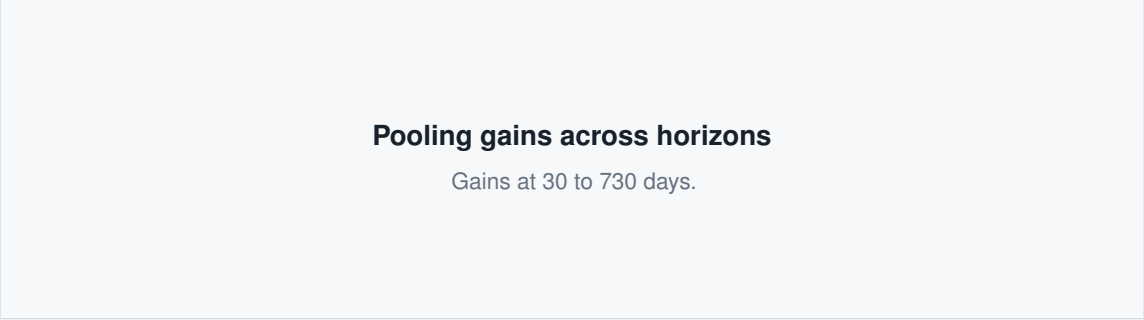


Figure S8. Pooling gains across horizons. Median paired AUC gain over the specialist by clinical domain at each horizon for the disease-pooled, outcome-pooled and joint models.

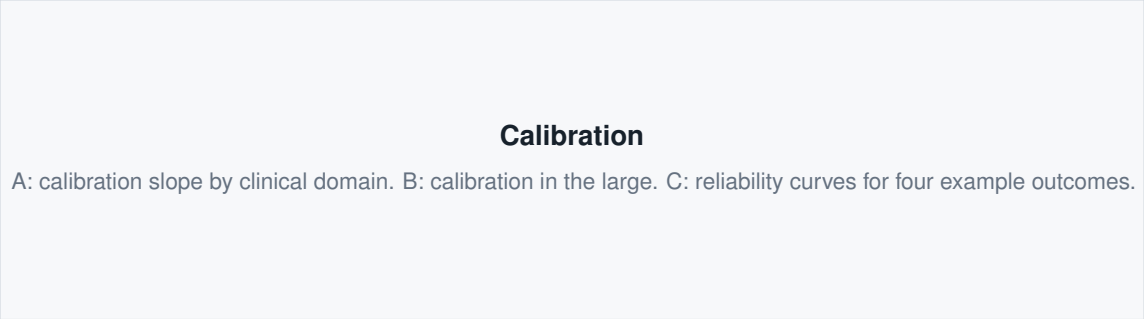


Figure S9. Calibration of the joint, routed, specialist and XGBoost models. A, Calibration slope at one year for the joint, routed, specialist and XGBoost models by clinical domain (median and interquartile range across cells, ideal 1; where an interval runs past the axis it is cut). **B**, Calibration in the large, observed minus mean predicted risk (ideal 0). **C**, Reliability curves for four example outcomes. Observed risks are weighted by the inverse probability of censoring and point size reflects the number of patients per bin; bins with fewer than 20 patients are not drawn. No model was recalibrated on test data.

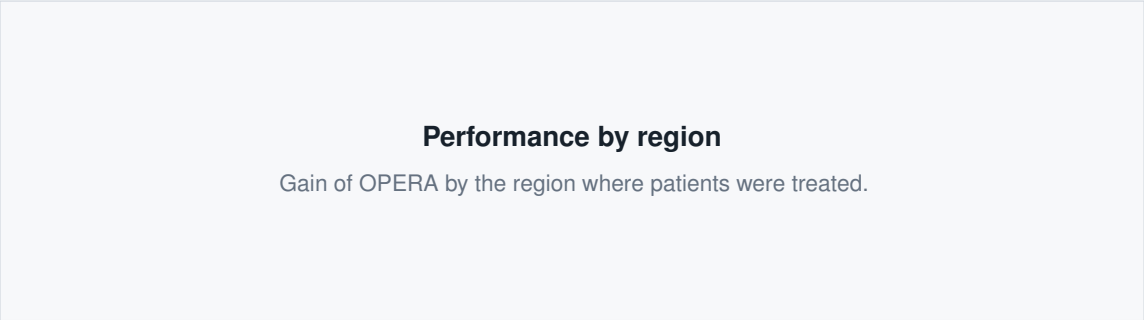


Figure S10. Performance by region. The models were trained on patients from all five regions, and test patients are grouped by the region of most of their pre-treatment contacts and every comparison is repeated within each region on the same fits. Maps show **A**, the one-year AUC gain of OPERA over per-task XGBoost, **B**, its one-year AUC, and **C**, its calibration in the large (observed minus predicted risk). Below, the same with 95% paired bootstrap intervals over equally weighted cells: **D**, gains over the specialist and XGBoost, **E**, AUC of OPERA and XGBoost, **F**, calibration in the large, and **G**, recorded events and laboratory results before treatment (median and interquartile range), which may explain regional differences. Region boundaries from geoBoundaries (CC BY 4.0) [108], with Bornholm as an inset.

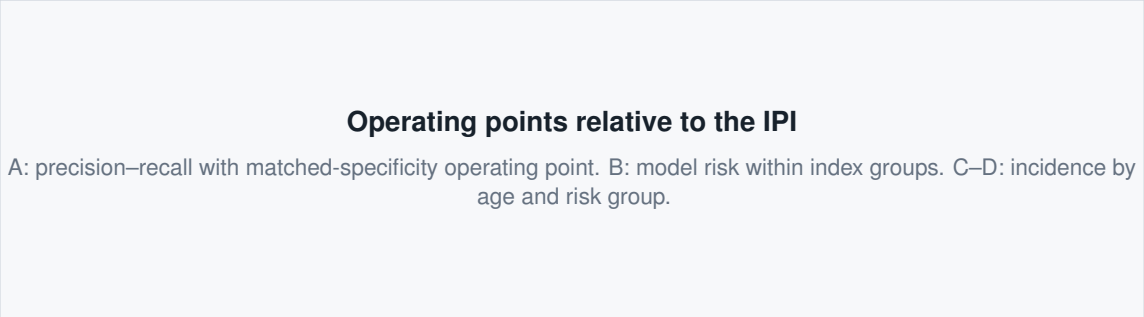


Figure S11. Operating points and reclassification relative to the IPI in DLBCL. A, Precision–recall curves for death within two years (patients with known status at two years) for the model and for each IPI category threshold. The square marks the IPI high-risk category. Dotted lines project it onto the model's curve: straight up gives the model's precision at the IPI's recall, and straight right its recall at the IPI's precision. The diamond marks the model at a threshold chosen on tuning data to match the IPI's specificity. AP, average precision. **B**, Predicted two-year risk of each patient within each IPI category, coloured by whether the patient died, with the model risk groups shaded. **C–D**, Two-year overall survival (95% bootstrap interval) and number of patients by age band and risk group for the model (**C**) and the IPI (**D**). Counts below five are shown as <5, and estimates are suppressed for cells with fewer than ten patients.

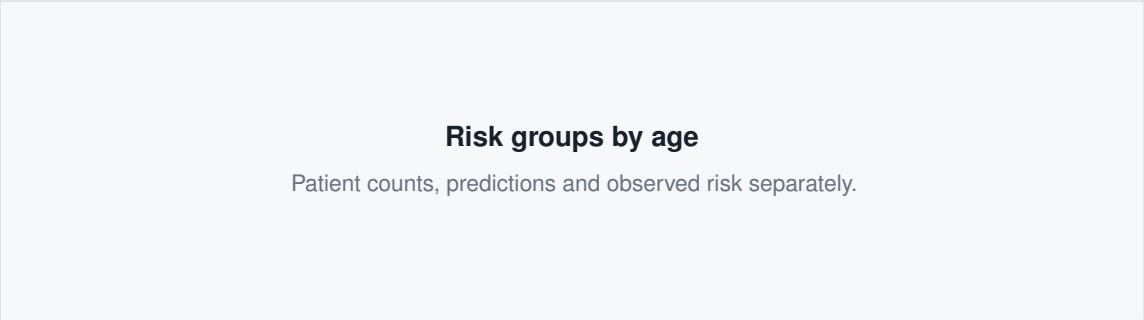


Figure S12. Model and IPI risk groups by age in DLBCL. Patients with DLBCL and a recorded IPI, grouped by model risk group (left) and IPI category (right). Rows show, by age band, the number of patients, the mean predicted two-year overall survival and the observed Kaplan–Meier estimate, each with its 95% bootstrap interval. Estimates are suppressed for cells with fewer than ten patients, but counts remain visible.

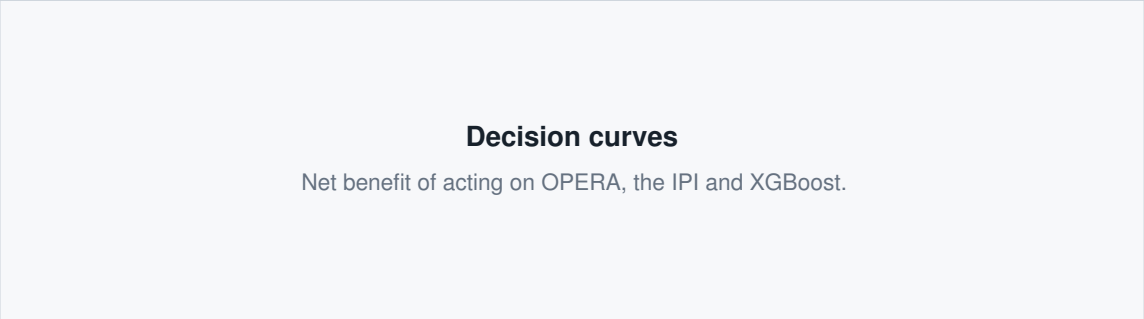


Figure S13. Decision curves for death within two years in DLBCL. A, Net benefit, as true positives per 1,000 patients net of false positives weighted by the threshold odds, of offering closer monitoring and supportive care to patients above each threshold of predicted two-year risk of death, for OPERA, the IPI (each category at its observed risk in tuning patients), per-task XGBoost, and acting for everyone or no one, in patients with an IPI recorded (shaded: 95% interval for OPERA). Censored patients are accounted for by inverse-probability weighting. **B**, Difference in net benefit between OPERA and each comparator, as net true positives per 1,000 patients, with 95% bootstrap intervals; values above zero favour OPERA. Dotted lines: the model's low- and high-risk cutoffs, fixed on tuning data by matching the IPI's low-risk observed risk and high-risk specificity, at which net benefit is also reported.

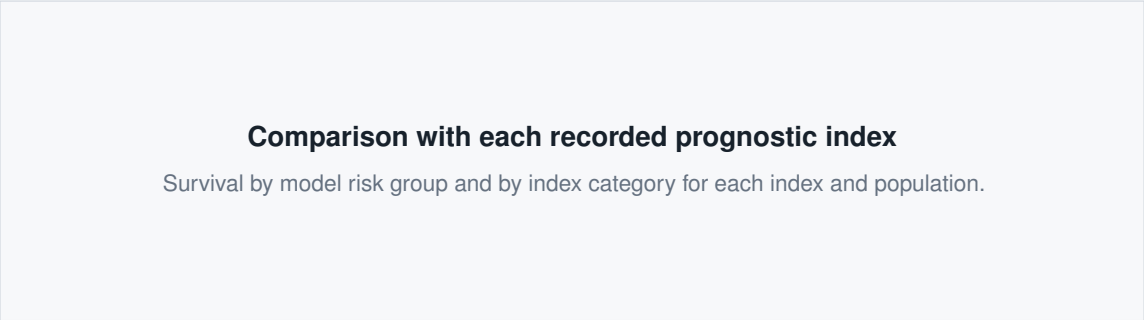


Figure S14. Model risk groups compared with each recorded prognostic index. One row per index and the population it applies to, on patients with that index recorded at diagnosis. Left, survival (or freedom from the endpoint) by model risk group. Middle, the same by index category, on the same patients. Right, estimates at two years with 95% CI (M, model, and I, index). Shaded bands are 95% bootstrap intervals and the dashed line marks two years. Model cutoffs were fixed on tuning data for each index, matching its low-risk observed risk and high-risk specificity. Every index recorded for at least 50 patients of a subtype is shown, and the NCCN-IPI in DLBCL. Group sizes, observed risk, sensitivity and specificity of both systems are in Supplementary Table [S]. **[Endpoint per index.]**

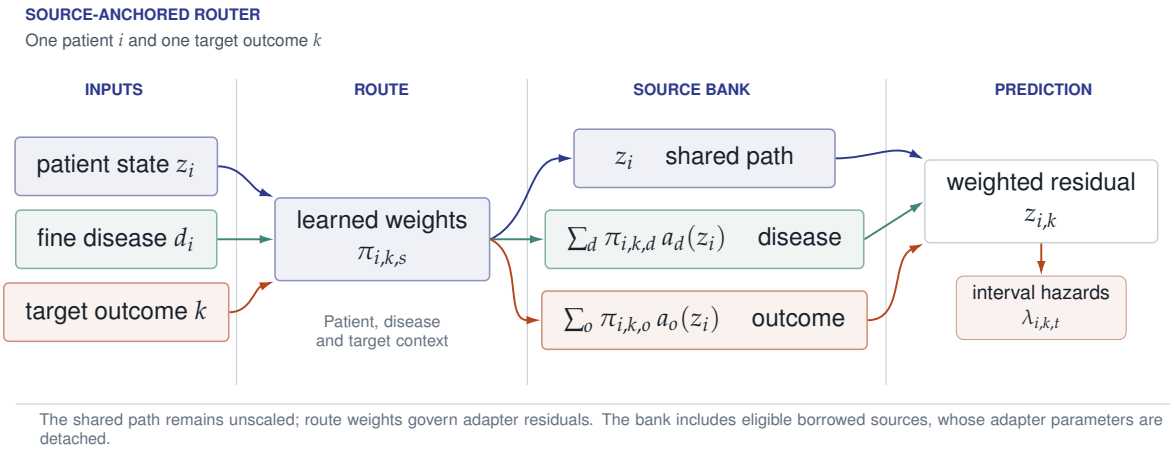


Figure S15. Source-anchored routing. For each patient and target outcome, the router mixes a shared representation with residual adapters, one per disease and, in the design screened on tuning data, one per outcome; the locked routed model uses disease adapters only. Its weights depend on the patient, disease and target. Sources with no training support are masked and low-event sources are shrunk. An adapter is updated by its own eligible supervision, while borrowed adapter parameters are detached. The mixed representation enters a shared linear head for interval hazards and cumulative incidence. This is a model schematic, not an empirical route-weight or performance plot.

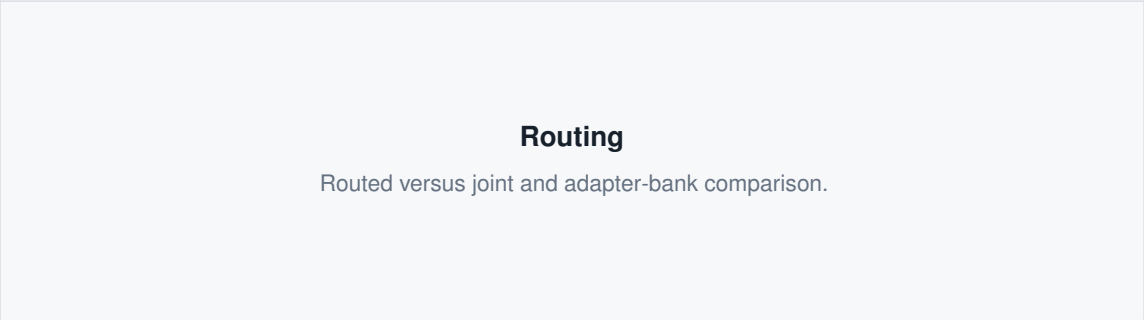


Figure S16. Routed versus joint model, and adapter-bank comparisons. **A**, Paired one-year AUC gain of the routed over the joint model by clinical domain, with the prespecified margin of practical equivalence shaded. **B**, Test-set panel gains of the four source banks, trained with the locked rank and query under the same protocol: the router trained with every source masked against the joint model (the noise floor of refitting, and a check that the architecture alone changes nothing), disease adapters, outcome adapters, each added to the other, both, and their interaction; and the controls (the joint model told the disease, the own-disease and own-outcome routers). The source bank was chosen on the tuning set (disease adapters only).

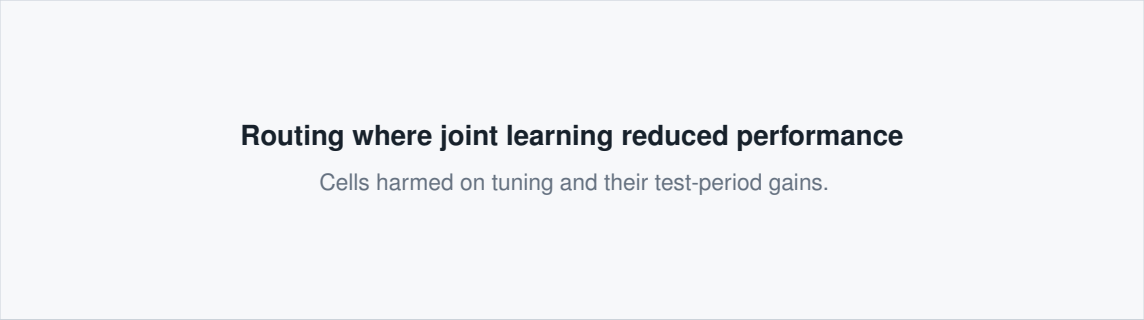


Figure S17. Routing in cells where joint learning reduced performance on the tuning set. Cells were classified on the tuning set by whether the joint model was worse than the specialist. **A**, Routed-versus-joint test gain against the tuning-set harm. **B**, Test-period gains of the joint and routed models over the specialist in each stratum.

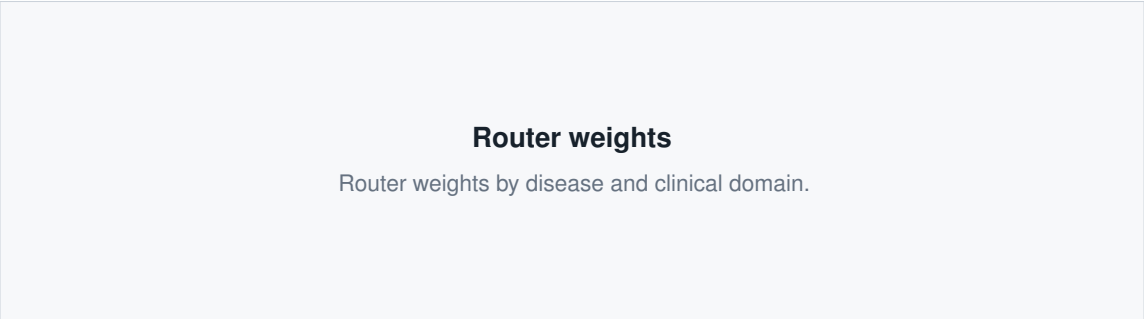


Figure S18. Router weights by disease and clinical domain. **A**, Mean routing weight on each disease adapter by patients' disease (the patient's own disease hatched). **B**, Outcome-adapter weights by clinical domain, and **C**, the share of the routed correction carried by the disease and outcome banks, both in the test-set model with both banks (the locked model routes over diseases only). **D**, Routing intensity across patients. Weights show where the router sends weight, not what that weight contributes (Fig. 8).

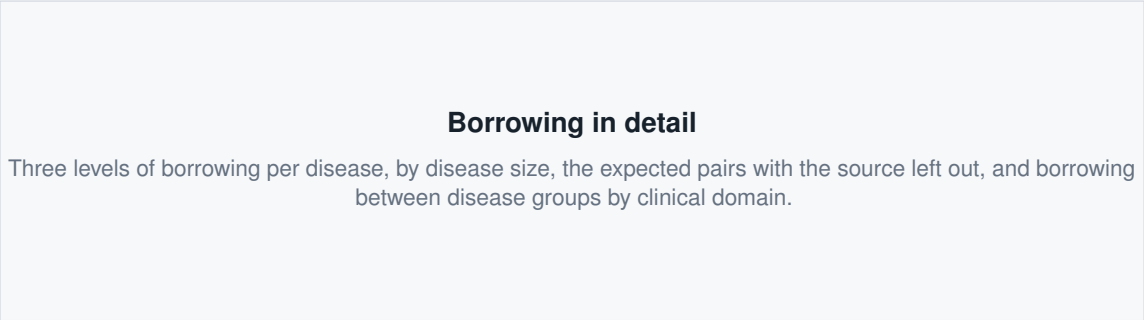


Figure S19. Borrowing in detail. **A**, For each target disease, over all outcomes and horizons: the AUC lost without borrowing (the fitted routed model with and without its borrowed corrections, as in Fig. 8B), the gain over the own-disease router (the router trained with only the patient's own disease adapter) and routing (against the joint model). The last two compare separately trained models and carry the noise of refitting; the loss without borrowing can exceed the trained gain because the encoder was trained alongside the borrowed corrections. **B**, The loss without borrowing and the gain over the own-disease router against the number of patients with the disease. **C**, The pairs of Fig. 8C with the source alone (filled) and the loss when only that source is left out of the full model (open). **D**, Mean partially pooled gain from a source disease alone between disease groups (rows: target, columns: source) in each of the 12 clinical domains, on one scale; diseases with few estimable outcomes are omitted, and the groups summarise for reading only (they do not enter the pooling). AUC points with paired patient-bootstrap 95% intervals; IPA differences are in **[Supplementary Table]**.

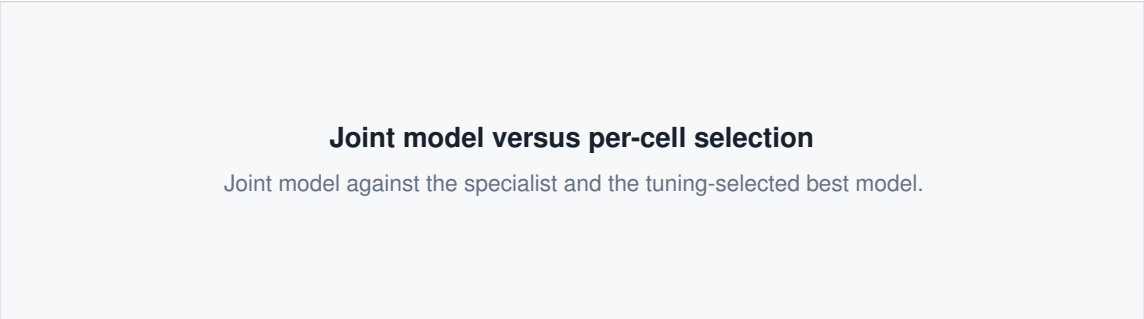


Figure S20. Joint model versus the best model selected for each cell. **A**, Test AUC of the joint model against the specialist in each cell, with point size reflecting training events. **B**, Against the best of the specialist, disease-pooled and outcome-pooled models chosen per cell on the tuning set. **C**, Which model was chosen, by clinical domain.

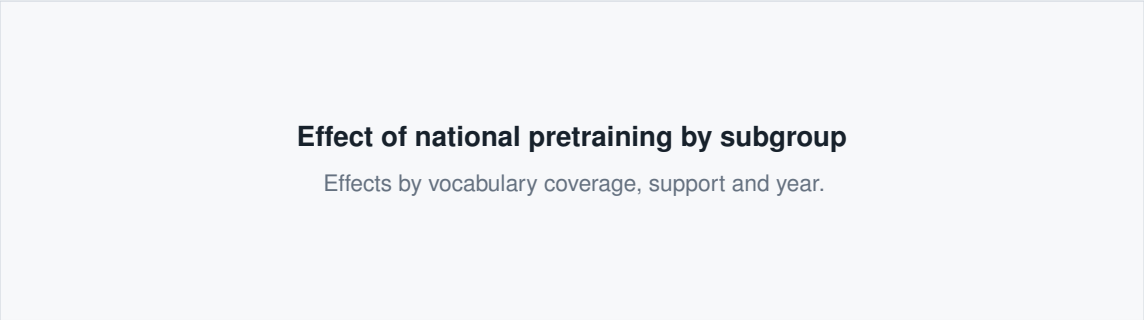


Figure S21. Effect of national pretraining by vocabulary coverage, training support and year of treatment. Patient-weighted improvement in one-year Brier score ($\times 100$, 95% CI) of the joint model initialised from the national encoder, used directly or after local adaptation, over the locally pretrained joint model, by the share of a patient's history covered by the national vocabulary, by the number of training events of the predicted combination, and by year of treatment.

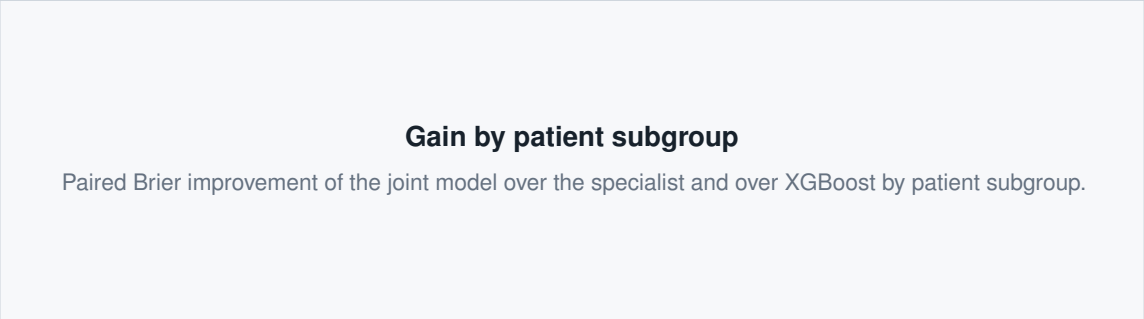


Figure S22. Gain from joint learning by patient subgroup. Patient-weighted improvement in one-year Brier score ($\times 100$, 95% bootstrap CI) of the joint model over the specialist and over XGBoost, pooled across cells, by subgroups defined at the start of treatment. Positive values favour the joint model.

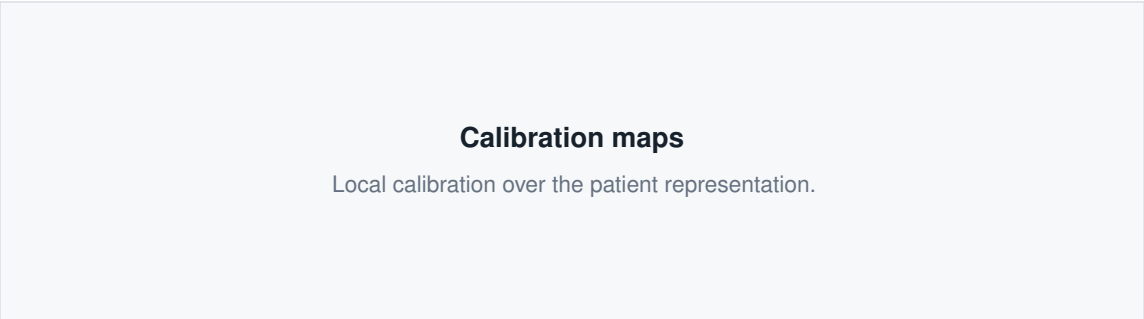


Figure S23. Local calibration across the patient representation. Two-dimensional projection of test patients' representations for example outcomes: event status, routing displacement, kernel-smoothed predicted risk and the IPCW-weighted local residual (observed minus predicted).



Figure S24. Observed outcome risk across OPERA's patient representation. A, Test patients on a two-dimensional projection of OPERA's patient representation (fitted on tuning patients), with the densest half of each disease group outlined. **B–F**, The same map coloured by the observed one-year risk of five outcomes from different clinical domains, estimated with censoring-weighted kernels and shown relative to each outcome's average. Regions with too few evaluable patients are blank. **G**, Observed minus predicted risk of OPERA for the outcome in **B**. All 69 outcomes are shown in Fig. S26–Fig. S31. The map is descriptive and carries no performance claim.

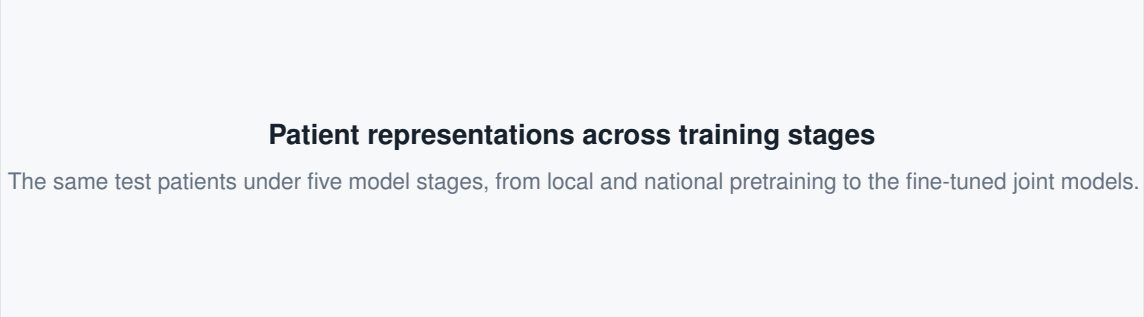


Figure S25. Patient representations at each training stage. The same test patients under five model stages: pretraining on the cohort, the joint model from that encoder, national pretraining, national pretraining adapted to the cohort, and the nationally initialised joint model. The first row outlines where each disease group lies, and the others show the local mean (Gaussian kernel) of age at treatment, the number of recorded events before treatment (log scale) and the IPI category among patients with a recorded IPI, left blank where too few patients are nearby. Each column's projection was fitted on tuning patients and applied unchanged to test patients, so coordinates are comparable within a column but not between columns. The maps are descriptive and carry no performance claim.

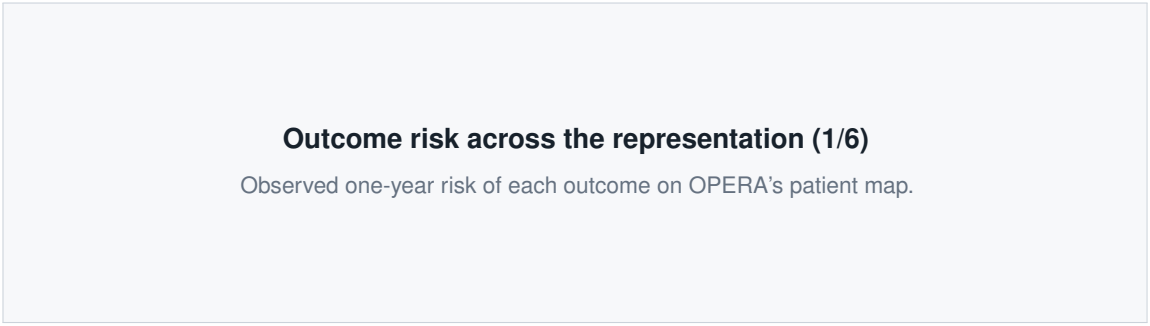


Figure S26. Observed one-year risk of each outcome across the patient representation (1 of 6). Every panel shows the same two-dimensional projection of OPERA's patient embedding (fitted on tuning patients) for test patients. Colour is the censoring-weighted kernel estimate of the observed one-year risk of one outcome, relative to that outcome's average (shown under its name): bright regions carry up to four times the average risk and dark regions a quarter. Regions with too few evaluable patients are blank. Outcomes are grouped by clinical domain, named above each group.

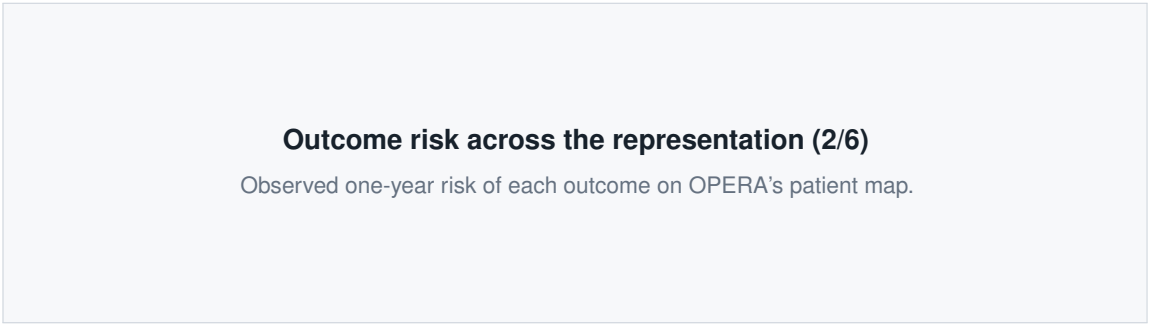


Figure S27. Observed one-year risk of each outcome across the patient representation (2 of 6). As Fig. S26, continued.

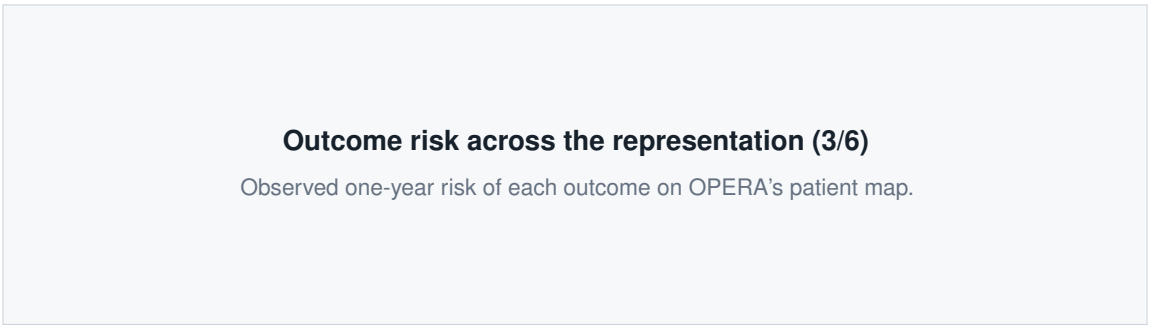


Figure S28. Observed one-year risk of each outcome across the patient representation (3 of 6). As Fig. S26, continued.

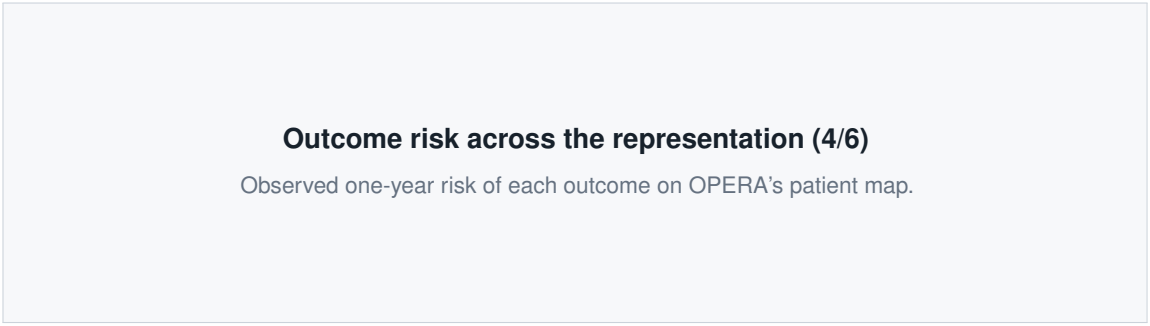


Figure S29. Observed one-year risk of each outcome across the patient representation (4 of 6). As Fig. S26, continued.

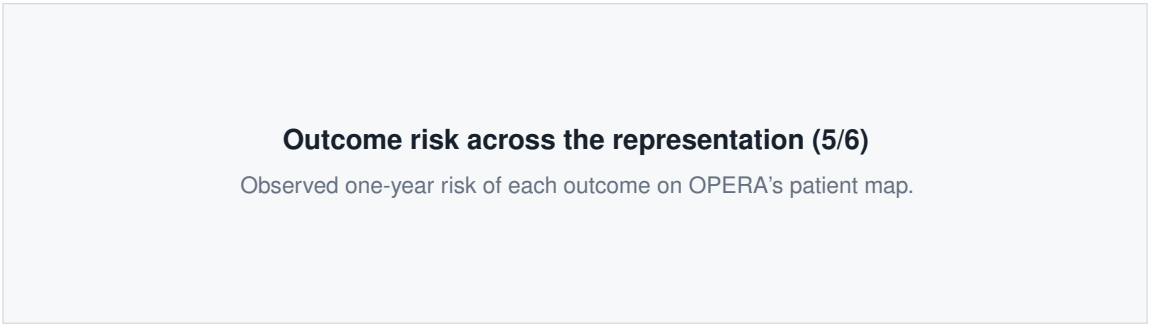


Figure S30. Observed one-year risk of each outcome across the patient representation (5 of 6). As Fig. S26, continued.

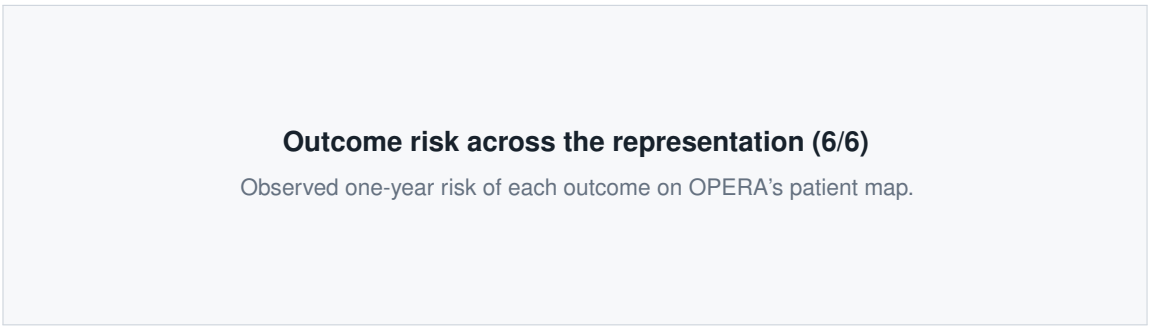


Figure S31. Observed one-year risk of each outcome across the patient representation (6 of 6). As Fig. S26, continued.

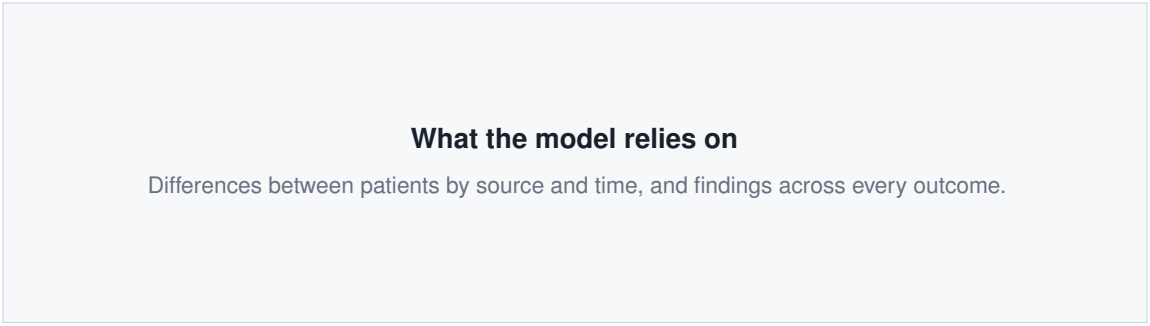


Figure S32. What the model relies on, for every outcome. Rows are outcomes grouped by clinical domain. Integrated gradients split each held-out patient's predicted log-odds at one year into a reference, which keeps the timing of care, age and subtype but replaces every recorded event by an average event, and the contribution of each event. **A**, The variance of the predictions between patients, divided into the part carried by the timing of care alone (grey) and the parts carried by what was recorded, by source. Shares are signed and add up to 100% per outcome. **B**, The same division by time before treatment. **C**, Mean change in predicted log-odds among the patients who have each finding, shown as the corresponding odds ratio. Boxed cells are the prespecified expected effects, dots mark 95% patient-bootstrap intervals that exclude zero, and grey cells had fewer than 20 patients with the finding.

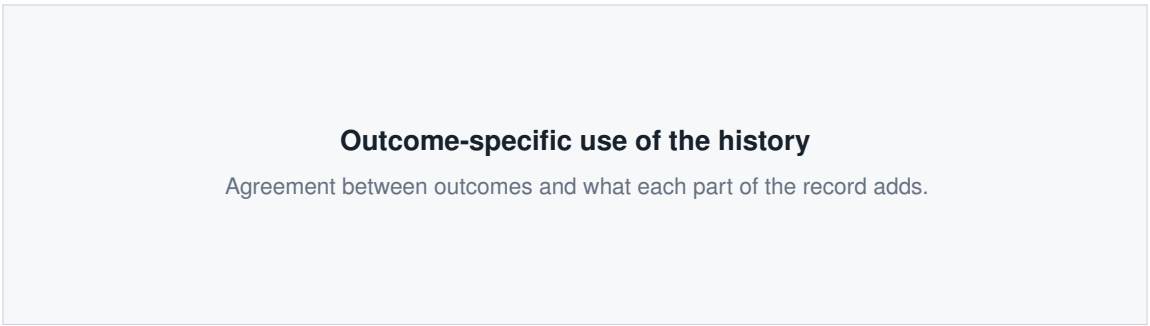


Figure S33. The model uses the same history differently for different outcomes. A, For every pair of outcomes, how much they agree on which episodes of a patient's history raise or lower risk (Spearman correlation of episode contributions within patients, averaged over patients). Episodes are events within one day of each other, and each clinical domain is outlined. **B**, Change in held-out AUC ($\times 100$) when the same fitted model reads only part of the record, averaged over all outcomes and over the outcomes of each domain. The model is not retrained, so this measures what it relies on, not what a model trained without that input could achieve.



Figure S34. One synthetic patient’s history read for six complications. Each track is one outcome from a different clinical domain. Circles are the episodes of care (events within one day of each other) that move its predicted risk, at their time before treatment, sized by the change in predicted log-odds and coloured by direction (sienna raises, indigo lowers). The episodes that matter most for any outcome are labelled with their main findings and timing. The predicted one-year risk is given at the right, with the risk for the same timing of care and average findings. For every patient, the episodes that most raise and lower each risk can be extracted in the same way and shown beside the prediction. The patient and its contributions are invented, and records of real patients stay in the secure environment.

Supplementary Methods: cohort, prediction time and outcome definitions

This supplement describes how the study cohort, the index date (start of first-line treatment), the temporal split and the 69 outcomes, from disease-course endpoints to treatment-related and acute-care outcomes, were defined in the Danish Lymphoid Cancer Research (DALY-CARE) data resource [1]. Reported cohort counts describe the analysis release (Supplementary Table D2). The R and Python code that implements every step is released with the paper.

Conventions. Dates are calendar dates without time of day unless stated otherwise. The *index date* is the start of the reconstructed first-line treatment episode (Section D2). *End of follow-up* is the earlier of death and last follow-up as recorded in the DALY-CARE patient table. The *administrative censoring date* is the earlier of end of follow-up and the data cutoff of 19 March 2026. Unless stated otherwise, an outcome event is the *first* qualifying date d with index date $< d \leq$ censoring date; events on the index date itself were never counted. For each outcome, patients for whom the outcome could not be ascertained were excluded from that outcome's risk set (masked supervision); exclusion was never interpreted as absence of the event.

How to read this supplement. Sections D1–D4 describe the data and who enters the study, including the reconstructed treatment index and disease assignment. Section D5 explains disease-course outcomes, particularly second-line treatment. Section D6 gives the clinical outcome dictionary; Section D8 explains who can contribute a label and when observation ends. The final sections distinguish panel selection and reporting groups from the clinical definitions and state their limitations.

Three questions for every outcome
What counts? The qualifying event and its threshold or code rule.
Who is eligible? Required baseline information and source coverage.
How long is it observable? Ascertainment, censoring and competing death.
An unavailable label is not a negative outcome.

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D1 Data sources and linkage

All sources were accessed within the DALY-CARE secure environment and linked at patient level by the DALY-CARE pseudonymised patient identifier. Supplementary Table D1 lists the sources and their use.

Supplementary Table D1: Data sources.

Source	Content used	Used for	Coverage and date handling
DALY-CARE patient table	Vital status, date of death or end of follow-up, sex, date of birth	Death, end of follow-up, demographics	End of follow-up is the earlier of death and last follow-up.
DALY-CARE haematological diagnoses	Harmonised haematological diagnoses with date, reliability rank and data source	Diagnosis anchor, disease classification, new haematological malignancy	Diagnoses from the Danish clinical quality registries (RKKP) were assigned the highest reliability rank.
Danish National Patient Registry: diagnoses	All ICD-10 diagnoses (SKS diagnosis codes)	New malignancy, cardiovascular, infection and fracture outcomes	LPR2 until 2019 and LPR3 thereafter; no era restriction.
DALY-CARE ICD-10 disease dictionaries	ICD-10 code patterns per haematological disease; SKS radiotherapy codes	Disease classification; radiotherapy as first-line treatment	Supplemented locally (Section D3).
Harmonised antineoplastic treatment data (version of 10 September 2026)	Drug code (ATC or SKS), drug text, indication code and text, source, start date	First- and second-line treatment episodes	Treatment-plan records were excluded; records from the LYFO, DaMyDa and DCLLR quality registries were regarded as reliable.
Curated CLL and MM treatment-line data (CLL version 4, MM version 5)	Dated treatment lines	First-line anchoring in CLL/MM; second-line events in CLL/SLL and MM	Assumed complete for CLL/SLL and MM patients.
Danish Lymphoma Registry (LYFO)	Date of second-line treatment	Second-line treatment in lymphoma	Data from the CLL (DCLLR) and myeloma (DaMyDa) registries were not used for second-line dating.
Laboratory results	Analytes identified by NPU codes, cleaned with the DALY-CARE standard procedure	Laboratory outcomes and laboratory ascertainment	Complete capture assumed from 1 January 2017 (Section D8).
DALY-CARE acute kidney injury data	Start dates of acute kidney injury episodes derived from creatinine by a separate DALY-CARE algorithm	Acute kidney injury outcome	Laboratory ascertainment rule.
Danish National Patient Registry: admissions, contacts and procedures (LPR2, LPR3)	Admissions and outpatient/emergency contacts (LPR2); physical contacts (LPR3); surgical and other procedures (SKS)	Procedure outcomes; acute hospitalisation	Procedures dated by the procedure start date, not the contact start.
Hospital Medicines Register (SMR)	In-hospital medicine administrations with ATC code and route	Intravenous antimicrobial-use outcome	Reporting starts 15 May 2018; outcome restricted to index dates on or after this date.
Danish National Prescription Registry	Dispensed prescriptions with ATC code and dispensing date	Prescription antimicrobial-use outcome	No era restriction.
International Prognostic Index (IPI)	IPI at diagnosis	Population description and prognostic benchmark	Not an outcome or sequence-model input.

Linkage of procedures. Procedure records were linked to their admission or contact through the full record identifier and dated by the start of the procedure rather than of the admission.

Data cutoff. The administrative data cutoff was 19 March 2026 for all endpoints. A death recorded after the cutoff was treated as censoring at the cutoff, not as an event.

D2 Study population and index date (first-line treatment)

First-line treatment is not recorded as such in any Danish register, so its start, the index date of every prediction, was reconstructed from the treatment records. The reconstruction has to separate lymphoma-directed treatment from three kinds of record that look alike in the data: drugs given for another cancer, drugs given as prophylaxis, maintenance or supportive care, and isolated administrations that do not amount to a course of therapy. Each of the choices below addresses one of these.

Where to look. Each patient's earliest DALY-CARE haematological diagnosis of any kind, including MGUS, MBL and unspecified codes, served as the diagnosis anchor, and treatment records from 30 days before to 3,650 days after it were eligible. Starting before the anchor allows for treatment that begins while the diagnosis is still being registered.

Which records count. A record needed a patient, a date and a drug code, or a drug text where no code was given. Records were then judged by their indication, in order of trust. Records from the clinical quality registries (LYFO, DaMyDa, DCLLR) were kept, as were records whose indication carried a haematological ICD-10 code (DC81–DC89, DC90–DC96, DD47). Records were discarded when the indication was a listed non-haematological cancer (DC18, DC19, DC20, DC25, DC34, DC43, DC50, DC56, DC61, DC71, DC78), when the indication text matched a list of non-haematological Danish terms, or when the drug is not used for lymphoid malignancy (e.g. fluorouracil, capecitabine, imatinib, azacitidine, hydroxycarbamide, cisplatin, carboplatin). Records without an informative indication were kept as uncertain rather than discarded, because much routine treatment is recorded without one.

From records to courses of therapy. Records belonged to one episode as long as consecutive treatment dates were at most 21 days apart, a gap rule that follows the cycle length of most regimens rather than a limit on duration; the episode started on its earliest record. An episode then had to look like therapy, and the evidence required depended on how specific its drugs were. The strongest evidence is a drug used almost only for lymphoid malignancy, and an episode containing one of these *anchor* agents qualified outright. The anchors were the CLL agents (fludarabine, cyclophosphamide, bendamustine, chlorambucil, obinutuzumab, ibrutinib, zanubrutinib, acalabrutinib, idelalisib, venetoclax, alemtuzumab; ATC and SKS BWHA/BOHJ codes); aggressive B-cell lymphoma agents and regimens (e.g. SKS codes for CHOP, COP and CHOEP; mitoxantrone, ifosfamide, carboplatin, gemcitabine, vincristine, melphalan, cytarabine, bortezomib, lenalidomide, thiotepa, carmustine); myeloma agents and regimens identified by drug text (e.g. daratumumab, carfilzomib, pomalidomide, autologous stem-cell transplantation), including the generic myeloma treatment categories recorded as drug text (complex or other cytostatic treatment, other antineoplastic agents, monoclonal antibodies and antibody–drug conjugates); SKS BWHA135 and the drug text "vcr"; and radiotherapy (SKS codes described as "strålebehandling", except listed prefixes, plus AFB04F3 and AFB03F3). Drugs that are also given for prophylaxis, maintenance or other diseases were kept out of the anchors on purpose, so that they could not confirm a course on their own: rituximab (ATC L01FA01, L01XC02), unspecified CD20 antibody (drug text and SKS BOHJ11), methotrexate (drug text and ATC L01BA01, L04AX03), gemcitabine and cisplatin identified by text, doxorubicin, etoposide, oxaliplatin, selinexor, basic cytostatic treatment, the generic cytostatic codes BWHA1, BWHA2 and BWHA4, and bevacizumab.

Without an anchor, more evidence was required the more easily the episode could be something else. A quality-registry record sufficed, unless the episode also contained one of the suspect agents above and fewer than two distinct codes. Anti-CD20 treatment (the codes above, or the drug texts rituximab, ofatumumab or CD20 antibody) on two or more dates sufficed, so that rituximab monotherapy, a genuine first-line treatment in indolent lymphoma, is not lost. An episode with any other suspect agent needed at least three treatment dates and two distinct codes, and any remaining episode at least two dates or two distinct codes.

Curated treatment lines. For patients in DALY-CARE's curated treatment-line data for CLL (version 4) and multiple myeloma (version 5), the curated first-line date was trusted over the reconstruction. An episode that contained this date, or lay within 30 days of it, qualified without the therapy-like criterion, and the episode closest to the curated date was chosen. The index date remained the start of that episode, not the curated date, so that all diseases share one definition of the index.

The first line. All other patients received their earliest qualifying episode, and patients without one had no index date and were not included.

Cohort flow. Supplementary Table D2 shows the cohort flow. Patients enter the outcome population only if they had a follow-up record and both end of follow-up and the data cutoff fell after the index date; a patient with a missing follow-up record, end of follow-up on or before the index date (including death on the index date) or an index date on or after the cutoff was therefore excluded from all outcomes. Twenty-six patients of the first-line cohort were excluded in this way: for all of them, death (17) or the end of follow-up (9) was recorded on or before the reconstructed index date.

Supplementary Table D2: Cohort flow.

Stage	Patients
Patients with a DALY-CARE haematological diagnosis (pretraining population)	78,668
No qualifying first-line treatment	40,033
First-line index date identified (pass 1)	38,635
Diagnosis assigned in pass 1 (diagnoses from 20 years before to 14 days after index)	37,357
Unassigned after pass 1	1,278
Additionally assigned in pass 2 (coding-lag reassessment, up to 90 days after index)	209
Unassigned after pass 2 (excluded)	1,069
Diagnosis assigned after both passes	37,566
IgM MGUS excluded	9
First-line cohort	37,557
No follow-up after the index date or index on/after the data cutoff	26
Supervised population	37,531
Training (index date up to 31 December 2021)	29,105
Tuning (index date in 2022)	1,980
Held-out test (index date from 1 January 2023)	6,446

D3 Disease classification

Diagnosis codes were mapped to disease groups using the DALY-CARE ICD-10 patterns with local refinements: SLL only by the exact code DC830; lymphoplasmacytic lymphoma by DC830B or DC880; nodal marginal-zone lymphoma by DC830C; splenic marginal-zone lymphoma by DC830D; the generic marginal-zone group was not used; and DD472A (IgM MGUS) was mapped to an exclusion group.

Pass 1 (contemporaneous diagnoses). Disease identity was taken from the diagnoses recorded around the start of treatment, from 20 years (7,300 days) before to 14 days after the index date. The long look-back captures an indolent disease diagnosed years before it needed treatment; the short forward window allows for codes registered during the first weeks of treatment. A single code can be a working diagnosis, so a disease group counted as confirmed only when supported by a diagnosis of reliability rank 3 or better (quality-registry diagnoses rank 1) or by at least two diagnoses at least 14 days apart. Catch-all labels (generic DALY-CARE, unspecified, unspecified B-cell lymphoma, unspecified lymphoma and unspecified leukaemia) could not be assigned, nor could MGUS and MBL, which by definition are not treated.

When several groups were confirmed, three principles decided between them. First, a specific label beat a broad one: the catch-all labels and the broad groups B-cell lymphoma, T-cell lymphoma, plasma-cell disorder and Hodgkin lymphoma gave way to any other confirmed specific label in the window, so that Hodgkin lymphoma was assigned only when no other specific lymphoma was confirmed. Second, an aggressive disease beat an indolent one, because first-line treatment in a patient with both is usually given for the aggressive disease: Richter transformation, DLBCL, BL, PTCL, ALCL, AITL, LBL and PCL took precedence over all other labels, including MCL. Third, remaining ties were broken by a ranking of aggressiveness (Richter transformation; then DLBCL, BL, PTCL, ALCL, AITL, LBL, PCL and MCL; then MM and solitary plasmacytoma; then CLL, SLL and LPL; then FL, MZL, EMZL and HCL; then others), and then by the most reliable, most recent and most specific diagnosis.

Transformation was recognised from the order of diagnoses. A patient with DLBCL confirmed in the window was labelled Richter transformation if CLL or SLL had been recorded anywhere in their history, confirmed or not, at least 90 days before their first DLBCL code, and transformed follicular lymphoma under the same rule for FL. The 90-day gap keeps a CLL or FL code registered during the same work-up as the DLBCL from being read as a transformation.

Pass 2 (coding-lag reassessment). A specific lymphoma diagnosis is often registered only after the diagnostic work-up is complete, which can be after treatment has started. Patients with an index date but no assignable disease after pass 1 therefore entered a second pass. Their earliest confirmed, assignable diagnosis anywhere in their history became a new anchor; first-line treatment was re-identified with the same algorithm in a window from 90 days before this anchor to 20 years after it (curated CLL/MM anchors still applied), and disease was re-assigned with a 10-year look-back and a 90-day forward window. Pass-2 patients therefore received a new index date, replacing their pass-1 index. This allowed a specific diagnosis registered up to 90 days after treatment start to define disease identity.

Exclusions. Patients still unassigned after pass 2, and patients assigned IgM MGUS, were excluded. Characteristics of the first regimen (for example radiotherapy alone) were recorded descriptively and did not lead to exclusion.

Disease taxonomy. The analysis used 24 disease subtypes nested in 10 disease groups (Supplementary Table D3).

Supplementary Table D3: Disease groups and subtypes.

Disease group	Subtypes
CLL/SLL	CLL, SLL
Indolent B-NHL	FL, extranodal MZL, nodal MZL, splenic MZL, LPL, B-cell lymphoma without further specification
DLBCL-like	DLBCL, Richter transformation (coded or derived), transformed FL
MCL	MCL
HL	HL
MM	MM, plasma-cell leukaemia, solitary plasmacytoma
T-NHL	AITL, ALCL, PTCL, T-cell lymphoma without further specification
HCL	HCL
BL/LBL	BL, LBL
Amyloidosis	Amyloidosis

D4 Temporal split and prediction time

Temporal split. Patients were assigned by index date to **training** (index date before 1 January 2022), **tuning** (1 January to 31 December 2022) or **held-out test** (from 1 January 2023). Each patient had the same index date and split in every outcome; composite outcomes required agreement of index date and split across components.

Prediction time. The model input for each patient comprised all records timestamped before 00:00 on the index date, followed by a prediction token. Treatment and laboratory records on the index date itself were therefore not model inputs, although an index-date laboratory draw counted towards laboratory ascertainment (Section D8). Because most records carry a date but no time, events on the index date cannot be ordered relative to treatment start; such events were not counted as outcomes.

Pretraining. The cohort encoder was pretrained with an exclusive calendar cutoff of 1 January 2022, applied in the same way, and its vocabulary was derived from pre-2022 events of pretraining patients only; the national encoder has its own vocabulary, fitted on national data with the same cutoff. Pre-2022 histories of patients later assigned to tuning or held-out test were included in pretraining; the evaluation therefore concerns future treatment episodes, not unseen individuals.

Disease identity at prediction time. Disease labels could rely on codes registered up to 14 days (pass 1) or 90 days (pass 2) after the index date. Models conditioned on disease therefore used information that was not necessarily available at the index date for these patients; a prespecified sensitivity analysis excludes the 209 patients whose disease was assigned after day 14 (Supplementary Methods, Prespecification).

D5 Disease-course endpoints

D5.1 Second-line treatment

Second-line treatment was identified from the same filtered treatment records as first-line treatment, with disease-specific rules.

Lymphoma groups (DLBCL-like, HL, BL/LBL, MCL, T-NHL, HCL, indolent B-NHL). (1) If the first-line episode spanned a single day, its end was imputed as start plus an expected regimen duration. (2) Treatment records dated more than a minimum gap after the first-line end were grouped into episodes (21-day gap

rule). (3) Episodes were excluded as autologous stem-cell transplant consolidation (transplant or conditioning codes within the consolidation window), consolidation (a share of first-line codes reappearing at or above an overlap threshold within the consolidation window), maintenance (only anti-CD20 or lenalidomide within the maintenance window), or single-date anti-CD20. (4) Remaining episodes had to be therapy-like (as for first line, except that quality-registry episodes needed $\geq$2 codes or $\geq$2 dates) *and* have at least moderate evidence of treatment for lymphoid malignancy: quality-registry source; an anchor agent with haematological indication; an anchor agent without suspect agents; or haematological indication with $\geq$2 dates *and* $\geq$2 distinct codes. Anti-CD20-only episodes on $\geq$2 dates were therefore counted as retreatment and qualified as second line when they met these evidence criteria (e.g. obinutuzumab, an anchor agent, or haematological indication with two distinct codes), except within the maintenance window in MCL and indolent B-NHL. The event date was the start of the earliest qualifying episode. (5) A second-line date recorded in LYFO after the index date replaced the derived date if it was earlier or if no derived date existed. The disease-specific parameters are listed in Supplementary Table D4.

Supplementary Table D4: Disease-specific parameters for second-line treatment in lymphoma groups. Durations and windows in days; – not applied.

Group	Imputed first-line duration	Minimum gap	Consolidation window	Maintenance window	Overlap threshold	Reliability
DLBCL-like	126	60	90	–	0.45	high
HL	120	60	90	–	0.45	high
BL/LBL	28	45	90	–	0.50	low
MCL	140	60	180	548	0.40	moderate
T-NHL	126	60	90	–	0.40	moderate
HCL	7	90	–	–	0.30	low
Indolent B-NHL	168	90	120	730	0.35	moderate

Ascertainment in lymphoma groups. Patients whose only post-first-line retreatment evidence was weak were excluded from the risk set rather than counted as event-free. Because second-line identification was unreliable in BL/LBL and HCL, these patients were excluded unless a LYFO second-line date created an event; BL/LBL, HCL and amyloidosis were consequently excluded from all second-line-dependent outcomes.

CLL/SLL and MM. The event was the earliest second-line date in the curated CLL or MM treatment-line data (matched on disease group), if later than the index date. CLL/SLL and MM patients without such a date were event-free, assuming completeness of the curated data. Amyloidosis was not included.

Censoring. Second-line treatment was censored at the earlier of the date of death and 19 March 2026; an event counted only if on or before this date, and patients censored on or before the index date were excluded. Death was not part of the endpoint and was handled as a competing event (Section D8). The endpoint is time to next qualifying treatment, not directly observed progression.

D5.2 Treatment failure and transformation

Treatment failure was the earlier of second-line treatment and death. **Treatment failure or transformation** was the earliest of second-line treatment, death and a new haematological malignancy. Transformation was thus operationalised as a new three-character ICD-10 block among C81–C96 and D45–D47 first recorded more than 31 days after the index date and not recorded on or before index +31 days; it is not histologically confirmed transformation and also captures unrelated second haematological neoplasms. Both composites inherited the second-line ascertainment rules and the exclusion of BL/LBL, HCL and amyloidosis (Section D10).

D5.3 Overall survival and new malignancy

Overall survival was the time to death on or before 19 March 2026, censored at the administrative censoring date. A new haematological malignancy and a new malignancy of any type were defined by the first post-index diagnosis in a three-character ICD-10 block never recorded on or before index +31 days; codes registered around treatment start, which typically reflect the presenting disease, were thereby not counted. Relapse coded in the same block therefore never counted, whereas a new block (for example, FL, C82, to DLBCL, C83) did.

Supplementary Table D5: Disease-course endpoints (6 outcomes). OP, outcome population (Section D8).

Outcome	Definition	Source	Ascertainment and censoring	Competing event
Overall survival	Death on or before 19 March 2026.	Patient table	OP; administrative.	None
Second-line treatment	As above.	Treatment data, curated CLL/MM lines, LYFO	Disease-specific; censored at death or cutoff; excludes BL/LBL, HCL, amyloidosis.	Death
Treatment failure	Earliest of second-line treatment and death.	Composite	Intersection of components; earliest component censoring.	None
Treatment failure or transformation	Earliest of second-line treatment, death and new haematological malignancy.	Composite	As above.	None
New haematological malignancy	New ICD-10 block among C81–C96, D45, D46, D47 after index +31 days.	DALY-CARE haematological diagnoses	OP; administrative.	Death
New malignancy (any)	New ICD-10 block among C00–C97, except C44 and C80, after index +31 days.	DNPR diagnoses	OP; administrative.	Death

D6 Treatment-related and acute-care outcomes

D6.1 Outcome classes

Each outcome belongs to one of the following classes.

Graded laboratory (DALY-CARE grading).
Laboratory values were graded by the DALY-CARE standardised adverse-event grading algorithms, whose upstream graders cite CTCAE v5 (correspondence with CTCAE in Section D7). The event was the first post-index measurement at or above the stated grade.

Local laboratory threshold.
As above, with the numeric threshold stated in the table.

Baseline-relative laboratory.
The event was defined relative to the patient's own pre-index value. The baseline was the median value on the latest day with a measurement in the 365 days strictly before the index date; patients without a gradable baseline were excluded. Many patients with lymphoid malignancy have abnormal values before treatment (for example, through hepatic or renal involvement); a fixed threshold would then record a pre-existing abnormality at the first post-index draw as an event, whereas a baseline-relative threshold requires worsening.

Registry diagnosis or procedure.
The first post-index ICD-10 diagnosis or SKS procedure code. For *new-diagnosis* outcomes, a post-index code counted only if its three-character block had never been recorded on or before the index date (or index + grace, where stated).

Medication-use proxy.
Repeated antimicrobial administration or dispensing (Section D6.2).

Composite.
Earliest of its components (Section D10).

Except for the baseline-relative definitions and the alkaline phosphatase tiers, laboratory events are the first post-index documentation of an abnormal value, not new onset or worsening: abnormal baseline values were not excluded. No outcome is a verified CTCAE v6 grade; grade labels denote the output of the grading algorithm ("CTCAE-aligned"). Prediction horizons were 30, 60, 90, 180, 365 and 730 days, all prespecified; pooled summaries weigh every horizon equally, and the main figures show 365 days. For every outcome in Supplementary Tables D6–D12, death was a competing event.

Abbreviations in Supplementary Tables D6–D12. *Sources*: DNPR-Dx, National Patient Registry diagnoses; DNPR-Proc, National Patient Registry procedures (LPR2 and LPR3); Lab, laboratory results; SMR, Hospital Medicines Register; Rx, National Prescription Registry. *Baseline*: Lab rule, shared laboratory ascertainment rule (Section D8.1); OP, all patients in the outcome population. *Ascertainment*: Admin, censored at the administrative censoring date; Lab, laboratory censoring (Section D8.1). *Class*: Grd, DALY-CARE grading; Loc, local threshold; Rel, baseline-relative; Reg, registry; Prx, medication-use proxy; Cmp, composite.

D6.2 Infection and antimicrobial use

Diagnosis-based infection outcomes used the first post-index code without exclusion of prior episodes, because infections are acute events and a new post-index episode is clinically relevant even after an earlier one. The medication-use proxies grouped administrations (SMR, intravenous route only) or dispensings (prescription registry) of antimicrobials (DALY-CARE ATC selection) into episodes separated by gaps of more than 7 days; an episode qualified if it spanned at least 1 day, that is, $\geq$2 administrations or dispensings on different days. The event was the first date of the first qualifying post-index episode. No prophylaxis exclusion was applied to intravenous administrations; for prescriptions, ATC J01EE01 (sulfamethoxazole–trimethoprim), J05AB01 (aciclovir), J05AB11 (valaciclovir) and the exact code P02 were excluded as typical prophylaxis. These outcomes measure repeated antimicrobial use, not confirmed infection, days of supply or treatment intent. Because SMR reporting began on 15 May 2018, absence of intravenous administrations before that date is not evidence that none occurred; the intravenous proxy and the composite that contains it were therefore restricted to patients with index date on or after 15 May 2018.

Supplementary Table D6: Infection and antimicrobial-use outcomes (6 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Sepsis	ICD-10 A40, A41; first post-index code.	DNPR-Dx	OP	Admin	Reg
Serious infection	ICD-10 A40, A41, B44, B45, B46, B59 (sepsis, invasive fungal infection, pneumocystosis) as one code set.	DNPR-Dx	OP	Admin	Reg
Antibiotic treatment (procedure)	SKS procedure codes beginning BPH; dated by procedure start.	DNPR-Proc	OP	Admin	Reg
Intravenous antimicrobial use	Qualifying episode of intravenous antimicrobial administration.	SMR	OP; index date $\geq$ 15 May 2018	Admin	Prx
Prescription antimicrobial use	Qualifying episode of antimicrobial dispensing, excluding prophylaxis codes.	Rx	OP	Admin	Prx
Infection diagnosis or antimicrobial use	Earliest of serious infection, prescription and intravenous antimicrobial use.	Composite	Intersection; index date $\geq$ 15 May 2018	Earliest component	Cmp

D6.3 Blood counts, cytopenia and transfusion

Supplementary Table D7: Blood count, cytopenia and transfusion outcomes (16 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Anaemia, grade $\geq$2; grade $\geq$3	Haemoglobin graded $\geq$2 / $\geq$3.	Lab	Lab rule	Lab	Grd
Neutropenia, grade $\geq$2; grade $\geq$3	Neutrophil count decreased, $\geq$2 / $\geq$3.	Lab	Lab rule	Lab	Grd
Leukopenia, grade $\geq$2; grade $\geq$3	White blood cell count decreased, $\geq$2 / $\geq$3.	Lab	Lab rule	Lab	Grd
Thrombocytopenia, grade $\geq$2; grade $\geq$3	Platelet count decreased, $\geq$2 / $\geq$3 (grade 3/4 boundary 25×10^9/L, CTCAE v5).	Lab	Lab rule	Lab	Grd
Lymphocytosis, grade $\geq$2; grade $\geq$3	Lymphocyte count increased, $\geq$2 / $\geq$3. In CLL largely reflects disease activity.	Lab	Lab rule	Lab	Grd
Eosinophilia	Eosinophil count increased, grade $\geq$1.	Lab	Lab rule	Lab	Grd
Haptoglobin decreased	Haptoglobin decreased (binary grading). Not a diagnosis of haemolysis.	Lab	Lab rule	Lab	Grd
Severe cytopenia	Earliest of neutropenia, thrombocytopenia and anaemia, each grade $\geq$3.	Composite	Intersection	Earliest component	Cmp
Red-cell transfusion	SKS BOQA0, BOQA00–BOQA05; dated by procedure start.	DNPR-Proc	OP	Admin	Reg
Platelet transfusion	SKS BOQC, BOQC3.	DNPR-Proc	OP	Admin	Reg
Any transfusion	Union of the red-cell and platelet code sets (single code-set outcome).	DNPR-Proc	OP	Admin	Reg

D6.4 Renal

Supplementary Table D8: Renal outcomes (5 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Acute kidney injury (first documented)	First post-index start of an acute kidney injury episode derived from creatinine by the DALY-CARE algorithm.	Lab	Lab rule	Lab	Grd
Creatinine increased, grade ≥2; grade ≥3	Creatinine graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Creatinine rise >1.5× pre-index baseline	Creatinine (µmol/L) >1.5 times baseline on any post-index day: the baseline arm of CTCAE v5 grade ≥2 alone, not an acute kidney injury diagnosis.	Lab	Lab rule; baseline value >0 (no reference-limit requirement)	Lab	Rel
Dialysis	SKS procedure codes beginning BJFD.	DNPR-Proc	OP	Admin	Reg

D6.5 Hepatic and coagulation

Supplementary Table D9: Hepatic and coagulation outcomes (9 outcomes). ULN, upper limit of normal.

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
ALT increased, grade ≥2; grade ≥3	ALT (U/L), with reference ULN >0. If baseline > its ULN: value >1.5 / >2.0 times baseline. Otherwise: value >3 / >5 times the ULN of the measurement.	Lab	Lab rule; gradable baseline (value >0, baseline ULN >0)	Lab	Rel
Alkaline phosphatase elevation, tier ≥2; tier ≥3	>2.5 / >5 times the sample ULN if the running mean of prior samples was normal, otherwise >2.5 / >5 times that prior mean; missing ULN set to 105 U/L. Time-varying baseline; CTCAE v6 grades ALP at grade 1 only.	Lab	Lab rule	Lab	Grd
Bilirubin increased, grade ≥2; grade ≥3	Bilirubin graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Hepatotoxicity	Earliest of ALT increased grade ≥3 (baseline-relative), AST increased grade ≥3 (a component only, not a panel outcome) and bilirubin increased grade ≥3.	Composite	Intersection, hence gradable ALT baseline	Earliest component	Cmp
INR increased, grade ≥2; grade ≥3	INR graded ≥2 / ≥3; anticoagulant therapy not considered.	Lab	Lab rule	Lab	Grd

D6.6 Electrolytes and metabolic

Supplementary Table D10: Electrolyte and metabolic outcomes (19 outcomes). LLN/ULN, lower/upper limit of normal of the measurement.

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Hyponatraemia, grade ≥2	Sodium <130 mmol/L and < LLN (LLN 136 mmol/L if missing). Symptoms not observed.	Lab	Lab rule	Lab	Loc
Sodium <125 mmol/L	Sodium <125 mmol/L and < LLN. Does not capture symptomatic hyponatraemia at 125 to <130 mmol/L.	Lab	Lab rule	Lab	Loc
Potassium <3.0 mmol/L	Potassium <3.0 mmol/L on any post-index day. Symptoms and intervention not observed.	Lab	Lab rule	Lab	Loc
Hypernatraemia, grade ≥2; grade ≥3	Sodium graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Hyperkalaemia, grade ≥2; grade ≥3	Potassium graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Hypercalcaemia, grade ≥2; grade ≥3	Calcium graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Hypocalcaemia, grade ≥2; grade ≥3	Calcium graded ≥2 / ≥3 (albumin correction not established).	Lab	Lab rule	Lab	Grd

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Hypomagnesaemia, grade $\geq$2	Magnesium (NPU02647) <0.5 mmol/L and < LLN (LLN 0.7 mmol/L if missing).	Lab	Lab rule	Lab	Loc
Hypertriglyceridaemia, grade $\geq$2; grade $\geq$3	Triglycerides graded $\geq$2 / $\geq$3; fasting status unknown.	Lab	Lab rule	Lab	Grd
Hypoalbuminaemia, grade $\geq$2; grade $\geq$3	Albumin graded $\geq$2 / $\geq$3.	Lab	Lab rule	Lab	Grd
Hyperphosphataemia	Phosphate > ULN (ULN 1.45 mmol/L if missing); NPU03096 used where the standard phosphate code was absent. Laboratory elevation only.	Lab	Lab rule	Lab	Loc
Hyperuricaemia	Urate > ULN; no event if the ULN was missing. Laboratory elevation only.	Lab	Lab rule	Lab	Loc
LDH increased	Lactate dehydrogenase > ULN; no event if the ULN was missing.	Lab	Lab rule	Lab	Loc

D6.7 Cardiovascular and thrombotic

New-diagnosis rule without grace period: a post-index code counted only if its three-character block had not been recorded on or before the index date. Blocks were evaluated separately, so that, for example, a prior I80 did not exclude a new I26.

Supplementary Table D11: Cardiovascular and thrombotic outcomes (4 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Atrial fibrillation	ICD-10 I48; new diagnosis.	DNPR-Dx	OP	Admin	Reg
Heart failure	ICD-10 I50; new diagnosis.	DNPR-Dx	OP	Admin	Reg
Stroke or TIA	ICD-10 I63, I64, G45; new diagnosis per block.	DNPR-Dx	OP	Admin	Reg
Venous thromboembolism	ICD-10 I26, I80, I81, I82; new diagnosis per block.	DNPR-Dx	OP	Admin	Reg

D6.8 Acute care and other events

Acute hospitalisation. The definition followed the Danish database of acute hospital contacts. In LPR2, inpatient admissions (acute or planned) were combined with outpatient and emergency-room contacts with an acute mode of admission (LPR2 patient type: inpatient, outpatient, or emergency room before 2014; from 2014 emergency-room contacts were registered as outpatient contacts with acute mode of admission). Missing admission and discharge hours were set to 09:00 and 21:00; outpatient and emergency contacts were treated as point events. In LPR3, physical attendances were used, acute if registered with acute priority; planned LPR3 contacts were retained only if they lasted $\geq$12 hours. Psychiatric specialties were excluded in both eras. Contacts separated by at most 4 hours were merged into one episode. An episode was *acute* if any of its contacts was acute, and a *hospitalisation* if it contained an LPR2 inpatient admission, lasted $\geq$12 hours or crossed midnight. The outcome comprised acute hospitalisations only, so a planned admission counted only if it was merged with an acute contact; planned admissions, which typically deliver treatment, were not outcome events. The event date was the start date of the first post-index acute hospitalisation.

Supplementary Table D12: Acute-care and other outcomes (4 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Acute hospitalisation	As described above.	DNPR admissions and contacts	OP	Admin	Reg
Emergency department stay	SKS procedure codes beginning NAB, excluding codes containing NAB1.	DNPR-Proc	OP	Admin	Reg
Respiratory support	SKS BGDA0, BGDA1, BGDA2, BGDA7.	DNPR-Proc	OP	Admin	Reg

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Bone fracture	ICD-10 S02, S12, S22, S32, S42, S52, S62, S72, S82, S92, T02, M80, M84.3, M84.4 (including subcodes); first post-index code, no exclusion of prior fractures.	DNPR-Dx	OP	Admin	Reg

D7 Correspondence with CTCAE

Laboratory outcomes were defined with reference to the NCI Common Terminology Criteria for Adverse Events, version 5.0 [2] and, where noted, 6.0 [3], using laboratory criteria only: the symptom, intervention and hospitalisation criteria of some grades cannot be observed in the data, and none changes a laboratory threshold below. An outcome is *derived* from CTCAE when its thresholds equal the laboratory thresholds of the stated grade, *analogous* when it follows the same grade but differs in the stated way, and has *no counterpart* otherwise. Graded outcomes come from the DALY-CARE graders (package commit a0742b0, every grader citing CTCAE v5); the version installed on the analysis server is to be confirmed. Table D13 lists every laboratory outcome; outcomes were not redefined. Where a DALY-CARE grader uses R's default interval rule, a value exactly at a cut-off falls in the more severe band ($\leq$ where CTCAE uses $<$), which matters for haemoglobin, reported to 0.1 mmol/L, and albumin, reported in whole g/L.

Supplementary Table D13: Correspondence of laboratory outcomes with CTCAE.

Outcome	CTCAE term and grade	Relation	Difference
Thrombocytopenia $\geq$2; $\geq$3	Platelet count decreased, v5	Derived	None; v6 changed only the grade 3/4 boundary.
Lymphocytosis $\geq$2; $\geq$3	Lymphocyte count increased, v5	Derived	None.
Haptoglobin decreased	Haptoglobin decreased, v5	Derived	None.
INR increased $\geq$2; $\geq$3	INR increased, v5	Derived	The arm for patients on anticoagulation is not applied.
Hyponatraemia $\geq$2	Hyponatremia, v5	Derived	None for grade $\geq$2.
Potassium $<$3.0 mmol/L	Hypokalemia grade $\geq$3, v5	Derived	Named by its threshold.
Hypernatraemia $\geq$2; $\geq$3	Hypernatremia, v5	Derived	None.
Hyperkalaemia $\geq$2; $\geq$3	Hyperkalemia, v5	Derived	None.
Hypomagnesaemia $\geq$2	Hypomagnesemia, v5	Derived	None.
Hypertriglyceridaemia $\geq$2; $\geq$3	Hypertriglyceridemia, v5	Derived	None (fasting not specified by CTCAE).
Hyperuricaemia	Hyperuricemia grade $\geq$1, v5	Derived	Unit conversion of urate to be confirmed.
LDH increased	Blood lactate dehydrogenase increased, v5	Derived	None.
ALT increased $\geq$2; $\geq$3	Alanine aminotransferase increased, v6	Derived	Equals v6 in both baseline arms (v5 uses $\times$3–5 of an abnormal baseline).
Anaemia $\geq$2; $\geq$3	Anemia, v5	Analogous	Values exactly at 6.2 or 4.9 mmol/L fall in the more severe grade.
Neutropenia $\geq$2; $\geq$3	Neutrophil count decreased, v5	Analogous	Values exactly at a cut-off fall in the more severe grade; v6 moved every grade down one step, so these are v6 grades $\geq$1 and $\geq$2.
Leukopenia $\geq$2; $\geq$3	White blood cell decreased, v5	Analogous	Values exactly at a cut-off fall in the more severe grade.
Hypoalbuminaemia $\geq$2; $\geq$3	Hypoalbuminemia, v5	Analogous	Values exactly at 30 or 20 g/L fall in the more severe grade.
Creatinine increased $\geq$2; $\geq$3	Creatinine increased, v5	Analogous	Arm relative to the upper limit of normal only; the arm relative to baseline is not applied.
Creatinine $>$1.5$\times$ baseline	Creatinine increased grade $\geq$2, v5	Analogous	Arm relative to baseline only; v6 restricts that arm to a baseline below the lower limit of normal.
Bilirubin increased $\geq$2; $\geq$3	Blood bilirubin increased, v5	Analogous	Arm relative to the upper limit of normal only.
Eosinophilia	Eosinophilia grade $\geq$1, v5	Analogous	The condition "above baseline" is not applied.
Hypercalcaemia $\geq$2; $\geq$3	Hypercalcemia, v5	Analogous	CTCAE grades corrected or ionised calcium; whether the total-calcium codes are albumin-corrected is to be confirmed (derived if so).
Hypocalcaemia $\geq$2; $\geq$3	Hypocalcemia, v5	Analogous	As for hypercalcaemia.
Sodium $<$125 mmol/L	Hyponatremia grade $\geq$3, v5	Analogous	Laboratory arm only; symptomatic values of 125–129 are not included.

Outcome	CTCAE term and grade	Relation	Difference
Hyperphosphataemia	Hyperphosphatemia grade 1, v5	Analogous	CTCAE gives no number for "laboratory finding only"; the upper limit of normal is used.
Alkaline phosphatase tier ≥2; ≥3	Alkaline phosphatase increased, v5	Analogous	v5 multiples applied to a running mean of earlier samples; v6 has grade 1 only, so these are named tiers, not grades.
Acute kidney injury	Acute kidney injury (clinical), v5	No counterpart	Episode algorithm from creatinine, close to KDIGO.
Severe cytopenia	–	No counterpart	Composite of grade ≥3 anaemia, neutropenia and thrombocytopenia.
Hepatotoxicity	–	No counterpart	Composite of ALT (v6), AST (v5) and bilirubin (v5) grade ≥3.

D8 Outcome-specific eligibility, ascertainment and censoring

Outcome population. All outcomes started from the first-line cohort restricted to patients with a follow-up record whose end of follow-up and data cutoff both fell after the index date (37,531 patients; Section D2).

D8.1 Laboratory ascertainment

Laboratory outcomes are only observable in patients whose blood is tested. Absence of an abnormal value in a patient without measurements is not evidence that the abnormality did not occur. All graded, local-threshold and baseline-relative laboratory outcomes and acute kidney injury therefore used a shared ascertainment rule based on the dates of *any* laboratory measurement (not only the outcome analyte). A patient was eligible if all three conditions held:

- *source coverage*: index date on or after 1 January 2017 (the date from which laboratory capture was regarded as complete), or any measurement between 365 days before the index date and the administrative censoring date;
- *baseline*: any measurement between 365 days before the index date and the index date inclusive;
- *entry*: any measurement between the index date and 30 days after it, inclusive, *or* death within 30 days of the index date with index date on or after 1 January 2017 (so that early death did not remove patients from the risk set).

A single measurement on the index date satisfied both the baseline and the entry condition. Observing an event did not override these criteria.

Laboratory censoring. Patients with an event were censored at the administrative censoring date. Event-free patients were censored at the earlier of their last post-index measurement of any analyte plus 180 days and the administrative censoring date, so that time without laboratory monitoring was not credited as event-free. The ascertainment status of each patient was recorded (before laboratory coverage, missing baseline, no post-index measurement, insufficient monitoring, eligible with event, eligible with competing death, eligible event-free).

Baseline-relative and supplementary threshold outcomes. The risk set was that of the corresponding graded outcome (creatinine increased grade ≥2; ALT increased grade ≥2; for potassium <3.0 mmol/L, hypokalaemia grade ≥3), further restricted to patients with a gradable pre-index baseline where required, and censored by the laboratory rule.

D8.2 Non-laboratory outcomes

Diagnosis, procedure, hospitalisation, medication-use and survival outcomes were administratively ascertained: all patients in the outcome population were eligible and censored at the administrative censoring date. The only calendar restriction was the SMR start date (index date on or after 15 May 2018) for the intravenous antimicrobial-use proxy and the infection composite that contains it. Diagnosis, procedure and hospitalisation outcomes were not restricted to one register era, so calendar trends in registry capture are part of these labels.

D8.3 Competing death and horizon labels

For a fixed horizon h, a patient was a case if the event occurred by h and before any competing death; a known non-case if follow-up reached h or a competing death occurred by h; and otherwise of unknown status (right-censored) and excluded from the fixed-horizon loss and from binary metrics. Death precludes the subsequent

occurrence of every non-fatal outcome; treating it as ordinary censoring would overstate the risk of these outcomes, whereas treating it as a known non-case keeps patients who die early in the denominator. A death counted as a competing event only if it occurred on or after the index date, on or before the outcome's censoring date and within the horizon. For laboratory outcomes, an event-free patient who died more than 180 days after the last measurement was therefore censored rather than counted as a competing death. Endpoints that include death (overall survival and both treatment-failure composites) had no competing event.

D9 Selection of the outcome panel

The panel was selected from 87 candidate outcomes using training and tuning data only. Each outcome needed at least 100 events within a year of the index date in training and 20 in tuning, and clinical judgement of the definition took precedence over support (Supplementary Table D14). Test-period counts played no part in the selection.

Supplementary Table D14: Selection of the outcome panel from 87 to 69 outcomes.

Step	n	Outcomes and reason
Candidate outcomes	87	
Definition not reliable	−6	Hypofibrinogenaemia grade ≥2 and grade ≥3 (sparse; assay and grading not verified); leukocytosis grade ≥2 (grading did not match the label, and the term is discouraged for disease-related rises); DIC-like coagulopathy, Hy's-law liver injury and tumour lysis syndrome (laboratory co-occurrence proxies without monitoring-based ascertainment).
Definition concerns, excluded	−5	Lymphopenia grade ≥2 and grade ≥3 (CTCAE v6 has no grade 2/3 tier); hypokalaemia grade ≥2 and grade ≥3 (local thresholds ≤3.0 and ≤2.5 mmol/L, not CTCAE v6); major adverse cardiovascular events (composite included atrial fibrillation, venous thromboembolism and heart failure).
Below support threshold	−9	AST increased grade ≥2 and grade ≥3; herpes zoster; hypermagnesaemia grade ≥3; hypoglycaemia grade ≥2 and grade ≥3; hypomagnesaemia grade ≥3; invasive fungal infection; myocardial infarction.
Met support, no concern	60	
Retained, renamed	+7	Named for what they measure: alkaline phosphatase elevation tier ≥2/≥3; sodium <125 mmol/L (for hyponatraemia grade ≥3); intravenous, prescription and combined infection outcomes as antimicrobial-use proxies; hyponatraemia grade ≥2. ALT increased and hepatotoxicity use the baseline-relative ALT definition.
Added	+2	Creatinine rise >1.5× pre-index baseline; potassium <3.0 mmol/L.
Final panel	69	

D10 Composite outcomes, endpoint groups and display groups

Composite rule. A patient was eligible for a composite only if eligible for every component (intersection of risk sets). The composite was censored at the earliest component censoring date, and the event was the earliest component event, counted only if on or before that censoring date. Components had to agree on index date and split. Death was an event only when overall survival was a component; otherwise death remained censoring in each component and was handled as a competing event. The composites were treatment failure, treatment failure or transformation, severe cytopenia, hepatotoxicity, and infection diagnosis or antimicrobial use. Any transfusion and serious infection were defined as single code-set outcomes rather than composites.

Endpoints. The training loss and the checkpoint criterion (the tuning loss) averaged the horizon-wise binary cross-entropy within each outcome and then equally over endpoints. Endpoints combined the grade variants of one endpoint (grade ≥2 and grade ≥3), the two alkaline phosphatase tiers, and hyponatraemia grade ≥2 with sodium <125 mmol/L, giving 52 endpoints for the 69 outcomes. Related diagnoses, proxies and composites were not merged.

Clinical domains. Each outcome was assigned once to one of 12 clinical domains: disease course and later malignancy; infection and antimicrobial use; erythroid findings and transfusion; neutrophil, white-cell and platelet findings; lymphocyte and eosinophil counts; renal injury and support; liver injury and coagulation; sodium and potassium; calcium and magnesium; metabolic and cell-turnover markers; cardiovascular and thrombotic events; and acute care and other events. Creatinine increase was placed under renal injury and support and hypokalaemia under sodium and potassium. Domains organise the figures, define the sensitivity

analysis that weighs every domain equally, and group outcomes in the borrowing analysis; they do not enter training.

D11 Limitations of the outcome definitions

1. **Proxies rather than adjudicated events.** Antimicrobial-use outcomes measure repeated antimicrobial administration or dispensing, not confirmed infection; the prescription proxy has no calendar restriction and intravenous prophylaxis was not excluded. Transformation is a new haematological ICD-10 block, not histologically confirmed transformation. Second-line treatment is time to next qualifying treatment, not observed progression.
2. **CTCAE-aligned, not CTCAE.** Laboratory grades derive from the DALY-CARE grading algorithms (all citing CTCAE v5) and local thresholds, some using reference limits with fixed fallback values. Symptoms, interventions and clinical context required by CTCAE for several grades are not observed; no outcome is a verified CTCAE v6 grade [2, 3].
3. **Prevalent versus incident abnormalities.** Except for the baseline-relative definitions, laboratory events are the first post-index documentation of an abnormal value, not incident toxicity.
4. **Analyte-agnostic laboratory ascertainment.** Eligibility and censoring use measurements of any analyte, so a patient can be eligible for an outcome whose analyte was rarely measured.
5. **Coding lag and retrospective disease identity.** Disease labels can use codes registered up to 14 days (pass 1) or 90 days (pass 2) after the index date and are used as model input.
6. **First-line identification.** The search is anchored at the first DALY-CARE diagnosis of any kind and limited to 10 years in pass 1; treatment records without indication are retained as uncertain; curated anchors exist for CLL and MM only.
7. **Registry era.** Diagnosis, procedure and hospitalisation outcomes are not restricted across LPR2 and LPR3, so changes in coding practice over calendar time are part of the labels; for LPR3 contacts without recorded clock times, the 12-hour criterion reduces to ending on a later calendar day. Diagnosis-based infection and fracture outcomes do not exclude continued coding of a pre-index episode.
8. **Second-line ascertainment.** Second-line treatment depends on the completeness of the curated CLL and MM data, excludes patients with only weak retreatment evidence (enriching the risk set for events) and is censored at death or the data cutoff rather than at end of follow-up as for other outcomes.
9. **Tumour lysis syndrome** is not part of the panel, because no laboratory definition could be ascertained reliably.

References

[1] Christian Brieghel, Mikkel Werling, Casper Møller Frederiksen, Mehdi Parviz, Thomas Lacoppidan, Tereza Faitova, Rebecca Svanberg Teglgaard, Noomi Vainer, Caspar da Cunha-Bang, Emelie Curovic Rotbain, Rudi Agius, and Carsten Utoft Niemann. The Danish Lymphoid Cancer Research (DALY-CARE) data resource: the basis for developing data-driven hematology. *Clinical Epidemiology*, 17:131–145, 2025. doi: 10.2147/CLEP.S479672.

[2] National Cancer Institute. *Common Terminology Criteria for Adverse Events (CTCAE), Version 5.0*, 2017. URL `https://dctd.cancer.gov/research/ctep-trials/for-sites/adverse-events/ctcae-v5-5x7.pdf`. Published 27 November 2017.

[3] National Cancer Institute. *Common Terminology Criteria for Adverse Events (CTCAE), Version 6.0*, 2025. URL `https://dctd.cancer.gov/research/ctep-trials/for-sites/adverse-events/ctcae-v6.pdf`. Published 22 July 2025.